Training dynamics of tree tensor networks: saddle-to-saddle, lazy and rich regimes

TTN regimes project*

Draft of October 10, 2026

Abstract

[Abstract to be written last: question, model (TTN regression with exact population loss), the regimes found and the parameters that select them, main new results with their status.]

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*Working draft; authors to be added.

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Conventions. Every claim carries a status tag: proven (proof given or cited), derived (derivation under stated assumptions), empirical (supported by experiments only) or conjecture (stated without sufficient derivation or evidence). Results we believe to be new are marked new, known results known with a reference; unmarked statements are definitions or standard background. Notation is defined in Section 2 and collected in `macros.tex`.

1 Introduction and summary of findings

[Skeleton. To be written once the results sections exist; intended content is in the TODO comments of this file.]

1.1 Question and motivation

[Which training regimes do tree tensor networks show under gradient methods, which parameters select them, and how far theory predicts them; the main focus is saddle-to-saddle dynamics.]

1.2 Summary of findings

[One table with one row per question (Q1–Q8 of the research plan): finding, status tag, new or known, and where it is shown.]

1.3 Relation to prior work

[What is known (deep linear networks, diagonal networks, tensor factorization, lazy versus rich, tensor networks for learning) and what is new here.]

1.4 Outline

[Roadmap of the report.]

2 Setup: model, data, teacher, training and diagnostics

This section fixes the model studied throughout and the measurements used to classify training regimes. Every experiment uses these definitions; deviations are stated where they occur.

2.1 Tree tensor network regression

Inputs. An input x consists of N sites. Site $i \in \{1, \dots, N\}$ carries a local feature vector $\phi_i(x) \in \mathbb{R}^{d_i}$ (by default $d_i = d$). Unless stated otherwise $\phi_i(x) = x^{(i)}$ with $x^{(i)} \sim \mathcal{N}(0, \Sigma_i)$ independent across sites, and the inputs are *whitened*, $\Sigma_i = I$. We write

$$\Phi(x) = \phi_1(x) \otimes \cdots \otimes \phi_N(x) \in \mathbb{R}^{d_1 \times \cdots \times d_N}, \qquad \Sigma = \Sigma_1 \otimes \cdots \otimes \Sigma_N.$$

Trees and cores. A tree tensor network (TTN) is specified by a rooted tree $\mathcal{T}$ with node set $\mathcal{V}$, $|\mathcal{V}| = K$, and an assignment of every site to exactly one node. For a node v, S_v is the (possibly empty) set of sites attached to v, $\mathrm{ch}(v)$ the set of its children and $p(v)$ its parent. Every non-root node v is joined to its parent by the edge (bond) e_v of *bond dimension* χ_v; for the root we set $\chi_{\mathrm{root}} = 1$. Node v carries a *core* G_v, a tensor with one *physical leg* of dimension d_i for every $i \in S_v$, one leg of dimension χ_c for every child $c \in \mathrm{ch}(v)$, and one leg of dimension χ_v for its parent bond (for

the root, the scalar output). The legs other than the parent leg are the *child legs*; their dimensions multiply to the *fan-in*

$$F_v = \prod_{i \in S_v} d_i \prod_{c \in \mathrm{ch}(v)} \chi_c,$$

so G_v can be viewed as an $F_v \times \chi_v$ matrix. The parameters are $\theta = (G_v)_{v \in \mathcal{V}}$, with $\sum_v F_v \chi_v$ entries in total.

Model. The tensor $W(\theta) \in \mathbb{R}^{d_1 \times \cdots \times d_N}$ is the contraction of all cores over all bonds, and the model is

$$f_\theta(x) = \langle W(\theta),\, \Phi(x) \rangle. \tag{1}$$

Equivalently, f_θ is computed by passing messages $m_v(x) \in \mathbb{R}^{\chi_v}$ from the leaves to the root,

$$m_v(x)_a = \sum_{(j_i),(b_c)} G_v\big[(j_i)_{i \in S_v}, (b_c)_{c \in \mathrm{ch}(v)}, a\big] \prod_{i \in S_v} \phi_i(x)_{j_i} \prod_{c \in \mathrm{ch}(v)} m_c(x)_{b_c}, \qquad f_\theta(x) = m_{\mathrm{root}}(x). \tag{2}$$

Two structural facts drive the dynamics. The model is multilinear in the site features, and W is multilinear in the cores (each core appears once), so $\theta \mapsto W(\theta)$ is a homogeneous polynomial of degree K: rescaling the cores, $G_v \mapsto c_v G_v$, rescales W by $\prod_v c_v$.

Bipartitions and bond spectra. Let A_v be the set of sites in the subtree rooted at v. The edge e_v splits the sites into A_v and its complement, and $W^{\langle e_v \rangle}$ denotes the matricization of W with the sites in A_v as row indices. Contracting the subtree below v and the rest of the network separately gives a factorization $W^{\langle e_v \rangle} = T_v C_v^\top$ with $T_v \in \mathbb{R}^{(\prod_{i \in A_v} d_i) \times \chi_v}$, so $\operatorname{rank} W^{\langle e_v \rangle} \le \chi_v$ (Hackbusch and Kühn, 2009; Grasedyck, 2010). The *bond spectrum* (Schmidt values) of an edge $e = e_v$ is the list $s_1^e \ge s_2^e \ge \cdots \ge s_{\chi_v}^e \ge 0$ of the χ_v largest singular values of $W^{\langle e \rangle}$. All nonzero singular values are among them, so $\sum_k (s_k^e)^2 = \|W\|_F^2$ on every edge. With anisotropic inputs we use function-space spectra, the singular values of $(\Sigma^{1/2} W)^{\langle e \rangle}$ with $\Sigma^{1/2} = \Sigma_1^{1/2} \otimes \cdots \otimes \Sigma_N^{1/2}$. Bond spectra are computed from the cores by QR sweeps, without forming W, to about 10^{-15} relative accuracy.

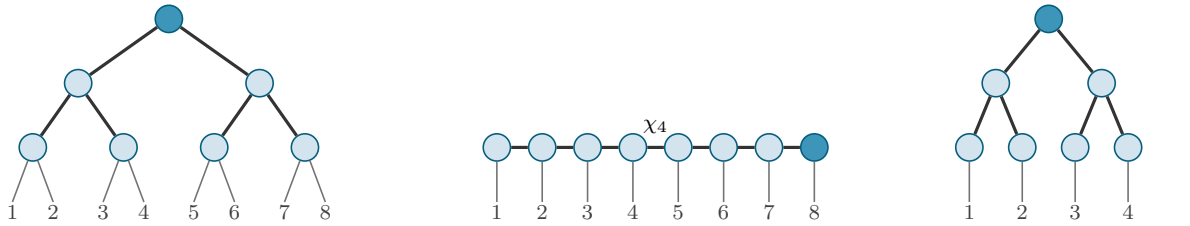
Why this model. It is the simplest genuine TTN. (i) With at least three sites and an internal bond, W is a genuine tensor rather than a matrix; a deep linear network is the special case of a chain with physical legs only at its two ends. (ii) Whitened inputs reduce the population loss to a Frobenius distance (Section 2.3), the exact analogue of the setting in which Saxe et al. (2014) solved deep linear networks, so that theory provides baselines. (iii) It is the regression version of the tensor-network models used in machine learning: the matrix product state (MPS) models of Stoudenmire and Schwab (2016) have the same form $\langle W, \bigotimes_i \phi_i(x) \rangle$, with non-Gaussian local features. (iv) With whitened inputs it is gradient flow on a fully observed hierarchical tensor factorization, whose implicit regularization was studied by Razin et al. (2022).

2.2 Tree topologies

Table 1 lists the built-in topologies and Figure 1 draws them. A binary tree and an MPS with the same N are the two extremes of tree shape at nearly equal K ($N - 1$ versus N) and are the default pair for saddle-to-saddle studies. The hierarchical Tucker (HT) format adds one leaf core per site; when the leaf bond dimension equals d this adds no expressivity, but it changes K and the number of cores that must change to learn a component. Any parent array with a site assignment defines a

Table 1: Built-in tree topologies for N sites (K cores). Depth is the number of edges on the longest root-to-leaf path. Binary-tree values are for $N = 2^m$; for other N the sites are split recursively into halves of sizes $\lceil N/2 \rceil$ and $\lfloor N/2 \rfloor$ until groups of at most two sites remain, each of which becomes a bottom core.

name	structure	K	depth
binary	balanced binary tree; each bottom core carries two adjacent sites, internal cores carry none	$N-1$	$\log_2 N - 1$
MPS	chain (caterpillar), the most unbalanced tree: core k carries site k and its parent is core $k+1$, so the root is core N	N	$N-1$
HT	hierarchical Tucker: balanced binary tree over N leaf cores, one site per leaf core; internal cores carry none	$2N-1$	$\lceil \log_2 N \rceil$
custom	any parent array and site assignment	–	–



(a) binary, $N = 8$, $K = 7$ **(b)** MPS, $N = 8$, $K = 8$ **(c)** HT, $N = 4$, $K = 7$

Figure 1: The built-in tree topologies (Table 1). Circles are cores G_v; the shaded core is the root, whose parent leg (dimension one, the scalar output) is not drawn. Thick lines are bonds: the edge e_v between core v and its parent has dimension χ_v (one label shown in (b), where cores are numbered $1, \ldots, 8$ from the left). Thin dangling lines are physical legs, labelled by their site i (dimension d_i).

custom tree. Unless stated otherwise every edge has the same nominal bond dimension χ, capped at the largest rank its bipartition allows,

$$\chi_v = \min\Big(\chi,\ \prod_{i \in A_v} d_i,\ \prod_{i \notin A_v} d_i\Big). \tag{3}$$

2.3 Teacher, data and loss

Teachers. The target is realizable, $f^*(x) = \langle W^*, \Phi(x) \rangle$. A *CP teacher* has

$$W^* = \sum_{r=1}^{R} s_r U_r, \qquad U_r = u_r^{(1)} \otimes \cdots \otimes u_r^{(N)}, \qquad \|u_r^{(i)}\| = 1, \qquad s_1 \geq \cdots \geq s_R > 0. \tag{4}$$

Its inner products with a TTN do not depend on the TTN's tree (see (9)), and the following special cases have exactly known bond spectra.

- *Odeco* (orthogonally decomposable): $\langle u_r^{(i)}, u_{r'}^{(i)} \rangle = \delta_{rr'}$ at every site, which requires $R \leq \min_i d_i$. By default the factors at each site are columns of an independent Haar-random orthogonal matrix.
- *Pattern teachers*: each site has an orthonormal basis $(\beta_1^{(i)}, \ldots, \beta_{d_i}^{(i)})$ and component r uses $u_r^{(i)} = \beta_{\pi_r(i)}^{(i)}$ for a pattern $\pi_r \colon \{1, \ldots, N\} \to \{1, \ldots, d\}$. Odeco teachers are the case where the

patterns differ at every site; with $\pi_1 = (1,1,1,1)$ and $\pi_2 = (2,2,1,1)$ the second component differs from the first only on the sites $D = \{1,2\}$ (a *local* component).

- *Random CP*: $u_r^{(i)} = g/\|g\|$ with independent $g \sim \mathcal{N}(0, I)$, so the components are generically non-orthogonal.

A *TTN teacher* gives W^* as a TTN on the student's tree or on another tree; a random TTN teacher is a fan-in initialization (Section 2.4) with $\sigma = 1$, rescaled so that $\|W^*\|_F = 1$. The overlap of the student with component r of a CP teacher is

$$a_r = \langle W, U_r \rangle_\Sigma / s_r; \tag{5}$$

for an odeco teacher with whitened inputs, $a_r = 0$ at $W = 0$ and $a_r = 1$ at $W = W^*$.

Proposition 2.1 (Teacher bond spectra)**.** proven *Let e be an edge whose two sides both carry sites (true for every edge of the trees of Table 1). (a) For an odeco teacher the bond spectrum of W^* on e is $(s_1, \dots, s_R)$, for every tree. (b) For a two-component pattern teacher whose components differ exactly on the site set D, the bond rank of W^* on e is 2 if e lies in the smallest subtree of $\mathcal{T}$ containing the nodes that carry the sites of D, and 1 otherwise.*

Proof. Let A be the set of sites on one side of e and write $x_r = \bigotimes_{i \in A} u_r^{(i)}$, $y_r = \bigotimes_{i \notin A} u_r^{(i)}$, so that $(W^*)^{\langle e \rangle} = \sum_r s_r x_r y_r^\top$. (a) Both $(x_r)_r$ and $(y_r)_r$ are orthonormal families, so this is a singular value decomposition. (b) If $D \subseteq A$ or $D \cap A = \emptyset$, then $y_1 = y_2$ or $x_1 = x_2$ and the rank is one. Otherwise the two components differ (by orthogonal basis vectors) on a site of $D \cap A$ and on a site of $D \setminus A$, so $x_1 \perp x_2$, $y_1 \perp y_2$ and the rank is two. Finally, e separates two sites of D exactly when it lies on the tree path between the nodes carrying them, that is, when it belongs to the smallest subtree containing those nodes. □

The number of cores in that subtree is the quantity k of the locality hypothesis tested in Section 4.

Population loss. Training minimizes the population square loss

$$L(\theta) = \tfrac{1}{2}\,\mathbb{E}_x\big[(f_\theta(x) - f^*(x))^2\big] = \tfrac{1}{2}\|W(\theta) - W^*\|_\Sigma^2, \qquad \|A\|_\Sigma^2 = \langle A, \Sigma A \rangle. \tag{6}$$

The second equality holds because, by independence across sites, $\mathbb{E}[\Phi \otimes \Phi] = \bigotimes_i \mathbb{E}[\phi_i \phi_i^\top] = \Sigma$ (as an operator on $\mathbb{R}^{d_1 \times \cdots \times d_N}$). With whitened inputs L is half the squared Frobenius distance. Only second moments enter: every local feature map with $\mathbb{E}[\phi_i \phi_i^\top] = I$ gives the same loss and the same dynamics (for example Hermite features of a scalar input, or $(1, \xi_i)$ with $\xi_i \sim \mathcal{N}(0,1)$), and a non-orthonormal feature map is equivalent to $\Sigma_i = \mathbb{E}[\phi_i \phi_i^\top] \neq I$.

Closed form. Expanding (6),

$$L(\theta) = \tfrac{1}{2}\langle W, W \rangle_\Sigma - \langle W, W^* \rangle_\Sigma + \tfrac{1}{2}\langle W^*, W^* \rangle_\Sigma, \tag{7}$$

and every term is an exact contraction, with no sampling noise:

- For two TTNs W, W' on the same tree (bond dimensions may differ), environments $E_v \in \mathbb{R}^{\chi_v \times \chi'_v}$ are computed from the leaves to the root,

$$(E_v)_{aa'} = \sum G_v[(j_i), (b_c), a]\, G'_v[(j'_i), (b'_c), a'] \prod_{i \in S_v} (\Sigma_i)_{j_i j'_i} \prod_{c \in \mathrm{ch}(v)} (E_c)_{b_c b'_c}, \tag{8}$$

  and $\langle W, W' \rangle_\Sigma = E_{\mathrm{root}}$. The cost is linear in K at fixed d and χ.

- For a CP teacher, $\langle W, U_r\rangle_\Sigma$ is the student evaluated at the site vectors $\Sigma_i u_r^{(i)}$, so for any tree

$$\langle W, W^*\rangle_\Sigma = \sum_r s_r\, f_\theta(\Sigma_1 u_r^{(1)}, \dots, \Sigma_N u_r^{(N)}), \qquad \langle W^*, W^*\rangle_\Sigma = \sum_{r,r'} s_r s_{r'} \prod_i u_r^{(i)\top} \Sigma_i u_{r'}^{(i)}, \quad (9)$$

  where $f_\theta(z_1, \dots, z_N)$ denotes (2) with $\phi_i(x)$ replaced by z_i.

- A TTN teacher on another tree is glued to the student along the physical legs; the resulting network has loops and is contracted exactly along an optimized contraction path.

For an odeco teacher with whitened inputs, (7) reads $L = \frac{1}{2}\|W\|_F^2 - \sum_r s_r^2 a_r + \frac{1}{2}\sum_r s_r^2$. A *partial fit* $W_S = \sum_{r\in S} s_r U_r$ that has learned exactly the components in S has loss $\frac{1}{2}\sum_{r\notin S} s_r^2$; in particular, learning the components in decreasing order of s_r passes through the levels

$$L_k = \tfrac{1}{2}\sum_{r>k} s_r^2, \qquad k = 0, \dots, R. \quad (10)$$

These are the candidate plateau levels examined in Section 4.

Anisotropic and correlated inputs. Anisotropic product covariances are handled exactly by (6)–(9); the default family has power-law site spectra, $\Sigma_i = c_\beta\, \mathrm{diag}(1^{-\beta}, 2^{-\beta}, \dots, d^{-\beta})$ with c_β fixed by $\mathrm{tr}\,\Sigma_i = d$, aligned with the teacher's factors or rotated away from them. Cross-site correlations break the product structure of Σ; they are studied with sampled data only (Section 6).

Finite samples. For sample-size questions, a training set $(x_j, y_j)_{j=1}^n$ has $x_j \sim \mathcal{N}(0, \Sigma)$ i.i.d. and $y_j = f^*(x_j) + \varepsilon_j$ with optional label noise $\varepsilon_j \sim \mathcal{N}(0, \varsigma^2)$. The empirical loss is $\hat{L}_n(\theta) = \frac{1}{2n}\sum_j (f_\theta(x_j) - y_j)^2$, and the test error is the population loss (6).

2.4 Initialization and its variants

Fan-in initialization. The default initialization is

$$G_v = \frac{\sigma_v}{\sqrt{F_v}}\, Z_v, \qquad Z_v \text{ with i.i.d.} \mathcal{N}(0,1) \text{ entries}, \quad (11)$$

with a single per-core scale $\sigma_v = \sigma$ unless stated otherwise. Each core is then approximately σ_v times an isometry (when $F_v \gg \chi_v$), so that with $\sigma_v = 1$ the messages (2) keep their size on the way up the tree.

Proposition 2.2 (Initial scales). proven *Under* (11), $\mathbb{E}\|W(\theta_0)\|_F^2 = \prod_v \sigma_v^2$ *exactly,* $\mathbb{E}\|G_v\|_F^2 = \sigma_v^2 \chi_v$, *and hence* $\mathbb{E}\|\theta_0\|^2 = \sum_v \sigma_v^2 \chi_v$.

Proof. Let T_v be the contraction of the subtree below v with its parent leg open, a matrix with χ_v columns. Conditionally on the children's subtrees, the independence and variance σ_v^2/F_v of the entries of G_v give $\mathbb{E}[T_v^\top T_v] = \frac{\sigma_v^2}{F_v}\prod_{i\in S_v} d_i \prod_{c\in \mathrm{ch}(v)} \mathrm{tr}(T_c^\top T_c)\, I_{\chi_v}$. The children's subtrees are independent, so induction from the leaves gives $\mathbb{E}[T_v^\top T_v] = (\prod_u \sigma_u^2) I_{\chi_v}$, the product running over the subtree of v, because $F_v = \prod_{i\in S_v} d_i \prod_c \chi_c$. At the root $\chi_{\mathrm{root}} = 1$ and T_{root} is W flattened. The second claim is $\mathbb{E}\|G_v\|_F^2 = F_v \chi_v\, \sigma_v^2/F_v$. □

We therefore report both the per-core scale σ and the function-space scale $\alpha = \sigma^K$, the root-mean-square norm of $W(\theta_0)$. Small σ makes $W(\theta_0)$ exponentially small in K, while $\|\theta_0\|^2 \approx \sigma^2 \sum_v \chi_v$ grows with the bond dimensions.

Variants.

(a) *Imbalance*: per-core scales $\sigma_v = \sigma c_v$ with a fixed profile (c_v), for example root versus the rest, or by depth, compared at fixed function-space scale $\prod_v \sigma_v$.

(b) *Centred model*: the target is $W^* + W(\theta_0)$, that is, $f_\theta - f_{\theta_0}$ is trained on f^* (Chizat et al., 2019). The initial function is removed exactly at any σ without changing the parameterization, so σ alone moves the model from small to large initialization. Overlaps are measured on $W - W(\theta_0)$. This is the main construction of the regime map (Section 3).

(c) *Two-copy model* (Woodworth et al., 2020): $W(\theta^+) - W(\theta^-)$ with $\theta^+(0) = \theta^-(0) = \theta_0$, which vanishes at initialization for every σ. It is represented as one TTN on the same tree with doubled bond dimensions $2\chi_v$ and block-diagonal cores: core v holds G_v^+ on the block where all its bond indices are in the "+" half and G_v^- on the "−" block, except that the root holds G_{root}^+ and $-G_{\text{root}}^-$. It is used as a cross-check of the centred model; see Remark 2.3.

(d) *Aligned (decoupled) initialization* for CP teachers (Proposition 2.4): cores diagonal in the teacher's basis, giving exactly solvable Saxe-type dynamics.

(e) *Per-core learning-rate multipliers* η_v (Section 2.5). A fixed forward multiplier c_v on core v (writing $G_v = c_v H_v$ and training H_v) is equivalent to the initial scale $c_v\sigma_v$ together with the multiplier $c_v^2\eta_v$. This is how NTK-type and μP-type parameterizations for large bond dimension (Yang and Hu, 2021) are expressed (Section 5).

Remark 2.3 (The two-copy model needs a gradient mask). proven Gradient flow on the doubled TTN does not preserve its block-diagonal structure. For a root with children c_1, c_2, the gradient of L with respect to the root entry pairing index b_1 of the + half of bond e_{c_1} with index b_2 of the − half of bond e_{c_2} is $\langle W - W^*,\, T_{c_1}^+[:, b_1] \otimes T_{c_2}^-[:, b_2]\rangle_\Sigma$ (up to the ordering of sites), which at initialization equals $-\langle W^*,\, T_{c_1}^+[:, b_1] \otimes T_{c_2}^-[:, b_2]\rangle_\Sigma$ and is generically nonzero; mixed blocks of non-root cores also receive gradient. The two-copy dynamics is therefore the masked flow $\dot\theta = -m \odot \nabla L(\theta)$, with m the 0/1 indicator of the diagonal blocks, which keeps the blocks and equals gradient flow of $\frac{1}{2}\|W(\theta^+) - W(\theta^-) - W^*\|_\Sigma^2$ in (θ^+, θ^-).

Proposition 2.4 (Aligned initialization). proven known (deep linear networks: Saxe et al., 2014) *Let W^* be odeco, let inputs be whitened, let $\chi_v \geq R$ for every edge, and let every subtree and every subtree's complement carry at least one site (true for the trees of Table 1). Initialize*

$$G_v = \sum_{r=1}^{R} q_{v,r} \Big(\bigotimes_{i \in S_v} u_r^{(i)} \Big) \otimes \Big(\bigotimes_{c \in \mathrm{ch}(v)} \varepsilon_r \Big) \otimes \varepsilon_{r(v)}, \tag{12}$$

with ε_r the r-th standard basis vector of the bond in question, $r(v) = r$ for $v \neq \text{root}$ and $r(\text{root}) = 1$. Under gradient flow with multipliers η_v, the cores keep the form (12), $W = \sum_r \Pi_r U_r$ with $\Pi_r = \prod_v q_{v,r}$, $a_r = \Pi_r / s_r$, $L = \frac{1}{2}\sum_r (s_r - \Pi_r)^2$, and

$$\dot q_{v,r} = \eta_v \, (s_r - \Pi_r) \prod_{u \neq v} q_{u,r}. \tag{13}$$

With $\eta_v = 1$ the differences $q_{v,r}^2 - q_{u,r}^2$ are conserved; a balanced start $q_{v,r}(0) = q_r(0)$ stays balanced, and $\dot q_r = q_r^{K-1}(s_r - q_r^K)$.

Proof. On the family (12), contracting the basis vectors on the bonds gives $W = \sum_r \Pi_r U_r$, and the U_r are orthonormal. The derivative of W with respect to an entry of G_v is the tensor product of

basis vectors on the physical legs of v, the columns $T_c[:, b_c]$ of the child subtrees and the column of the environment above v indexed by the parent index a. On the family these columns are multiples of $\bigotimes_{i\in A_c} u_{b_c}^{(i)}$ (and of the analogous product over the sites outside A_v), so the inner product with $W - W^* = \sum_r (\Pi_r - s_r) U_r$ vanishes unless all bond indices equal the same r (with $a = 1$ at the root). The gradient therefore lies in the span of the family, and its component along $q_{v,r}$ is $(\Pi_r - s_r) \prod_{u\neq v} q_{u,r}$. Finally, with $\eta_v = 1$, $\frac{\mathrm{d}}{\mathrm{d}t} q_{v,r}^2 = 2(s_r - \Pi_r)\Pi_r$ is the same for every v. □

For $K \geq 3$ a balanced mode started at $q_r(0) \ll s_r^{1/K}$ reaches $a_r = \frac{1}{2}$ at

$$t_r = \frac{q_r(0)^{-(K-2)}}{(K-2)\, s_r} + O(s_r^{-(2-2/K)}), \tag{14}$$

and for $K = 2$ at $t_r = \log(s_r/q_r(0)^2)/(2s_r) + O(q_r(0)^2/s_r^2)$ (direct integration of the balanced ODE) proven. With $q_r(0) \propto \sigma$ the plateau before the first escape grows like $\sigma^{-(K-2)}$; this time scale is why the integration below must be adaptive and in double precision.

2.5 Training dynamics

Gradient flow is the reference dynamics,

$$\dot\theta = -D\, \nabla L(\theta), \tag{15}$$

where D is diagonal with entry $\eta_v m_j$ for every parameter j of core v: the η_v are the per-core learning-rate multipliers (default 1) and $m \in \{0,1\}^{\dim\theta}$ is an optional mask (Remark 2.3; default all ones). Here t always denotes gradient-flow time. It is integrated with the adaptive Dormand–Prince 5(4) pair (Dormand and Prince, 1980) (first-same-as-last, local extrapolation) in double precision, with relative and absolute tolerances 10^{-8} and 10^{-12} by default. A step of size h is accepted when the root-mean-square over components of the embedded error estimate, each divided by $10^{-12} + 10^{-8}\max(|\theta_j|, |\theta_j^{\mathrm{new}}|)$, is at most one; the next step is $h \min\{5, \max\{0.2, 0.9\,\mathrm{err}^{-1/5}\}\}$, with no increase right after a rejection, and the first step follows Hairer et al. (1993, Sec. II.4). Step sizes thus follow the local time scale $\|\theta\|/\|\dot\theta\|$, so that long plateaus are cheap. Diagnostics are recorded at $t = 0$ and at 200 log-spaced times from $t_{\min} = 10^{-3}\min(\tau_0, t_{\max})$ to $t_{\max}$, where $\tau_0 = \|\theta_0\|/\|\dot\theta(0)\|$; steps are shortened to land on these times exactly. A run stops at $t_{\max}$, when L falls below a prescribed floor, after a maximum number of steps, or if the step size underflows. Accuracy is checked with the conservation laws (Section 2.6) and against the closed-form aligned solutions (Proposition 2.4).

Gradient descent with learning rate η, $\theta_{k+1} = \theta_k - \eta D \nabla L(\theta_k)$, is recorded at $t = \eta k$, so it approaches gradient flow at equal t as $\eta \to 0$. A run has diverged when L becomes non-finite or exceeds $10^6 L(\theta_0)$.

Stochastic gradient descent uses the minibatch loss $\hat L_B(\theta) = \frac{1}{2B}\sum_{b=1}^{B}(f_\theta(x_b) - y_b)^2$ in the same update, again recorded at $t = \eta k$. Online SGD draws fresh inputs $x_b \sim \mathcal{N}(0, \Sigma)$ with $y_b = f^*(x_b)$ at every step; with a fixed training set, batches are taken from a new random permutation in every epoch. Optional label noise is drawn afresh at every step.

Precision. All computations are in double precision: with $\sigma = 10^{-2}$ and $K = 7$, $\|W(\theta_0)\|_F \approx 10^{-14}$ and the gradients are of order 10^{-12}, which single precision cannot resolve over long plateaus.

2.6 Symmetries and conservation laws

For a bond $e = e_v$ joining core v to its parent $p = p(v)$, let $G_{u,(e)}$ denote core $u \in \{v, p\}$ matricized with its e-leg as rows (a $\chi_v \times F_v$ matrix for $u = v$). For every invertible $B \in \mathrm{GL}(\chi_v)$ the *gauge transformation*

$$G_{v,(e)} \mapsto B\, G_{v,(e)}, \qquad G_{p,(e)} \mapsto B^{-\top} G_{p,(e)} \tag{16}$$

leaves W, and hence any loss that depends on θ only through W, unchanged.

Proposition 2.5 (Conservation laws). proven known (deep linear networks: Arora et al., 2018; Du et al., 2018) *Under gradient flow $\dot{G}_u = -\eta_u \nabla_{G_u} L$ of any loss that depends on θ only through W (population or empirical), the $\chi_v \times \chi_v$ matrix*

$$Q_e = \frac{1}{\eta_v} G_{v,(e)} G_{v,(e)}^\top - \frac{1}{\eta_p} G_{p,(e)} G_{p,(e)}^\top \tag{17}$$

is constant in time, for every bond $e = e_v$ with $p = p(v)$.

Proof. Write $X = G_{v,(e)}$, $Y = G_{p,(e)}$ and take $B = I + \epsilon M$ in (16), so that $X \mapsto X + \epsilon M X$ and $Y \mapsto Y - \epsilon M^\top Y + O(\epsilon^2)$. Invariance of L at first order in ϵ gives $\langle \nabla_X L\, X^\top - Y\,(\nabla_Y L)^\top,\, M \rangle = 0$ for every M, so $C := \nabla_X L\, X^\top = Y (\nabla_Y L)^\top$. Along the flow, $\frac{\mathrm{d}}{\mathrm{d}t}(XX^\top)/\eta_v = -(C + C^\top) = \frac{\mathrm{d}}{\mathrm{d}t}(YY^\top)/\eta_p$. □

This is the tree version of the balancedness of deep linear networks and an instance of the general correspondence between continuous symmetries and conserved quantities of gradient flow (Kunin et al., 2021). Consequences:

- *Norm balance*: $\mathrm{tr}\, Q_e = \|G_v\|_F^2/\eta_v - \|G_p\|_F^2/\eta_p$. With equal multipliers, the differences $\|G_u\|_F^2 - \|G_{u'}\|_F^2$ between any two cores are conserved (the tree is connected), so all cores grow together.
- *Balancedness at small initialization*: under (11), $\mathbb{E}\, Q_e(0) = (\sigma_v^2/\eta_v - \sigma_p^2 \chi_p/(\eta_p \chi_v))I$, so with small σ the cores remain balanced up to $O(\sigma^2)$ while they grow to $O(1)$. Imbalanced initializations or multipliers set Q_e to other values, a second knob for the regime map (Kunin et al., 2024).
- The masked two-copy flow conserves the Q_e of each copy (its gauge group is the block-diagonal subgroup). Gradient descent and SGD conserve Q_e only up to $O(\eta^2)$ per step.
- *Integrator check*: the conservation drift

$$\mathrm{drift}(t) = \frac{\max_e \|Q_e(t) - Q_e(0)\|_{\max}}{\max_e \|Q_e(0)\|_{\max} + \max_u \|G_u(t)\|_F^2/\eta_u}, \tag{18}$$

  with $\|\cdot\|_{\max}$ the largest absolute entry, measures the integration error relative to the size of the terms that cancel in Q_e. It must stay below 10^{-6} under gradient flow.

2.7 Regime diagnostics

Regimes are assigned from measured quantities, never by eye. The thresholds below are conventions; every phase diagram reports how its boundaries move when they change.

Raw quantities (Table 2) are recorded on the logarithmic time grid of Section 2.5, with extra points where needed. The tangent kernel uses a fixed probe set of $P = 128$ inputs $x_1, \ldots, x_P \sim \mathcal{N}(0, \Sigma)$ and the Jacobian $J_t = \partial(f_\theta(x_1), \ldots, f_\theta(x_P))/\partial\theta$ at $\theta(t)$. The gauge directions at θ are the tangent vectors of the orbits (16), $(\delta G_{v,(e)}, \delta G_{p,(e)}) = (M G_{v,(e)}, -M^\top G_{p,(e)})$ for all bonds e and $M \in \mathbb{R}^{\chi_v \times \chi_v}$; $H_\perp$ is the Hessian restricted to their orthogonal complement.

Table 2: Raw diagnostics recorded along every run.

symbol	definition
$L(t)$, $\ell(t)$	exact population loss (6); normalized loss $\ell(t) = L(t)/L(0)$
$s_k^e(t)$	bond spectrum of $W(t)$ across edge e; function-space version when $\Sigma \neq I$
$s_k^{e,*}$	bond spectrum of W^*; equal to the weights s_k on every edge for an odeco teacher (Proposition 2.1)
$a_r(t)$	overlap (5) with teacher component r (CP teachers); of $W - W(\theta_0)$ in the centred model
$\delta_\theta(t)$	$\|\theta(t) - \theta(0)\|/\|\theta(0)\|$, also per core
Θ_t	tangent kernel $J_t J_t^\top$ on the probe set; for small N also the tensor-space operator $\sum_v (\partial W/\partial G_v)(\partial W/\partial G_v)^\top$
$\delta_\Theta(t)$	relative kernel change $\|\Theta_t - \Theta_0\|_F/\|\Theta_0\|_F$
$\mathcal{A}(t)$	kernel-target alignment $y^\top \Theta_t y/(\|\Theta_t\|_F \|y\|^2)$, with $y_p = f^*(x_p)$ on the probe set
drift(t)	conservation drift (18), an integrator check
$H(t)$, $H_\perp(t)$	Hessian of L at $\theta(t)$, and its restriction to the complement of the gauge directions

Learning events. For a CP teacher, event r occurs at the first time t_r at which $a_r(t)$ crosses $\frac{1}{2}$, and its width is $w_r = t(a_r = 0.9) - t(a_r = 0.1)$. For a general teacher, event (e, k) occurs when $s_k^e(t)$ first crosses half of its final value, with widths defined in the same way from the crossings of 10% and 90% of the final value; since one component typically raises several bond spectra at once, events closer than their widths are merged. The *effective bond rank* is $\mathrm{rk}_e(t) = \#\{k : s_k^e(t) > 0.1\, s_{\min}^{e,*}\}$, where $s_{\min}^{e,*}$ is the smallest nonzero teacher Schmidt value on e; the threshold-free alternative is the entropy rank $\mathrm{rk}_e^H = \exp(-\sum_k p_k \log p_k)$ with $p_k = (s_k^e)^2/\sum_j (s_j^e)^2$.

Plateaus and the saddle-to-saddle signature. Number the events in time order, $t_1 < t_2 < \cdots$, with widths w_i, and define the *separation ratios*

$$\rho_i = \frac{t_{i+1} - t_i}{\max(w_i, w_{i+1})}. \tag{19}$$

Between two events with $\rho_i > 3$ the loss sits on a plateau at the level of a partial fit. The plateau midpoint t_i^{mid} is the time of minimal speed $\|\dot\theta(t)\|$ in $[t_i, t_{i+1}]$, and the Hessian is evaluated there. A run shows the *saddle-to-saddle (S2S) signature* when

(i) there are at least two events,

(ii) every separation ratio satisfies $\rho_i > 3$, and

(iii) across a sweep, plateau durations grow without bound as the initialization scale tends to zero (like $\log(1/\sigma)$ or a power of $1/\sigma$).

Plateaus are judged in linear time between events, not only on logarithmic time axes: an escape that needs two cores takes a time of order $\log(1/\sigma)$, which can be invisible on a log-time axis after a long first plateau.

Regime assignment. Each run, trained up to time T, receives one label by the rules of Table 3.

Remark 2.6 (Reading of the rules). "The loss converges" means $L(T) < 10^{-3} L(0)$, the complement of "stalled". Labels are checked in the order diverged, stalled, then the three converged regimes. Condition (iii) is a property of a sweep: a single run is labelled S2S on the basis of (i)–(ii), and

Table 3: Regime assignment for one run trained up to time T (the S2S signature is defined in Section 2.7). Threshold sensitivity: the kernel threshold 0.1 is varied over $\{0.05, 0.1, 0.2\}$ and the separation threshold 3 over $\{2, 3, 5\}$.

regime	rule
lazy	the loss converges and $\delta_\Theta(T) < 0.1$ (so δ_θ is small as well)
saddle-to-saddle	$\delta_\Theta(T) \geq 0.1$ and the S2S signature (i)–(iii)
rich, non-incremental	$\delta_\Theta(T) \geq 0.1$ and the loss converges, but some $\rho_i \leq 3$ (overlapping transitions)
stalled	$L(T) \geq 10^{-3} L(0)$ within the time budget (for example a long initial plateau)
diverged	GD and SGD only: L non-finite or above $10^6 L(0)$

a region of a phase diagram is reported as S2S only if (iii) holds there. A converged run with $\delta_\Theta(T) \geq 0.1$ and a single (merged) event is counted as rich, non-incremental.

Saddle type at a plateau. At a plateau midpoint, a *strict saddle* has $\lambda_{\min}(H_\perp) < -\epsilon$ with $\epsilon = 10^{-3} \max_j |\lambda_j(H_\perp)|$; a *higher-order (non-strict) saddle* has no eigenvalue below $-\epsilon$ although the loss later decreases. We report the number of negative eigenvalues, $\lambda_{\min}$ (also unnormalized) and $\lambda_{\min}/\lambda_{\max}$.

Reporting rules. Every scaling claim uses at least five initialization scales spanning at least one decade and at least five seeds per point, reports mean ± standard error and a fitted exponent with its standard error, and overlays the theoretical prediction. Every regime boundary comes with the threshold sensitivity check of Table 3. Every number states the dynamics that produced it (gradient flow with its tolerances, GD with its step size, or SGD with its step size and batch size).

2.8 What the results may depend on

- *Population versus finite samples.* Plateaus and saddle sequences are properties of the population gradient flow; statements about implicit bias and generalization need finite samples and are tested separately.
- *Multilinear features.* There is no pointwise nonlinearity; orthonormal local feature maps all give the same population dynamics, and non-orthonormal ones are equivalent to anisotropic Σ_i.
- *Product input measure.* It makes the population loss a tree contraction; cross-site correlations break this and are studied with sampled data.
- *Placement of physical legs.* Binary trees (two sites per bottom core) and HT (one leaf core per site) differ in K and in how many cores must change to learn a component, which is expected to change plateau-time exponents.
- *Scalar output.* Multi-output regression or classification would add an output leg to the root core.
- *Fan-in initialization.* It fixes the relative scales of the cores. Exponents should not depend on it, but prefactors and the order of learning might, which is why imbalanced and aligned initializations are also studied.

3 Map of regimes

[Skeleton. Phase diagrams of the training regimes over initialization scale, bond dimension, number of cores and core imbalance, with predicted boundaries and threshold sensitivity. Waits for the regime-map results.]

3.1 Protocol

[Centred model as the one-knob family; labels from Table 3; seeds, sweeps, time budgets.]

3.2 Initialization scale and bond dimension

[Phase diagram $\sigma \times \chi$ for binary trees with $N = 4, 8$.]

3.3 Number of cores and function-space scale

[Phase diagram $\sigma \times K$; boundaries in both σ and $\alpha = \sigma^K$.]

3.4 Imbalance between cores

[Does the conserved imbalance Q_e move the TTN between regimes, as in deep linear networks?]

3.5 Large bond dimension

[NTK-type versus μP-type parameterizations as χ grows.]

3.6 Threshold sensitivity and cross-checks

[How the boundaries move with the kernel threshold $\{0.05, 0.1, 0.2\}$ and the separation threshold $\{2, 3, 5\}$; two-copy cross-check.]

3.7 Summary: which parameter moves which boundary

[Table: parameter, boundary moved, direction, fitted versus predicted.]

4 The saddle-to-saddle regime

[Skeleton. Theory and experiments for gradient flow from small initialization: plateau sequence, escape times and their dependence on each parameter, locality of later escapes, learning order, saddle types.]

4.1 Phenomenology

[Representative runs: loss in linear and logarithmic time, overlaps, bond spectra on every edge, effective ranks; S2S signature (Section 2.7).]

4.2 An exactly solvable case: aligned initialization

[Proposition 2.4 in experiments: TTN gradient flow against the exact mode solutions for binary, MPS and HT trees.]

4.3 Escape from the origin

[First escape time t_1 versus σ, χ and K, against the prediction.]

4.4 Later saddles and locality

[Escape times after the first plateau depend on where the new component lives in the tree.]

4.5 Order of learning

[Which component is learned first, how random the order is, and how a wrong order is corrected.]

4.6 Dependence on each parameter

[Summary table: initialization scale, number of cores and tree shape, bond dimension, physical dimension, teacher spectrum, imbalance, learning rate; effect on plateau levels, durations and order; status of each statement.]

5 Other regimes

[Skeleton. The regimes other than saddle-to-saddle, and how discrete steps, gradient noise and finite samples modify the gradient-flow picture.]

5.1 Lazy regime

[Large initialization in the centred and two-copy models: linearized dynamics, kernel at initialization, time rescaling by $\sigma^{2(K-1)}$.]

5.2 Rich, non-incremental regime

[Intermediate scales where features move but transitions overlap; what sets the crossover.]

5.3 Stalled runs

[Runs that do not converge within the budget: long initial plateaus at small σ and large K, and the relation to barren plateaus.]

5.4 Large bond dimension: NTK versus μP

[Per-core scalings for an infinite-bond-dimension limit with and without feature learning.]

5.5 Discrete steps, gradient noise and finite samples

[Gradient descent versus gradient flow, the stability limit and edge of stability; SGD noise and plateau durations; sample complexity of the saddle-to-saddle and lazy regimes.]

5.6 Optimizers used in practice

Tensor-network models are mostly trained with Adam (Kingma and Ba, 2015), with ALS and DMRG sweeps (White, 1992; Holtz et al., 2012a; Stoudenmire and Schwab, 2016) or with Riemannian methods (Uschmajew and Vandereycken, 2013; Novikov et al., 2018). Which features of the gradient-flow picture of Section 4 survive them? Table 4 collects the answers, Figure 2 shows them from one initial point, and the evidence follows.

Setup (details in Section B.6). Exact population loss, whitened inputs, double precision. Binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$, odeco teacher $(3, 2, 1)$: all methods start from the same 42 fan-in initializations (seven $\sigma \in [0.002, 0.128]$, six seeds). Deep trees, $N = 8$: binary ($K = 7$) and MPS ($K = 8$), with the same teacher and with a pattern teacher E8 whose weaker components need $k = 6$ and $k = 2$ cores on the MPS (Proposition 2.1); $\sigma \in \{0.3, 0.8\}$, plus seven $\sigma \in [0.25, 0.8]$ (binary) and eight $\sigma \in [0.01, 0.3]$ (MPS); six seeds; gauge-balanced initializations, and fan-in ones for the gauge-dependent methods (fan-in gradient flow stalls on the odeco MPS, Proposition A.28). Methods as listed in Table 4, with GD and heavy ball ($\beta = 0.9$) at effective steps $\eta/(1-\beta) \le 0.27\eta_c$ (Section A.6) and Adam's $\epsilon = 10^{-8}$ unless stated. Clocks: t for gradient flow (LSODA, events by root finding), ηn for GD, $\eta n/(1-\beta)$ for heavy ball, steps for the other gradient methods, sweeps for ALS and DMRG; exponents p ($T \propto \sigma^{-p}$) are per-seed fits, mean ± s.e.

GD and heavy ball reproduce gradient flow. empirical new (for TTNs) GD with $\eta = 0.13\eta_c$ on binary $N = 4$ has gradient flow's exponents to within 0.003 and its order from all 42 initial points; its first-escape delay grows by 0.0199 ± 0.0000 per e-fold of $1/\sigma$, the asymptotic slope $\eta(K-1)/2 = 0.020$ that follows from Proposition A.40. Heavy ball's delay grows $19.0 = (1+\beta)/(1-\beta)$ times faster (conjecture: the momentum analogue), hence $p = 0.970 \pm 0.002$. On the deep trees ($\sigma = 0.8$) GD and heavy ball escape after 1.001–1.006 and 1.01–1.11 times gradient flow's time.

Adam: same order, but the first escape forgets σ. derived new (for TTNs) With $\epsilon = 0$ Adam's step $-\eta\hat{m}_n/\sqrt{\hat{v}_n}$ is invariant under a common rescaling of past gradients known (Kingma and Ba, 2015), and near the origin ∇L is homogeneous of degree $K-1$; so the iterates from $c\theta_0$ with step $c\eta$ are c times those from θ_0 with step η, Adam sees σ only through σ/η, and the number n_1 of steps to the first escape stays bounded as $\sigma \to 0$ (in the sign-flow limit $\eta \to 0$ known (Balles and Hennig, 2018), ηn_1 tends to a constant). Measured empirical: $p = 0.009$–0.013 on binary $N = 4$ for four learning rates, RMSProp alike (Figure 3); $p = 0.049 \pm 0.022$ and 0.145 ± 0.033 ($\eta = 10^{-3}$, 10^{-2}; small but not zero) on binary $N = 8$, where gradient flow has 5.00 ± 0.02 ($\sigma \le 0.4$, prediction 5; fit computed from the committed runs in the review of this experiment). The selection survives empirical: Adam and RMSProp pick gradient flow's first component in 94–100% of the resolved deep-tree runs and keep E8's locality order on the MPS (the weaker $k = 2$ component before the stronger $k = 6$ one, in 4–5 of 6 seeds, as gradient flow). On binary $N = 4$ the first escape is as wide as its own time, so 79–100% of the runs fail $\rho_i > 3$ and are rich, non-incremental by Table 3.

Adam's ϵ brings the gradient-flow law back. derived new Where $|\partial_j L| \ll \epsilon$ Adam is gradient flow with $t = \eta n/\epsilon$: from a small σ it crawls along the gradient-flow path until the gradient reaches ϵ, then escapes as in the sign regime,

$$n_1 \simeq \frac{\epsilon}{\eta} T_{\mathrm{GF}}(\sigma) + n_{\mathrm{sign}}(\eta), \tag{20}$$

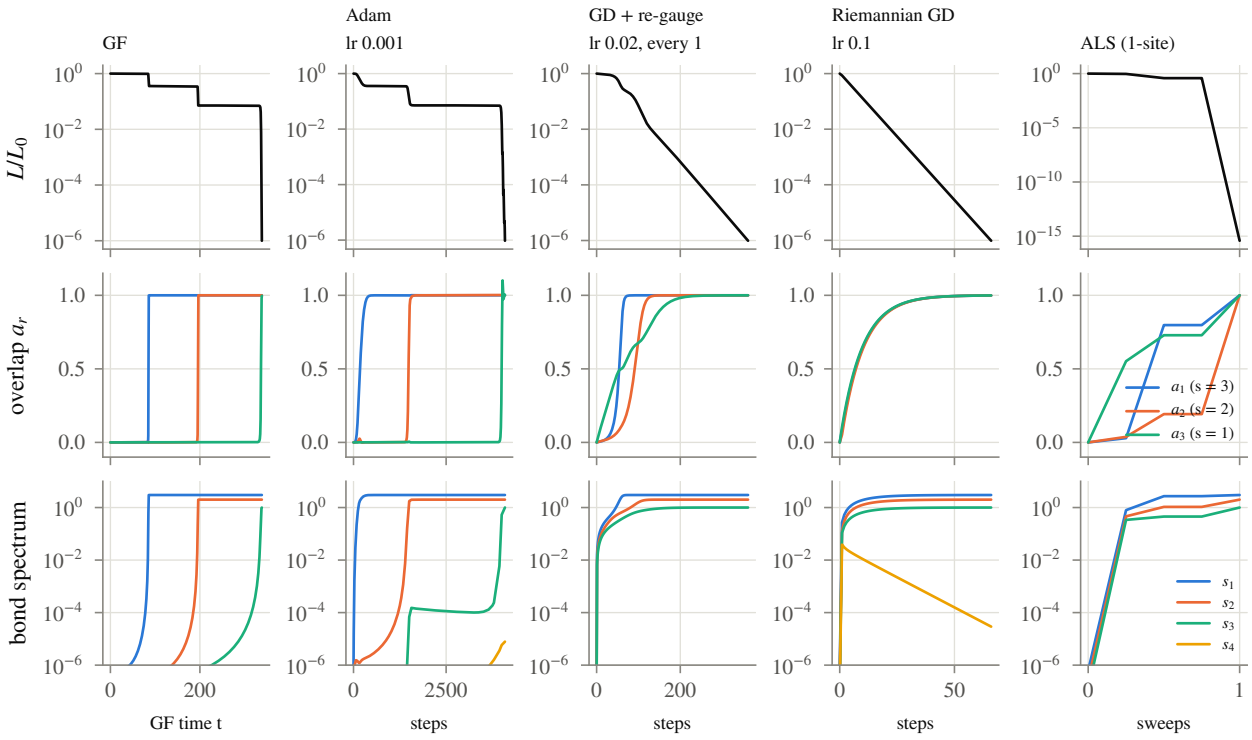


Figure 2: Five optimizers from one initial point: binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$, odeco teacher $(3, 2, 1)$, fan-in initialization, $\sigma = 0.008$, seed 0. Columns: gradient flow, Adam ($\eta = 10^{-3}$), GD re-gauged at every step ($\eta = 0.02$), Riemannian GD ($\eta = 0.1$), one-site ALS, each on its own clock. Rows: normalized loss, overlaps a_r, bond singular values of the cut $\{1, 2\}|\{3, 4\}$. Gradient flow and Adam learn one component at a time in the same order, with plateaus at the partial-fit levels $L_1/L_0 = 0.36$ and $L_2/L_0 = 0.07$; the other three have none. This is the only one of the 60 ALS and DMRG runs in which no two components cross $a_r = \frac{1}{2}$ in the same micro-step.

so the law $T_{\mathrm{GF}} \propto \sigma^{-(K-2)}$ returns, with prefactor ϵ/η, below the crossover $\sigma_\times$ where $\epsilon T_{\mathrm{GF}} = \eta n_{\mathrm{sign}}$. Test empirical on the odeco MPS with $N = 8$ (eight $\sigma \in [0.01, 0.3]$, six seeds, $\eta = 10^{-3}$, 10^{-2}): per seed, from that seed's exact gradient-flow escape at $\sigma = 0.3$ times $(0.3/\sigma)^6$ and its own escape with $\epsilon = 10^{-30}$, with no fitted parameter, measured/predicted is 0.9989–1.0003 ($\eta = 10^{-3}$, $\sigma = 0.02$–0.03) and 1.0009–1.0062 ($\eta = 10^{-2}$, $\sigma = 0.015$–0.02) for every run that escaped within $2 \cdot 10^5$ steps, and (20) predicts which runs do in all 60 cases with $\sigma \leq 0.05$; near $\sigma_\times \approx 0.05$–$0.1$ the deviations are systematic (-5% to $+17\%$). The crossover depends on η and on the tree: on binary $N = 4$, (20) puts it near 10^{-8} (derived, not tested), far below the $2 \cdot 10^{-4}$ at which the initial gradient falls under ϵ. Only $K = 8$ was tested.

Later plateaus and minima under Adam. empirical Later gaps depend weakly on σ on binary $N = 4$ ($p = 0.0$–0.4). On binary $N = 8$ they grow as σ decreases, up to $8 \cdot 10^4$ steps at $\sigma = 0.25$, but per-seed local slopes range from -0.4 to 5.5, so over half a decade no law is established (gradient flow: 4.72 and 4.85, prediction 5); with $\sigma \leq 0.4$, 83–100% of these runs are S2S. Adam and RMSProp break the conservation of Q_e and end at imbalanced minima (balance index $\|Q_e\|_F/(\|G_{v,(e)}G_{v,(e)}^\top\|_F + \|G_{p,(e)}G_{p,(e)}^\top\|_F) = 0.21$–$0.96$), 1.01–6.1 times sharper than the balanced value (51), the more so the larger the step; gradient flow, GD and heavy ball end within 0.2% of it.

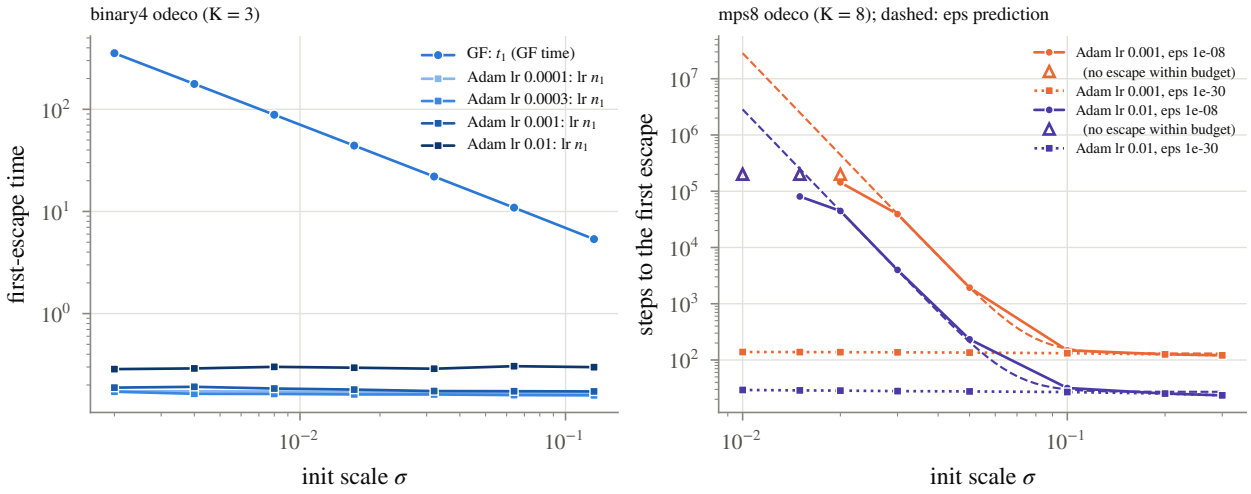


Figure 3: Adam's first escape. Left: binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$, odeco teacher $(3, 2, 1)$, fan-in initialization; medians over six seeds of gradient flow's first-escape time t_1 and of Adam's ηn_1 ($\epsilon = 10^{-8}$). Gradient flow follows σ^{-1}; Adam does not depend on σ, and $\eta n_1 \to 0.17$ as $\eta \to 0$ (0.30, 0.18, 0.16, 0.17 for $\eta = 10^{-2}$ to 10^{-4}). Right: MPS, $N = 8$ ($K = 8$), same teacher, balanced initialization; median steps to the first escape over the runs that escaped within $2 \cdot 10^5$ steps, for $\epsilon = 10^{-8}$ (solid) and 10^{-30} (dotted); triangles, drawn at the budget, mark σ with runs that did not escape; dashed: (20) with median inputs. Without ϵ the escape takes 121–139 ($\eta = 10^{-3}$) and 23.5–29.3 steps ($\eta = 10^{-2}$) at every σ; with $\epsilon = 10^{-8}$ the law σ^{-6} returns below $\sigma \approx 0.05$–0.1.

No plateaus: sweeps and Riemannian GD. proven new (elementary) In the canonical form centred at core v the environment of v is an isometry, so the least-squares core is $\nabla_{G_v}\langle W, W^*\rangle$, a function of the other cores' frames, which are Q factors of QR decompositions and do not change when a core is multiplied by a positive number: every ALS and DMRG iterate is independent of σ. Riemannian GD, $W \leftarrow \mathrm{HSVD}_\chi(W - \eta P_{T(W)}(W - W^*))$ with $P_{T(W)}$ the projector onto the tangent space of the fixed-rank manifold, maps W_0 at $\eta = 1$ to $\mathrm{HSVD}_\chi(P_{T(W_0)}W^*)$, and that tangent space is the same at cW_0: it is exactly σ-free too. Measured empirical: ALS and DMRG fit the teacher within one sweep, with two components crossing $a_r = \frac{1}{2}$ in the same micro-step in 59 of 60 runs: no resolved order. Riemannian GD needs 2–7 steps; it fits all components in one step on binary $N = 4$ but stages them on binary $N = 8$, the weakest strictly last in 5 of 6 seeds for both teachers (3 and 2 of 6 on the MPS).

Re-gauged and Gram-preconditioned GD. Re-gauging moves all of $\|W\|_F \sim \sigma^K$ into the root, whose environment becomes an isometry, so the root fits the teacher's projection onto the current frames in a time $O(1)$ for any σ derived; on deep trees random frames keep only a fraction $\prod_e \chi/d^{n_e}$ of each component over the root's edges ($2.4 \cdot 10^{-3}$ on binary $N = 8$, 0.2 on $N = 4$), so they must rotate and learning is sequential again, with σ-free event times (empirical, $|p| \le 0.02$). Gram preconditioning moves each core towards its own least-squares solution, $\sim \sigma^{-(K-1)}$ times larger than the core, so its first step overshoots the loss by $\sim \sigma^{-2K(K-1)}$ (derived; up to $4 \cdot 10^{34}$); it has no flat plateau, but its time depends on σ (empirical, Table 4).

Finite samples. empirical With $n = 45$ and 60 samples on binary $N = 4$ (four data draws, six initializations, $\sigma \in \{0.01, 0.1\}$, paired comparisons), gradient flow and Adam reach the same population loss: at $n = 60$ every run that met the training tolerance recovered the teacher; at $n = 45$

Table 4: What survives which optimizer (Section 5.6). Binary $N = 4$ ($K = 3$): seven σ over 1.8 decades; binary $N = 8$ ($K = 7$) and MPS $N = 8$ ($K = 8$): two to eight σ over at most half a decade (1.5 decades for Adam's ϵ); six seeds each. "Gauge": the group (16). p: exponent of the first-escape time, $T \propto \sigma^{-p}$. Order: agreement with gradient flow from the same initial point, or counts over independent seeds for methods that give one trajectory per seed. Cost: median steps (sweeps, micro-steps) to $L < 10^{-3}L_0$, binary $N = 4$ at $\sigma = 0.008$, deep trees at $\sigma = 0.3$.

method	update equivariant under	plateaus	first escape vs. σ	learning order	cost
gradient flow	orthogonal maps of θ; conserves Q_e	saddle-to-saddle (S2S)	$p = 1.007$ ($K = 3$), 5.00 ($K = 7$, $\sigma \leq 0.4$); predicted $K - 2$	reference	$3.6 \cdot 10^3$–$1.1 \cdot 10^4$ evaluations
GD, heavy ball (steps $\leq 0.27\eta_c$)	as gradient flow; Q_e drifts	as gradient flow	$p = 1.005$, 0.970 ($K = 3$); delay $\propto \ln(1/\sigma)$	as gradient flow (100%)	T_{GF}/η ($1.7 \cdot 10^4$, $K = 3$)
Adam, RMSProp	coordinate permutations, sign flips; gradient rescaling ($\epsilon = 0$)	sequential; rich ($K = 3$), S2S ($K = 7$, $\sigma \leq 0.4$)	σ-free, $p = 0.00$–0.15; below $\sigma_\times$: (20)	first component as gradient flow: 94–100%; full order 87–98% ($K = 3$)	$1.7 \cdot 10^2$–$2.1 \cdot 10^5$
re-gauged GD	gauge (cores reset to the root-centred canonical form)	none ($K = 3$); σ-free ($K = 7$, 8)	σ-free, $p = 0.00$	largest first: 6/6 ($K = 3$), 2/6 and 5/6 ($K = 7$)	$2 \cdot 10^2$–$2 \cdot 10^4$; odeco MPS: budget
Riemannian GD ($\eta = 1$)	gauge (acts on W)	none	σ-free (exact)	one step ($K = 3$); weakest last: 5/6 ($K = 7$)	2–7, each a dense Jacobian
Gram-precond. GD	gauge, up to damping (per-core Gauss–Newton blocks)	overshoot, then return	not σ-free: $p = 0.65$ ($\eta = 0.01$); deep: 147–632 vs. 26–35 steps	mostly unresolved	$1.5 \cdot 10^2$–$4.3 \cdot 10^3$; odeco MPS: 9/24 converge
ALS, DMRG	gauge; invariant under per-core rescaling	none	σ-free (exact, proven)	none resolved (59/60 ties)	≤ 1 sweep (3–11 micro-steps)

none did, and Adam was better in 8–13 of 21–23 pairs per setting. Only a threshold count differs (a fourth bond singular value above $0.1\, s^{e,*}_{\min}$: Adam with $\eta = 10^{-2}$ in 35 of 44 runs, gradient flow in 5); the entropy rank does not.

Interpretation: which invariance removes which feature. Two properties of an update decide what survives (Table 4, second column). The first is how its step scales with the gradient, which near the origin is $O(\sigma^{K-1})$ while the parameters must move by their own size: a step proportional to the gradient needs a time $\sigma^{-(K-2)}$, a step invariant under its rescaling (Adam) does not, and an absolute scale in the preconditioner, such as Adam's ϵ, brings the law back derived; Adam still follows the gradient's sign, which one component dominates, so learning stays sequential and ordered (conjecture). The second is the metric. Under gradient flow W moves through $JJ^\top$ ($J = \partial W/\partial\theta$), which vanishes like $\sigma^{2(K-1)}$ at the origin, whereas Riemannian GD (in continuous time the Gauss–Newton flow $\dot\theta = J^+(W^* - W)$) and the exact block minimizations of ALS and DMRG move W at a rate of order one. These methods are equivariant under the whole gauge group (16), including the rescaling of a core against its neighbour, and see W_0 only through its frames, so they keep neither plateaus (derived) nor σ (proven); plateaus at singular points of a parametrization, and

the better behaviour of the natural gradient there, are known for multilayer perceptrons known (Amari et al., 2006). Equivariance alone is not enough: Gram preconditioning is gauge-equivariant up to damping derived but moves all cores at once from their current values. Balance and the flattest minimum (Proposition A.39) rest on the conserved Q_e, the charges of the gauge symmetry under Euclidean gradient flow (Proposition 2.5); their measured relative drift is 10^{-11}, 10^{-5} and $3 \cdot 10^{-4}$ under gradient flow, GD and heavy ball (momentum flows lose most conservation laws, Marcotte et al., 2024), while Adam, equivariant only under coordinate permutations and sign flips, breaks them at order one empirical. A law for Adam's later plateaus is open (Section 7).

6 Effects of input covariance and teacher structure

[Skeleton. Starting from whitened inputs and an odeco teacher, one complication at a time: does it change the saddle-to-saddle picture qualitatively, quantitatively, or not at all?]

6.1 Anisotropic covariance aligned with the teacher

[Prediction: component r behaves as if its weight were $\Lambda_r s_r$, $\Lambda_r = \prod_i \lambda_{r,i}$.]

6.2 Misaligned covariance

[Teacher factors that are not covariance eigenvectors.]

6.3 Cross-site correlations

[One correlated input model (shared latent factor), with sampled data and small steps.]

6.4 Non-orthogonal CP teachers

[Saddle sequence and plateau levels versus the overlap between components.]

6.5 Random TTN teachers

[Are the visited saddles the hierarchical-SVD truncations of the teacher, and in which order do bond ranks grow?]

6.6 Teacher and student on different trees

[Which bonds grow first when the teacher's correlation structure does not match the student's tree; connection to locality (Section 4.4).]

6.7 Summary

[Table: complication, why it was checked, what changed, and whether the baseline theory (possibly after a substitution such as effective weights) predicts it.]

7 Open questions

[Skeleton. Questions left open by this work, each with what would settle it.]

References

Emmanuel Abbe, Enric Boix-Adserà, and Theodor Misiakiewicz. SGD learning on neural networks: leap complexity and saddle-to-saddle dynamics. In *Proceedings of Thirty Sixth Conference on Learning Theory*, volume 195 of *Proceedings of Machine Learning Research*, pages 2552–2623, 2023.

El Mehdi Achour, François Malgouyres, and Sébastien Gerchinovitz. The loss landscape of deep linear neural networks: a second-order analysis. *Journal of Machine Learning Research*, 25(242): 1–76, 2024.

Shun-ichi Amari, Hyeyoung Park, and Tomoko Ozeki. Singularities affect dynamics of learning in neuromanifolds. *Neural Computation*, 18(5):1007–1065, 2006. doi:10.1162/neco.2006.18.5.1007.

Animashree Anandkumar, Rong Ge, Daniel Hsu, Sham M. Kakade, and Matus Telgarsky. Tensor decompositions for learning latent variable models. *Journal of Machine Learning Research*, 15 (80):2773–2832, 2014.

Sanjeev Arora, Nadav Cohen, and Elad Hazan. On the optimization of deep networks: Implicit acceleration by overparameterization. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 244–253, 2018.

Sanjeev Arora, Nadav Cohen, Wei Hu, and Yuping Luo. Implicit regularization in deep matrix factorization. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Alexander Atanasov, Blake Bordelon, and Cengiz Pehlevan. Neural networks as kernel learners: The silent alignment effect. In *International Conference on Learning Representations*, 2022.

Lukas Balles and Philipp Hennig. Dissecting Adam: The sign, magnitude and variance of stochastic gradients. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 404–413, 2018.

David G. T. Barrett and Benoit Dherin. Implicit gradient regularization. In *International Conference on Learning Representations*, 2021.

Blake Bordelon and Cengiz Pehlevan. Self-consistent dynamical field theory of kernel evolution in wide neural networks. In *Advances in Neural Information Processing Systems*, volume 35, pages 32240–32256, 2022. doi:10.52202/068431-2336.

Dennis Chemnitz and Maximilian Engel. Characterizing dynamical stability of stochastic gradient descent in overparameterized learning. *Journal of Machine Learning Research*, 26(134):1–46, 2025.

Feng Chen, Daniel Kunin, Atsushi Yamamura, and Surya Ganguli. Stochastic collapse: How gradient noise attracts SGD dynamics towards simpler subnetworks. In *Advances in Neural Information Processing Systems*, volume 36, pages 35027–35063, 2023. doi:10.52202/075280-1522.

Lénaïc Chizat, Edouard Oyallon, and Francis Bach. On lazy training in differentiable programming. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Guillaume Corlouer, Avi Semler, Alexander Strang, and Alexander Gietelink Oldenziel. Stochastic gradient descent in the saddle-to-saddle regime of deep linear networks. arXiv preprint arXiv:2604.06366, 2026.

Lieven De Lathauwer, Bart De Moor, and Joos Vandewalle. On the best rank-1 and rank-$(R_1, R_2, \ldots, R_N)$ approximation of higher-order tensors. *SIAM Journal on Matrix Analysis and Applications*, 21(4):1324–1342, 2000. doi:10.1137/S0895479898346995.

J. R. Dormand and P. J. Prince. A family of embedded Runge–Kutta formulae. *Journal of Computational and Applied Mathematics*, 6(1):19–26, 1980. doi:10.1016/0771-050X(80)90013-3.

Simon S. Du, Wei Hu, and Jason D. Lee. Algorithmic regularization in learning deep homogeneous models: Layers are automatically balanced. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Lars Eldén and Berkant Savas. Perturbation theory and optimality conditions for the best multilinear rank approximation of a tensor. *SIAM Journal on Matrix Analysis and Applications*, 32(4):1422–1450, 2011. doi:10.1137/110823298.

Abraham Frandsen and Rong Ge. Optimization landscape of Tucker decomposition. *Mathematical Programming*, 193(2):687–712, 2022. doi:10.1007/s10107-020-01531-z.

Zach Furman, Stephan Wäldchen, Yangda Bei, and Liam Hodgkinson. Benign loss landscapes can coexist with worst-case hardness. arXiv preprint arXiv:2609.13057, 2026.

Rong Ge, Yunwei Ren, Xiang Wang, and Mo Zhou. Understanding deflation process in over-parametrized tensor decomposition. In *Advances in Neural Information Processing Systems*, volume 34, pages 1299–1311, 2021.

Behrooz Ghorbani, Song Mei, Theodor Misiakiewicz, and Andrea Montanari. Limitations of lazy training of two-layers neural networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Avrajit Ghosh, Soo Min Kwon, Rongrong Wang, Saiprasad Ravishankar, and Qing Qu. Learning dynamics of deep matrix factorization beyond the edge of stability. In *International Conference on Learning Representations*, pages 17753–17800, 2025.

Gauthier Gidel, Francis Bach, and Simon Lacoste-Julien. Implicit regularization of discrete gradient dynamics in linear neural networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Daniel Gissin, Shai Shalev-Shwartz, and Amit Daniely. The implicit bias of depth: How incremental learning drives generalization. In *International Conference on Learning Representations*, 2020.

Lars Grasedyck. Hierarchical singular value decomposition of tensors. *SIAM Journal on Matrix Analysis and Applications*, 31(4):2029–2054, 2010. doi:10.1137/090764189.

Erdong Guo and David Draper. Neural tangent kernel of matrix product states: Convergence and applications. arXiv preprint arXiv:2111.14046, 2021.

W. Hackbusch and S. Kühn. A new scheme for the tensor representation. *Journal of Fourier Analysis and Applications*, 15(5):706–722, 2009. doi:10.1007/s00041-009-9094-9.

Ernst Hairer, Syvert P. Nørsett, and Gerhard Wanner. *Solving Ordinary Differential Equations I: Nonstiff Problems*. Springer, Berlin, Heidelberg, second revised edition, 1993. doi:10.1007/978-3-540-78862-1.

Thomas Hofmann. The map behind the flow: Finite-step gradient descent as a dynamical system. arXiv preprint arXiv:2607.04993, 2026.

Sebastian Holtz, Thorsten Rohwedder, and Reinhold Schneider. The alternating linear scheme for tensor optimization in the tensor train format. *SIAM Journal on Scientific Computing*, 34(2): A683–A713, 2012a. doi:10.1137/100818893.

Sebastian Holtz, Thorsten Rohwedder, and Reinhold Schneider. On manifolds of tensors of fixed TT-rank. *Numerische Mathematik*, 120(4):701–731, 2012b. doi:10.1007/s00211-011-0419-7.

Arthur Jacot, François Ged, Berfin Şimşek, Clément Hongler, and Franck Gabriel. Saddle-to-saddle dynamics in deep linear networks: Small initialization training, symmetry, and sparsity. arXiv preprint arXiv:2106.15933, 2022.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

Eleftherios Kofidis and Phillip A. Regalia. On the best rank-1 approximation of higher-order supersymmetric tensors. *SIAM Journal on Matrix Analysis and Applications*, 23(3):863–884, 2002. doi:10.1137/S0895479801387413.

Itai Kreisler, Mor Shpigel Nacson, Daniel Soudry, and Yair Carmon. Gradient descent monotonically decreases the sharpness of gradient flow solutions in scalar networks and beyond. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 17684–17744, 2023.

Akshay Kumar and Jarvis Haupt. Early directional convergence in deep homogeneous neural networks for small initializations. *Transactions on Machine Learning Research*, 2025a.

Akshay Kumar and Jarvis Haupt. Towards understanding gradient flow dynamics of homogeneous neural networks beyond the origin. *Journal of Machine Learning Research*, 26(269):1–83, 2025b.

Akshay Kumar and Jarvis Haupt. Learning neural networks by neuron pursuit. arXiv preprint arXiv:2509.12154, 2026.

Daniel Kunin, Javier Sagastuy-Brena, Surya Ganguli, Daniel L. K. Yamins, and Hidenori Tanaka. Neural mechanics: Symmetry and broken conservation laws in deep learning dynamics. In *International Conference on Learning Representations*, 2021.

Daniel Kunin, Allan Raventós, Clémentine Dominé, Feng Chen, David Klindt, Andrew Saxe, and Surya Ganguli. Get rich quick: exact solutions reveal how unbalanced initializations promote rapid feature learning. In *Advances in Neural Information Processing Systems*, volume 37, pages 81157–81203, 2024. doi:10.52202/079017-2580.

Daniel Kunin, Giovanni Luca Marchetti, Feng Chen, Dhruva Karkada, James B. Simon, Michael R. DeWeese, Surya Ganguli, and Nina Miolane. Alternating gradient flows: A theory of feature learning in two-layer neural networks. In *Advances in Neural Information Processing Systems*, volume 38, pages 4951–4998, 2025. doi:10.52202/085713-0156.

Qianxiao Li, Cheng Tai, and Weinan E. Stochastic modified equations and adaptive stochastic gradient algorithms. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 2101–2110, 2017.

Qianxiao Li, Cheng Tai, and Weinan E. Stochastic modified equations and dynamics of stochastic gradient algorithms I: Mathematical foundations. *Journal of Machine Learning Research*, 20(40): 1–47, 2019.

Zhiyuan Li, Yuping Luo, and Kaifeng Lyu. Towards resolving the implicit bias of gradient descent for matrix factorization: Greedy low-rank learning. In *International Conference on Learning Representations*, 2021.

Kaifeng Lyu, Jikai Jin, Zhiyuan Li, Simon S. Du, Jason D. Lee, and Wei Hu. Dichotomy of early and late phase implicit biases can provably induce grokking. In *International Conference on Learning Representations*, 2024.

Sibylle Marcotte, Rémi Gribonval, and Gabriel Peyré. Keep the momentum: Conservation laws beyond Euclidean gradient flows. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 34790–34821, 2024.

Paolo Milanesi, Hachem Kadri, Stéphane Ayache, and Thierry Artières. Implicit regularization in deep tensor factorization. In *2021 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8, 2021. doi:10.1109/IJCNN52387.2021.9533690.

Rotem Mulayoff and Tomer Michaeli. Unique properties of flat minima in deep networks. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7108–7118, 2020.

Rotem Mulayoff and Tomer Michaeli. Exact mean square linear stability analysis for SGD. In *Proceedings of Thirty Seventh Conference on Learning Theory*, volume 247 of *Proceedings of Machine Learning Research*, pages 3915–3969, 2024.

Alexander Novikov, Mikhail Trofimov, and Ivan Oseledets. Exponential machines. *Bulletin of the Polish Academy of Sciences: Technical Sciences*, 66(6):789–797, 2018. doi:10.24425/bpas.2018.125926.

Andres Potapczynski, Shikai Qiu, Marc Finzi, Christopher Ferri, Zixi Chen, Micah Goldblum, C. Bayan Bruss, Christopher De Sa, and Andrew Gordon Wilson. Searching for efficient linear layers over a continuous space of structured matrices. In *Advances in Neural Information Processing Systems*, volume 37, pages 3857–3881, 2024. doi:10.52202/079017-0126.

Shikai Qiu, Andres Potapczynski, Marc Anton Finzi, Micah Goldblum, and Andrew Gordon Wilson. Compute better spent: Replacing dense layers with structured matrices. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 41698–41716, 2024.

Divit Rawal and Michael R. DeWeese. A theory of saddle escape in deep nonlinear networks. arXiv preprint arXiv:2605.01288, 2026.

Noam Razin, Asaf Maman, and Nadav Cohen. Implicit regularization in hierarchical tensor factorization and deep convolutional neural networks. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 18422–18462, 2022.

Andrew M. Saxe, James L. McClelland, and Surya Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. In *International Conference on Learning Representations*, 2014.

Edwin Stoudenmire and David J. Schwab. Supervised learning with tensor networks. In *Advances in Neural Information Processing Systems*, volume 29, pages 4799–4807, 2016.

Zhenfeng Tu, Santiago Aranguri, and Arthur Jacot. Mixed dynamics in linear networks: Unifying the lazy and active regimes. In *Advances in Neural Information Processing Systems*, volume 37, pages 106059–106104, 2024. doi:10.52202/079017-3365.

André Uschmajew and Bart Vandereycken. The geometry of algorithms using hierarchical tensors. *Linear Algebra and its Applications*, 439(1):133–166, 2013. doi:10.1016/j.laa.2013.03.016.

Yuqing Wang, Minshuo Chen, Tuo Zhao, and Molei Tao. Large learning rate tames homogeneity: Convergence and balancing effect. In *International Conference on Learning Representations*, 2022.

Frederiek Wesel and Kim Batselier. Tensor network-constrained kernel machines as Gaussian processes. In *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pages 2161–2169, 2025.

Steven R. White. Density matrix formulation for quantum renormalization groups. *Physical Review Letters*, 69(19):2863–2866, 1992. doi:10.1103/PhysRevLett.69.2863.

Blake Woodworth, Suriya Gunasekar, Jason D. Lee, Edward Moroshko, Pedro Savarese, Itay Golan, Daniel Soudry, and Nathan Srebro. Kernel and rich regimes in overparametrized models. In *Proceedings of the Thirty Third Conference on Learning Theory*, volume 125 of *Proceedings of Machine Learning Research*, pages 3635–3673, 2020.

Lei Wu, Chao Ma, and Weinan E. How SGD selects the global minima in over-parameterized learning: A dynamical stability perspective. In *Advances in Neural Information Processing Systems*, volume 31, pages 8279–8288, 2018.

Greg Yang and Edward J. Hu. Tensor programs IV: Feature learning in infinite-width neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 11727–11737, 2021.

Greg Yang, James B. Simon, and Jeremy Bernstein. A spectral condition for feature learning. arXiv preprint arXiv:2310.17813, 2024.

Yedi Zhang, Andrew Saxe, and Peter E. Latham. Saddle-to-saddle dynamics explains a simplicity bias across neural network architectures. In *International Conference on Learning Representations*, pages 57507–57551, 2026.

Liu Ziyin, Botao Li, Tomer Galanti, and Masahito Ueda. Type-II saddles and probabilistic stability of stochastic gradient descent. arXiv preprint arXiv:2303.13093, 2024.

A Derivations

This appendix gives the derivations behind the theoretical statements of the main text, organized by topic: conservation laws (Section A.1), the escape from the origin (Section A.2), the saddles visited after it (Section A.3), teachers that are not orthogonally decomposable (Section A.4), input covariance (Section A.5), discrete steps and gradient noise (Section A.6), and the boundaries between regimes (Section A.7). The short proofs of Section 2 are not repeated.

Conventions. Notation is that of Section 2. Unless stated otherwise inputs are whitened ($\Sigma = I$), gradient flow has unit multipliers ($\eta_v = 1$), so that $L = \frac{1}{2}\|W - W^*\|_F^2$, and the teacher is a CP teacher (4). For a pattern teacher (Section 2.3) and two of its components j, r, $D_{jr} = \{i : \pi_j(i) \neq \pi_r(i)\}$ is the set of sites where they differ, $N(D_{jr})$ the set of cores carrying a site of D_{jr} (the *terminal* cores), $m_{jr} = |N(D_{jr})|$, and $\mathrm{St}(N(D_{jr}))$ the smallest subtree of $\mathcal{T}$ containing $N(D_{jr})$ (its Steiner subtree; compare Proposition 2.1(b)). At a learned saddle the learned component is j, with per-core amplitude $q_j = s_j^{1/K}$ (the balanced amplitude of Proposition 2.4), and r is a component still to be learned. A ratio of two teacher weights, smaller over larger, is written ϱ; it is never a separation ratio ρ_i. $\sigma_k(M)$ is the k-th singular value of a matrix M and always carries its argument; σ without argument is the initialization scale. "LO" means leading order in the small quantities named in each statement, and $\hat{\theta} = \theta/\|\theta\|$. A statement that combines a proven identity with a derived consequence carries both tags, each at its part. The theory notes also use the label *heuristic* (a plausible argument whose assumptions are not controlled); the report has no such tag, so these statements are tagged conjecture and their argument is called heuristic. "Checked" refers to the numerical checks reported in the theory notes; the experimental tests are in Sections 3 to 6.

A.1 Conservation laws, gauge and invariant subspaces

Proposition 2.5 fixes the differences of the core Gram matrices across every bond. Three further facts are used below: the common value of the norm changes, the invariance of unused bond channels, and the exact change of Q_e under discrete steps.

Lemma A.1 (Equal norm shift). proven known (deep linear networks: Arora et al., 2018; Du et al., 2018) *Let $L(\theta) = \mathcal{L}(W(\theta))$ for a differentiable $\mathcal{L}$. Under gradient flow (15) (no mask), for every core v,*

$$\frac{1}{\eta_v}\frac{\mathrm{d}}{\mathrm{d}t}\|G_v\|_F^2 = -2\,\langle \nabla_W \mathcal{L},\, W\rangle, \tag{21}$$

which is the same for all cores. For the population loss (6) the right-hand side is $2(\langle W, W^*\rangle_\Sigma - \|W\|_\Sigma^2)$; for the leading-order flow $\dot{\theta} = \nabla g$ of Section A.2 it is $2g$.

Proof. Since W is linear in G_v,

$$\langle G_v,\, \nabla_{G_v} L\rangle = \langle \nabla_W \mathcal{L},\, W\rangle, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\|G_v\|_F^2 = 2\langle G_v,\, \dot{G}_v\rangle = -2\eta_v \langle G_v,\, \nabla_{G_v} L\rangle. \qquad \square$$

Proposition 2.5 gives only the differences $\|G_u\|_F^2 - \|G_{u'}\|_F^2$; the lemma gives the common shift $y(t)$ in $\|G_v(t)\|_F^2 = \|G_v(0)\|_F^2 + y(t)$ (unit multipliers). Along the leading-order flow $\dot{g} = \|\nabla g\|^2 \geq 0$, so $y(t) = 2\int_0^t g$ is convex: *all cores shrink, by the same amount, and only while $g < 0$*, that is, only if $\langle W_0, W^*\rangle < 0$ (half of all initializations, by the symmetry $G_v \mapsto -G_v$), and the shrinking stops when g changes sign. The identity holds for any block of variables in which g is multilinear, for example the reduced cores of Proposition A.11 (checked to 10^{-8}–10^{-11} relative on 2200 trajectories).

Lemma A.2 (Unused bond channels are invariant). proven new (elementary) *Let $e = e_v$ with $p = p(v)$, and let $\mathcal{B}$ be a set of channels of e such that every entry of G_v and of G_p whose e-index lies in $\mathcal{B}$ vanishes. Then, for every loss of the form $\mathcal{L}(W(\theta))$ (any input law, population or empirical), the gradient vanishes on these entries, so gradient flow, gradient descent and SGD keep them at zero. In particular, a critical point of the problem with bond dimensions $r_e \leq \chi_e$, padded with zeros, is a critical point of the problem with bond dimensions χ_e.*

Proof. The derivative of W with respect to an entry of G_v with e-index $b \in \mathcal{B}$ contracts the other cores with G_p fixed at e-index b, and that slice of G_p vanishes. The same argument applies to G_p. □

Lemma A.3 (Conservation laws under discrete steps). proven known (deep linear networks: Du et al., 2018; scalar form: Hofmann, 2026) *Write $g_v = \nabla_{G_{v,(e)}} L$ and $g_p = \nabla_{G_{p,(e)}} L$ for the gradients, matricized like the cores in (17). One step $\theta \mapsto \theta - \eta \nabla L$ (unit multipliers) changes the conserved quantity of bond e by exactly*

$$Q_e^+ - Q_e = \eta^2 (g_v g_v^\top - g_p g_p^\top). \tag{22}$$

The same holds for SGD with the minibatch gradient. For one aligned component (Proposition 2.4) with amplitudes $q_{v,1}$ and product Π_1, the scalar $Q = q_{v,1}^2 - q_{p,1}^2$ obeys $Q^+ = Q\,(1 - \eta^2 (s_1 - \Pi_1)^2 \Pi_1^2 / (q_{v,1}^2 q_{p,1}^2))$: every step contracts the imbalance.

Proof. With $X = G_{v,(e)}$ and $Y = G_{p,(e)}$, the proof of Proposition 2.5 gives $g_v X^\top = Y g_p^\top =: C$ for every loss that depends on θ through W, minibatch losses included. In $(X - \eta g_v)(X - \eta g_v)^\top - (Y - \eta g_p)(Y - \eta g_p)^\top$ the terms linear in η are $-\eta(C + C^\top) + \eta(C^\top + C) = 0$. For the aligned component, $g_v = -(s_1 - \Pi_1)\Pi_1 / q_{v,1}$, and likewise for p. □

The change is negligible on plateaus, where $(s_1 - \Pi_1)\Pi_1 \approx 0$, and large during oscillations at large steps (Section A.6).

Expected imbalance at initialization. By Section 2.6, fan-in initialization with a common scale has $\mathbb{E}\, Q_e(0) = \sigma^2 (1 - \chi_p / \chi_v) I$ for $e = e_v$, where χ_p is the bond dimension of the parent's own parent edge ($\chi_{\text{root}} = 1$). Edges into the root are therefore child-heavy, $\mathbb{E}\, Q_e = \sigma^2 (1 - 1/\chi_v) I \succ 0$; internal edges of equal bond dimension are balanced in expectation; and the leaf edges of an HT tree with $\chi_v = d = 2$ below a parent with $\chi_p = 3$ are parent-heavy, $\mathbb{E}\, Q_e = -\frac{1}{2}\sigma^2 I$. Fluctuations around these means have both signs at any σ. They matter for the stalls of Proposition A.28 and Section A.7.

A.2 Escape from the origin

Setting. Whitened inputs, a CP teacher, and gradient flow from the fan-in initialization $\theta_0 = \sigma Z$ (Equation (11); Z denotes the initialization at $\sigma = 1$). Section A.5 extends what follows to product covariances by the substitution $W^* \to \Sigma W^*$. Write

$$L(\theta) = \tfrac{1}{2}\|W^*\|_F^2 - g(\theta) + \tfrac{1}{2}\|W(\theta)\|_F^2, \qquad g(\theta) := \langle W(\theta),\, W^* \rangle. \tag{23}$$

g is homogeneous of degree K and linear in each core, while $\|W\|_F^2$ has degree $2K$. The *leading-order (LO) flow* $\dot\theta = \nabla g(\theta)$ drops a term of relative size $\|W\|_F / \|W^*\|_F$, which is $O(\sigma^K)$ on the first plateau. Let t^* be its blow-up time and $\nabla_{\mathbb{S}} g(\hat\theta) = \nabla g(\hat\theta) - K g(\hat\theta)\hat\theta$ the gradient of g on the unit sphere. The first event t_1 is defined in Section 2.7.

Proposition A.4 (Spherical–radial decomposition). proven known (directional convergence: Kumar and Haupt, 2025a) *Along the LO flow, with the time change $\mathrm{d}\tau = \|\theta\|^{K-2} \mathrm{d}t$,*

$$\frac{\mathrm{d}\hat\theta}{\mathrm{d}\tau} = \nabla_{\mathbb{S}} g(\hat\theta), \qquad \frac{\mathrm{d}\ln\|\theta\|}{\mathrm{d}\tau} = K\, g(\hat\theta). \tag{24}$$

The direction follows the gradient flow of g on the sphere, which involves neither $\|\theta\|$ nor σ. The conserved quantities obey $Q_e(\hat\theta(\tau)) = Q_e(\theta_0) / \|\theta(\tau)\|^2$.

Proof. Euler's identity gives $\langle\theta, \nabla g(\theta)\rangle = Kg(\theta)$, and ∇g is homogeneous of degree $K-1$, so $\nabla g(\theta) = \|\theta\|^{K-1}\nabla g(\hat{\theta})$. Hence $\frac{\mathrm{d}}{\mathrm{d}t}\|\theta\| = \langle\hat{\theta}, \nabla g(\theta)\rangle = K\|\theta\|^{K-1}g(\hat{\theta})$ and $\frac{\mathrm{d}}{\mathrm{d}t}\hat{\theta} = (\dot{\theta} - \hat{\theta}\langle\hat{\theta}, \dot{\theta}\rangle)/\|\theta\| = \|\theta\|^{K-2}\nabla_{\mathbb{S}}g(\hat{\theta})$. Finally, Q_e is quadratic in θ. □

As $\sigma \to 0$ the escape needs $\|\theta\|$ to grow by a factor of order $1/\sigma$, i.e. a τ of order $\ln(1/\sigma)$, so the escape direction is $\lim_{\tau\to\infty}\hat{\theta}(\tau)$, a KKT point of g on the sphere (Kumar and Haupt, 2025a, Thm 1, which applies verbatim to TTNs with $K \geq 3$). The conservation laws bind while $\|\theta\| \approx \|\theta_0\|$, which is the phase ($\tau = O(1)$) that decides both the escape time and the KKT point reached; they do not change the balanced form of the limit (Proposition A.9).

Corollary A.5 (Blow-up time). proven new (per-initialization form for TTNs; the exponent $K-2$ is known in analogues: Jacot et al., 2022; Gissin et al., 2020) *With* $G(\tau) = \int_0^\tau g(\hat{\theta})\,\mathrm{d}\tau'$,

$$t^* = \|\theta_0\|^{-(K-2)} \int_0^\infty e^{-(K-2)KG(\tau)}\,\mathrm{d}\tau. \tag{25}$$

For $\theta_0 = \sigma Z$, $t^* = \sigma^{-(K-2)}\,\hat{t}_1(Z)/s_1$, *where the prefactor* $\hat{t}_1(Z) := s_1\,t^*|_{\sigma=1}$ *depends on the initialization direction and on* W^*/s_1 *but not on* σ. *The exponent is exactly* $K-2$ *and counts every core, with or without sites.*

Proof. By (24), $\|\theta(\tau)\| = \|\theta_0\|e^{KG(\tau)}$ and $\mathrm{d}t = \|\theta\|^{-(K-2)}\mathrm{d}\tau$. The LO vector field is homogeneous of degree $K-1$, so $\theta_\sigma(t) = \sigma\,\theta_1(\sigma^{K-2}t)$, and replacing W^* by cW^* divides all times by c. □

There is no separate additive alignment time: the whole integral is alignment, because $g(\hat{\theta})$ starts near $\langle W_0, W^*\rangle/\|\theta_0\|^K \approx 0$ and grows gradually.

Proposition A.6 (From the blow-up to the first event). derived new (for TTNs) *Assume that at the end of the LO phase the direction* $\hat{\theta}$ *has converged to the balanced KKT point of the first component learned. Then, as* $\sigma \to 0$,

$$t_1 = t^* + \delta_K\,s_1^{-(2-2/K)} + o(1), \qquad \delta_K = -\frac{2^{(K-2)/K}}{K-2} + \int_0^{2^{-1/K}} \frac{b\,\mathrm{d}b}{1-b^K}, \tag{26}$$

with $\delta_3 = -0.850$, $\delta_4 = -0.266$, $\delta_5 = -0.047$, $\delta_6 = 0.072$, $\delta_7 = 0.148$ *and* $\delta_8 = 0.201$. *Hence* $t_1\sigma^{K-2}s_1/\hat{t}_1(Z) = 1 + O(\sigma^{K-2})$ *for every initialization.*

Derivation. Once $\|\theta\| = O(1)$ the remaining time does not depend on σ and is set by the direction at that moment. Along a balanced KKT point the flow is the balanced mode $\dot{q} = q^{K-1}(s_1 - q^K)$ of Proposition 2.4, whose half-time is $q_0^{-(K-2)}/((K-2)s_1) + \delta_K s_1^{-(2-2/K)} + O(q_0^2)$; the first term is the LO blow-up time. (With $b = q\,s_1^{-1/K}$ one has $\dot{b} = s_1^{2-2/K}b^{K-1}(1-b^K)$ and $1/(b^{K-1}(1-b^K)) = b^{-(K-1)} + b/(1-b^K)$.) □

Checked per initialization: deviations from gradient flow at most 0.8% ($K = 3, 4$, $\sigma \leq 0.03$), with the residual equal to the δ_K term to two digits; for $K = 6$–8 at $\sigma = 0.15$–0.3 the residual is 2–30 times the δ_K term (the direction has not converged), so a 1% test needs $\sigma^{K-2} \leq 10^{-3}$.

Remark A.7 (Per-initialization prediction). proven at LO. The prefactor is computed per initialization by integrating, in τ, the state (u, ℓ, T) with $u(0) = \theta_0/\|\theta_0\|$ and $\ell(0) = T(0) = 0$: $\mathrm{d}u/\mathrm{d}\tau = \nabla g(u) - Kg(u)u$ (renormalizing u), $\mathrm{d}\ell/\mathrm{d}\tau = Kg(u)$, $\mathrm{d}T/\mathrm{d}\tau = e^{-(K-2)\ell}$, up to $\ell = 40/(K-2) + 5$, and adding the tail $e^{-(K-2)\ell}/((K-2)Kg(u))$. Then $t^* = \|\theta_0\|^{-(K-2)}T$ for the actual teacher (any number of components), and the dominant component of the final u is the first one learned as $\sigma \to 0$. This is the parameter-free prediction of t_1 tested in Section 4.3.

Proposition A.8 (Maximum of g on the sphere). proven new (for TTNs) *Group the sites by the core that carries them (cores without sites form no group), and let* $\|W^*\|_{\text{inj}} = \max\{\langle W^*, \bigotimes_v Y_v\rangle : \|Y_v\|_F = 1\}$ *be the injective norm of* W^* *seen as a tensor with one leg per group. Then*

$$g(\theta) \le \|W^*\|_{\text{inj}} \prod_v \|G_v\|_F \le K^{-K/2}\|W^*\|_{\text{inj}}\|\theta\|^K, \tag{27}$$

with equality on the unit sphere at the balanced bond-rank-one configurations $G_v = K^{-1/2}\, Y_v \otimes \varepsilon_1 \otimes \cdots \otimes \varepsilon_1$ *built from a maximizing* (Y_v)*. For odeco teachers* $\|W^*\|_{\text{inj}} = s_1$ *for every grouping; in general the grouping matters.*

Proof. First inequality, by induction on the number of cores. Take a leaf ℓ with parent p and the SVD $G_\ell = \sum_k \mu_k Y_k \otimes z_k$ across (physical legs | bond leg). Then $g = \sum_k \mu_k \langle W'_k, W^*[Y_k]\rangle$, where W'_k is the network without ℓ and with G_p contracted with z_k on the ℓ-leg, and $W^*[Y_k]$ is W^* contracted with Y_k on the sites of ℓ. By induction, and since $\|W^*[Y_k]\|_{\text{inj}} \le \|W^*\|_{\text{inj}}$, the sum is at most $\|W^*\|_{\text{inj}} \prod_{v\neq\ell,p}\|G_v\|_F \sum_k \mu_k \|G_p \times_\ell z_k\|_F$, and Cauchy–Schwarz with $\sum_k \|G_p \times_\ell z_k\|_F^2 \le \|G_p\|_F^2$ closes the step. The second inequality is the AM–GM inequality $\prod_v \|G_v\|_F \le (\sum_v \|G_v\|_F^2/K)^{K/2}$. □

Example: for $W^* = (I/\sqrt{d}) \otimes (I/\sqrt{d})$ on the site pairs $(0,1)$ and $(2,3)$, the binary tree with $N = 4$ reaches $g = 3^{-3/2}$ on the unit sphere and the MPS only $4^{-2}/d$ (checked).

Proposition A.9 (KKT points are balanced). proven new (for TTNs) *Every KKT point* θ^* *of* g *on the unit sphere with* $g^* := g(\theta^*) > 0$ *has* $Q_e = 0$ *on every bond and* $\|G_v\|_F^2 = 1/K$ *for every core.*

Proof. At a KKT point $\nabla g(\theta^*) = \lambda\theta^*$, and Euler's identity gives $\lambda = Kg^* > 0$. g is invariant under the gauge transformations (16), so $\langle \nabla g, \delta\theta\rangle = 0$ for every gauge direction $\delta\theta = (MG_{v,(e)}, -M^\top G_{p,(e)})$, i.e. $\lambda \operatorname{tr}\left((G_{v,(e)}G_{v,(e)}^\top - G_{p,(e)}G_{p,(e)}^\top)M\right) = 0$ for all M; hence $Q_e = 0$. Since g is linear in each core, $\langle G_v, \nabla_{G_v} g\rangle = g^*$, i.e. $\lambda\|G_v\|_F^2 = g^*$. □

Numerically located KKT points satisfy the proposition to 10^{-15}. Consequences: the near-origin escape direction is balanced, hence of minimal norm in its fibre (Furman et al., 2026, Prop. F.10); and the imbalance of a random initialization, conserved in absolute size ($O(\sigma^2)$) while $\|\theta\|^2$ grows, does not affect the limit direction at LO.

Local maxima at component configurations. For a pattern teacher with distinct weights let θ_j be the balanced bond-rank-one configuration of component j on the unit sphere: per-core amplitude $K^{-1/2}$, the pattern of j on the physical legs and channel 1 on every bond (the aligned initialization of Proposition 2.4 with component j only), so that $g(\theta_j) = s_j K^{-K/2}$.

Proposition A.10 (Second-order structure at a component). derived new (for general trees; for binary $N = 4$, a deep linear network, it is known: Jacot et al., 2022; Achour et al., 2024) *(a) If some component* r *has* $m_{jr} = 1$*, then* θ_j *is not a critical point of* g *on the sphere. (b) Otherwise, assume that the components* r *with* $m_{jr} = 2$ *have pairwise different pairs of terminal cores. Each such* r*, with terminal cores* v, w*, contributes* n_{jr} *eigenvalues* $K^{1-K/2}(s_r - s_j)$ *to the Riemannian Hessian of* g *at* θ_j*, where* $n_{jr} = \chi_e$ *if* v *and* w *share the bond* e *and* $n_{jr} = 1$ *otherwise. All other eigenvalues are negative, except the* $\sum_e(\chi_e - 1)$ *zero modes along the gauge orbit. Hence* θ_j *is a strict local maximum modulo gauge iff no* $m_{jr} = 1$ *and every* r *with* $m_{jr} = 2$ *has* $s_r < s_j$*; otherwise it has* $\sum_{r:\, m_{jr}=2,\, s_r>s_j} n_{jr}$ *unstable directions. Ties* $s_r = s_j$ *are decided at higher order; partners that share a terminal core are treated by Proposition A.22.*

Derivation. The Riemannian Hessian is $\nabla^2 g - Kg(\theta_j)I = \nabla^2 g - s_j K^{1-K/2} I$ on the tangent space. Since g is linear in each core, $\nabla^2 g$ only has cross terms between two cores $v \neq w$, and such a term contracts W^* with the pattern of j on every site outside $S_v \cup S_w$: only components with $D_{jr} \subseteq S_v \cup S_w$ contribute. If $m_{jr} = 1$, $\nabla g(\theta_j)$ has a component along the pattern of r on the terminal core. Component j's own cross terms vanish on the gauge orbit and give $-2s_j K^{1-K/2}$ off it; every direction not coupled to another component has eigenvalue $-s_j K^{1-K/2}$. For $m_{jr} = 2$, rotate G_v and G_w toward the pattern of r on their sites, with channel c on the bond joining them if they are adjacent and channel 1 on every other bond. The cross term is $s_r K^{1-K/2}$ for every channel c if the cores are adjacent, and only for $c = 1$ otherwise (every bond on the path between them meets channel 1 of a third core). Each surviving channel gives the block $K^{1-K/2}\left(\begin{smallmatrix} -s_j & s_r \\ s_r & -s_j \end{smallmatrix}\right)$, with eigenvalues $K^{1-K/2}(s_r - s_j)$ and $-K^{1-K/2}(s_r + s_j)$. □

Checked on 19 pattern-teacher cases and 7 odeco settings: numbers and values of the positive eigenvalues as predicted in every case. For odeco teachers m_{jr} is the number of cores carrying sites: $m = N$ on MPS and HT trees, $m = 3, 4$ on binary trees with $N = 6, 8$, and $m = 2$ on binary $N = 4$, whose bottom cores are not adjacent ($n = 1$). Every odeco component is therefore a local maximum on MPS, HT and binary $N \geq 6$ trees (*tensor-like*: one basin per component), while on binary $N = 4$ only the largest one is (*matrix-like*). Pattern teachers whose components differ on exactly two cores are matrix-like on every tree. Spherical flows from 200 random initializations ended at component configurations in every case empirical; whether teachers that are not odeco have local maxima of bond rank larger than one is open (Section A.4), and even best rank-one approximation has non-global local optima (Kofidis and Regalia, 2002).

Proposition A.11 (Only the teacher slices move)**.** proven new (for TTNs) *For an odeco teacher,* $\nabla_{G_v} g = \sum_r s_r \left(\bigotimes_{i \in S_v} u_r^{(i)}\right) \otimes E_v^{(r)}$, *where* $E_v^{(r)}$ *contracts all other cores with* $u_r^{(i)}$ *on their sites. Hence, under the LO flow, the part of* G_v *that is orthogonal on its physical legs to every* $\bigotimes_{i \in S_v} u_r^{(i)}$ *is frozen. The moving variables are the reduced cores* $H_v^{(r)}$ *(G_v contracted with the* $u_r^{(i)}$ *on its physical legs; for a core without sites, the whole core, shared by all components), and* $g = \sum_r s_r \mathcal{W}_{\mathrm{red}}(H^{(r)})$, *with* $\mathcal{W}_{\mathrm{red}}$ *the scalar contraction of the reduced tree. Under (11) the* $H_v^{(r)}$ *have i.i.d.* $\mathcal{N}(0, \sigma^2/F_v)$ *entries, independent across* r*: a fan-in-initialized scalar TTN with per-core scale* $\sigma d^{-|S_v|/2}$. *If every core carries a site (MPS), different components have disjoint moving variables and evolve independently at LO.*

Proof. The physical legs of $\nabla_{G_v} g$ lie in the span of the teacher vectors, and g depends on G_v only through the $H_v^{(r)}$. Contracting a core with i.i.d. Gaussian entries with orthonormal vectors gives independent Gaussians of the same variance. □

The same holds for pattern teachers, with one slice per distinct pattern on each core. Most of $\|\theta_0\|$ sits in frozen slices: $\mathbb{E}\|H_v^{(r)}\|_F^2 = \sigma^2 \chi_v d^{-|S_v|}$ against $\mathbb{E}\|G_v\|_F^2 = \sigma^2 \chi_v$.

Corollary A.12 (The Kumar–Haupt time is a lower bound)**.** proven *(bounds)* empirical *(sizes)* new (the escape time t_{KH} itself is Kumar and Haupt, 2025b) *Let the teacher be odeco and let* r_{m} *be the norm of the moving variables at initialization. Then, for every initialization,*

$$t^* \;\geq\; \frac{K^{K/2-1}}{(K-2)s_1}\, r_{\mathrm{m}}^{-(K-2)} \;\geq\; \frac{K^{K/2-1}}{(K-2)s_1}\, \|\theta_0\|^{-(K-2)} =: t_{\mathrm{KH}}, \tag{28}$$

with equality iff the initial direction already maximizes g. *Here* $t_{\mathrm{KH}} = 1/(K(K-2)g_*\|\theta_0\|^{K-2})$, $g_* = s_1 K^{-K/2}$, *is the escape time of Kumar and Haupt (2025b) from an initialization exactly along*

a maximizer. Under fan-in initialization the median of t^*/t_{KH} is 2.5–5.2 (binary, $N=4$, $d=2$–4), 5.2–18 (MPS, $N=4$), 29–377 (MPS, $N=6$), 182–12378 (MPS, $N=8$), 94–4463 (binary, $N=8$) and 35–270 (HT, $N=4$), growing with d, χ and K.

Proof. By Proposition A.8, $g \le g_*$ on the unit sphere, so $\ln\|\theta(\tau)\| \le \ln\|\theta_0\| + Kg_*\tau$ and (25) gives $t^* \ge \|\theta_0\|^{-(K-2)}/((K-2)Kg_*)$. Frozen variables do not enter the dynamics of the moving ones, which is the LO flow of the same g restricted to them; the same argument with their norm r_{m} gives the first inequality. □

Proposition A.13 (Escape along an imbalanced product path). derived known (the trichotomy: Rawal and DeWeese, 2026, Thm 4.2; new: the TTN reduction and the constants) *If the escape follows a rank-one path whose per-core amplitudes $x_v > 0$ obey $\dot{x}_v = s\prod_{w\neq v} x_w$, then the $x_v^2 - x_w^2$ are conserved and, with $x_v^2 = c_v^2 + y$,*

$$\hat{t} = \frac{1}{2s}\int_0^\infty \prod_{v=1}^{K} (y + c_v^2)^{-1/2}\mathrm{d}y. \tag{29}$$

Let m internal cores start at amplitude γ and the other $K-m$ (the ends) at $c_e \gg \gamma$. As $\gamma \to 0$: for $m=0$ and equal amplitudes c, $\hat{t} = c^{-(K-2)}/((K-2)s)$ (Saxe et al., 2014); for $m=1$ and ends at c, $\hat{t} \to \frac{1}{2}\mathrm{B}(\frac{1}{2}, \frac{K-2}{2})c^{-(K-2)}/s$ (B is Euler's beta function), a finite limit; for $m=2$, $\hat{t} = \ln(1/\gamma)/(s\prod_e c_e) + O(1)$; for $m \ge 3$, $\hat{t} \simeq \gamma^{-(m-2)}/((m-2)s\prod_e c_e)$.

The integral follows from Lemma A.1: $\dot{y} = 2s\prod_w (y + c_w^2)^{1/2}$. The binary tree with $N=4$ and the MPS with $N=3$ have $m=1$, the MPS with $N=4$ has $m=2$, the binary tree with $N=8$ and the HT tree with $N=4$ have $m=3$, and the MPS has $m = N-2$. The escape time is set by the path amplitudes of the internal cores, while $\|\theta_0\|$ is dominated by frozen slices and by internal cores whose norm is spread over $\sim \chi^2$ directions; this is why t_{KH} fails most for large K. That the end cores keep their directions during the escape, which reduces the dynamics to such a path, is empirical (median cosine with the initial direction 0.92–1.00 at half the escape time).

Binary tree with $N=4$. derived (given that the end directions are frozen) new. The reduced network has end vectors $a, b \in \mathbb{R}^{\chi_b}$ with $\mathcal{N}(0, \sigma^2/d^2)$ entries, $\chi_b = \min(\chi, d^2)$, and a root matrix C with $\mathcal{N}(0, \sigma^2/\chi_b^2)$ entries, and $g = s_1 a^\top C b$. With frozen end directions, (29) with $m=1$ and typical end norms $\sigma\sqrt{\chi_b}/d$ gives the median $t_1 \approx \pi d/(2s_1\sigma\sqrt{\chi_b})$ for $d \le \chi_b \le d^2$, i.e. $t_1 \approx \pi/(2s_1\sigma)$ at full bond for every d, whereas $t_{\mathrm{KH}} \propto (2d^2+1)^{-1/2}$. Measured medians are 0.80–0.95 times the formula for $d \le \chi_b < 2d$ and 0.92–1.00 times it for $\chi_b \ge 2d$; at full bond $t_1\sigma s_1 = 1.55$–1.57 for $d = 2, 3, 4$ empirical. The log-spread is $\mathrm{sd}\ln\hat{t}_1 \approx 1/(2\sqrt{\chi_b})$ derived.

Learning order. Which component is learned first is decided in the spherical phase ($\tau = O(1)$) of Proposition A.4, by the basin of the initial direction. Two cases follow from Proposition A.10.

Proposition A.14 (Race on trees whose cores all carry sites). proven *(at LO)* new *Let the teacher be odeco and the tree an MPS. By Proposition A.11 the components evolve independently at LO, and component r blows up at $\sigma^{-(K-2)}\hat{t}_{1,r}/s_r$, where the $\hat{t}_{1,r}$ are i.i.d. copies of the prefactor of Corollary A.5 for a single component of unit weight. The first component learned is $\arg\min_r \hat{t}_{1,r}/s_r$, and for two components with $\varrho = s_2/s_1 < 1$*

$$P(\textit{larger first}) = P\Big(\frac{\hat{t}_{1,2}}{\hat{t}_{1,1}} > \varrho\Big) = \frac{1}{2} + \frac{1}{2}P\Big(\Big|\ln\frac{\hat{t}_{1,2}}{\hat{t}_{1,1}}\Big| < \ln\frac{1}{\varrho}\Big), \tag{30}$$

which does not depend on σ. In the spherical flow the basin of component r is exactly the set of initial directions with $\hat{t}_{1,r}/s_r < \hat{t}_{1,r'}/s_{r'}$ for all $r' \neq r$.

Proof. Independence of the components at LO and exchangeability of the $\hat{t}_{1,r}$; the law of $\ln(\hat{t}_{1,2}/\hat{t}_{1,1})$ is symmetric. □

The race agreed with the direct two-component LO flow and with full gradient flow on 100 of 100 initializations (MPS, $N = 4$) and with three components to 0.06% in time. Further, empirical: $P(\text{larger first}) \approx F_{\mathcal{N}}(\ln(1/\varrho)/S_{\hat{t}})$, with $F_{\mathcal{N}}$ the standard normal distribution function and $S_{\hat{t}}$ the standard deviation of $\ln(\hat{t}_{1,2}/\hat{t}_{1,1})$, within ± 0.05 in all tensor-like settings computed; $\mathrm{sd}\ln\hat{t}_1 \approx c\sqrt{K-2}$ on MPS with $c = 0.27$–0.49. Hence conjecture: $P(\text{larger first}) \to \frac{1}{2}$ as $K \to \infty$ at fixed ϱ, d, χ. The spread falls as the reduced cores get more entries but saturates once χ reaches its cap, so $P \to 1$ needs $d \to \infty$ as well as large χ (conjecture, heuristic; compare the largest-first order of over-parameterized tensor decomposition, Ge et al., 2021). When some cores carry no site (binary $N = 8$, HT), the components share the internal cores and the race is approximate: it predicts the first component of the direct LO flow on 94–100% of initializations empirical.

Proposition A.15 (Matrix-like pairs)**.** *Let the smaller component j of a pair have a larger partner r with $m_{jr} = 2$, and let $n = n_{jr}$ and $\varrho = s_j/s_r$. (a)* derived known (binary $N = 4$: Jacot et al., 2022, Thm 2) *θ_j is a saddle of the spherical flow with n unstable directions while θ_r is a local maximum (Proposition A.10), so $P(\text{smaller first}) \to 0$ as $\sigma \to 0$. (b)* conjecture *(heuristic)* new *As $\sigma \to 0$,*

$$P(\text{smaller first}) \propto \sigma^{\,n(1-\varrho)/\varrho}. \tag{31}$$

Heuristic argument for (b). A trajectory that reaches $\|\theta\| = O(1)$ near θ_j must pass close to it. There the deviations along the n unstable directions grow in τ at rate $K^{1-K/2}(s_r - s_j)$, while $\mathrm{d}\ln\|\theta\|/\mathrm{d}\tau = K^{1-K/2}s_j$; the deviations therefore grow like $(\|\theta\|/\|\theta_0\|)^{(1-\varrho)/\varrho}$. At the escape $\|\theta\|/\|\theta_0\| \sim 1/\sigma$, so the trajectory escapes near θ_j only if all n initial deviations are below $\sim \sigma^{(1-\varrho)/\varrho}$, which for a smooth density of initial deviations has probability $\propto \sigma^{n(1-\varrho)/\varrho}$. □

Checks of (b): in a leading-order emulation (MPS, $N = 4$, pattern teachers with $n = 1, \dots, 4$) the fitted slopes $\mathrm{d}\ln P/\mathrm{d}\ln\sigma$ are ordered as n and are 0.67–1.03 (typically 0.8) of $n(1-\varrho)/\varrho$ over $\sigma = 10^{-1}$–10^{-8}, steepening toward it at small σ. Full gradient flow (binary $N = 4$, $d = \chi = 2$, $\varrho = 0.7$) gives $P(\text{smaller first}) = 0.155,\ 0.064,\ 0.024$ at $\sigma = 0.1,\ 0.01,\ 0.001$ (local slopes 0.38, 0.43 against 0.43) empirical. Gradient flow puts the smaller component first on only 59–70% of the initializations on which the emulation does, because the term $\frac{1}{2}\|W\|_F^2$ slows the component that is ahead; whether gradient flow has the same exponent as the LO flow is open. When the wrong order occurs, it is corrected through a strict saddle (Remark A.24). These predictions are tested in Section 4.5.

A.3 Saddles after the first escape

Setting and assumptions. After the first escape, gradient flow passes near a *learned saddle* θ_S, a critical point that represents the partial fit $W_S = \sum_{j\in S} s_j U_j$ of Section 2.3. We assume: (A1) whitened inputs; (A2) a pattern teacher with distinct patterns, so that the U_r are orthonormal; (A3) θ_S is balanced ($Q_e = 0$ on every bond) and has bond ranks $r_e = \mathrm{rank}(W_S)^{\langle e\rangle}$, padded with zeros to χ_e (a critical point by Lemma A.2; gradient flow from small initialization has $Q_e = O(\sigma^2)$, so plateaus are balanced to that order); (A4) $\chi_e \geq r_e + 1$ on every bond whose rank must grow; (A5) leading order in the entries that vanish at θ_S, whose size σ_{eff} on the plateau is small. In a basis of each bond in which the learned channels come first, entries of θ whose bond indices are all learned are *old* and the others *new*; $c_e = \chi_e - r_e$ is the number of new channels of bond e. Let $\mathcal{R} = W^* - W_S$ be the residual and $L_S = \frac{1}{2}\|\mathcal{R}\|_F^2$. With one learned component j, balancedness on bonds of rank one forces equal core norms, so every core of θ_S is q_j times the pattern of j on its sites times channel 1 on its bonds, $q_j = s_j^{1/K}$ proven.

Expansion around the saddle. For a set $\mathcal{J}$ of cores and perturbations $\delta = (\delta_v)_v$, let $W_{\mathcal{J}}[\delta]$ be the tensor obtained by replacing the cores in $\mathcal{J}$ by the δ_v and keeping the others at θ_S. Multilinearity gives exactly $W(\theta_S + \delta) = \sum_{\mathcal{J}} W_{\mathcal{J}}[\delta]$, with $W_\emptyset = W_S$, and

$$L(\theta_S + \delta) = L_S - \sum_{\mathcal{J}\neq\emptyset} \langle \mathcal{R},\, W_{\mathcal{J}}[\delta]\rangle + \tfrac{1}{2}\Big\| \sum_{\mathcal{J}\neq\emptyset} W_{\mathcal{J}}[\delta]\Big\|_F^2. \tag{32}$$

Lemma A.16 (Routes). proven new *(i) If δ_v has only new entries, then $W_{\{v\}}[\delta_v] = 0$, so new entries carry no Gauss–Newton cost at second order. (ii) For new entries, $W_{\mathcal{J}}[\delta] \neq 0$ requires every new index of every δ_v to sit on a bond inside $\mathcal{J}$ and to be matched there; hence every bond leaving $\mathcal{J}$ carries a learned index, and every connected component of $\mathcal{J}$ has at least two cores. For a connected set $\mathcal{C}$ (a* route*),*

$$W_{\mathcal{C}}[\delta] = \mathcal{W}_{\mathcal{C}}(\delta) \otimes \bigotimes_{f\in\partial\mathcal{C}} T_f, \tag{33}$$

where $\mathcal{W}_{\mathcal{C}}(\delta)$ is the small TTN formed by the δ_v (internal bond dimensions c_e), $\partial\mathcal{C}$ is the set of bonds leaving $\mathcal{C}$, and T_f is the contraction of the learned network beyond f, a map from the r_f learned indices of f to tensors on the sites H_f beyond f. Every proper subset $\mathcal{C}' \subsetneq \mathcal{C}$ has $W_{\mathcal{C}'}[\delta] = 0$: a route pays no Gauss–Newton cost at the order of its gain.

Proof. A new index of δ_v on a bond meets, in the neighbouring core, either a perturbation of that core or the zero block of a learned core (Lemma A.2). In a proper subset of a route some new index meets such a zero block. □

Disconnected sets $\mathcal{J}$ also contribute at the same degree, but each connected component $\mathcal{J}_i$ alone already gives $W_{\mathcal{J}_i} \neq 0$ through learned boundary indices, so it pays the Gauss–Newton cost $\frac{1}{2}\|W_{\mathcal{J}_i}\|_F^2$ at the same order as the gain. For two blocks this makes the quadratic form positive semidefinite iff $s_j \geq s_r$, the condition of Proposition A.22 derived (checked on MPS, $N = 5$); that disconnected sets never descend before the single-core rotations is shown only for this case.

Definition A.17 (Local residual). For a route $\mathcal{C}$, $\mathcal{R}_{\mathcal{C}} := \langle \mathcal{R}, \bigotimes_{f\in\partial\mathcal{C}} T_f\rangle$ (contraction over all sites beyond $\partial\mathcal{C}$) is a tensor with one mode per core of $\mathcal{C}$, made of its physical legs and its learned boundary indices; then $\langle \mathcal{R}, W_{\mathcal{C}}[\delta]\rangle = \langle \mathcal{R}_{\mathcal{C}}, \mathcal{W}_{\mathcal{C}}(\delta)\rangle$.

Proposition A.18 (Escape degree). derived known (mechanism: Kumar and Haupt, 2026, Thm 8, and the accumulated utility of Kunin et al., 2025; new: the degree is set by the tree) *The new entries first lower L at order*

$$k^* = \min\{|\mathcal{C}| :\ \mathcal{C} \text{ a route},\ \mathcal{R}_{\mathcal{C}} \neq 0\}. \tag{34}$$

At LO (old entries frozen at θ_S) the new entries of a minimizing route follow the gradient flow of a k^*-core TTN that fits the target $\mathcal{R}_{\mathcal{C}}$ from small initialization: the escape from the saddle is the first-escape problem of Section A.2 for the route, with $K \to k^*$ and $W^* \to \mathcal{R}_{\mathcal{C}}$. This needs the old entries to sit at a local minimum of the fixed-rank problem; it fails at the strict saddles of Proposition A.22, where the old entries move first.

In Kumar and Haupt (2026) dormant weights of lowest homogeneity k stay small for a time $\propto \delta^{-(k-2)}$, δ their initial size, provided the active weights sit at a local minimum; the TTN content is which entries are dormant (the new entries of routes) and their degree k^*.

Table 5: Route degree $k = |\mathrm{St}(N(D))|$ (Proposition A.19) for a two-component pattern teacher whose components differ on the sites D (numbered from 0), on the trees of Table 1; $m = |N(D)|$ is 2 unless shown. $k = 1$: no plateau; $k = 2$: strict saddle, plateau $\propto \ln(1/\sigma)$; $k \geq 3$: higher-order saddle, plateau $\propto \sigma^{-(k-2)}$.

D ($N = 4$)	binary $K = 3$	MPS $K = 4$	HT $K = 7$	D ($N = 8$)	binary $K = 7$	MPS $K = 8$	HT $K = 15$
$\{0,1\}$	1 ($m = 1$)	2	3	$\{0,1\}$	1 ($m = 1$)	2	3
$\{1,2\}$	3	2	5	$\{1,2\}$	3	2	5
$\{0,2\}$	3	3	5	$\{3,4\}$	5	2	7
$\{0,3\}$	3	4	5	$\{0,4\}$	5	5	7
$\{0,1,2,3\}$	3	4 ($m = 4$)	7 ($m = 4$)	$\{2,5\}$	5	4	7
				$\{1,6\}$	5	6	7
				$\{0,7\}$	5	8	7
				$\{0,\ldots,3\}$	3	4 ($m = 4$)	7 ($m = 4$)
				all	7 ($m = 4$)	8 ($m = 8$)	15 ($m = 8$)

Proposition A.19 (Steiner identity). proven *(identity and upper bound on the order)* derived *(order $= k^*$)* new (descent along a whole connected rank-deficient component: Furman et al., 2026, Lemma D.8) *Let $S = \{j\}$. For every route $\mathcal{C}$,*

$$\mathcal{R}_{\mathcal{C}} = q_j^{K-|\mathcal{C}|} \sum_{r:\, N(D_{jr}) \subseteq \mathcal{C}} s_r \, U_r|_{\mathcal{C}}, \tag{35}$$

where $U_r|_{\mathcal{C}}$ is U_r restricted to the sites of $\mathcal{C}$. Hence $k^* = \min_r |\mathrm{St}(N(D_{jr}))|$; the route of component r has degree $k = |\mathrm{St}(N(D_{jr}))|$ and coupling

$$c_r = s_r \, q_j^{K-k} = s_r \, s_j^{(K-k)/K}. \tag{36}$$

The bonds whose rank must grow to add r are exactly the bonds of $\mathrm{St}(N(D_{jr}))$, so k is one plus their number, and $k = 1$ iff D_{jr} lies on a single core. With route entries proportional to t, $L - L_S = -c_r t^k + O(t^{2k})$: the saddle has order at most k^*. That no curve descends faster is shown only up to order three (Proposition A.21); that the order equals k^* is derived (Proposition A.18).

Proof. The learned network beyond a boundary bond f is $T_f = q_j^{n_f} U_j|_{H_f}$, with n_f cores and sites H_f, and $\sum_f n_f = K - |\mathcal{C}|$. Since $\mathcal{R} = \sum_{r \neq j} s_r U_r$, contracting U_r with T_f gives $q_j^{n_f} \prod_{i \in H_f} \langle u_r^{(i)}, u_j^{(i)} \rangle$, which is $q_j^{n_f}$ if $\pi_r = \pi_j$ on H_f and 0 otherwise. So U_r survives iff no site of D_{jr} lies beyond $\mathcal{C}$, i.e. iff $N(D_{jr}) \subseteq \mathcal{C}$, and a connected $\mathcal{C}$ containing $N(D_{jr})$ contains its Steiner subtree. The rank statement is Proposition 2.1(b) for the components j, r. For the descent curve, let δ_v be t times the pattern of r on S_v, with a new channel on the bonds inside $\mathcal{C}$ and channel 1 on $\partial\mathcal{C}$. Then $W_{\mathcal{C}}[\delta] = t^k q_j^{K-k} U_r$, every other $W_{\mathcal{J}}[\delta]$ vanishes by Lemma A.16, and $L - L_S = -c_r t^k + \frac{1}{2} q_j^{2(K-k)} t^{2k}$. □

Table 5 lists k for two-component pattern teachers. Furman et al. (2026) bound the order by the number of cores touching a connected component of rank-deficient bonds, which at a rank-one saddle with $\chi \geq 2$ is the whole tree (K cores), whereas k^* can be as small as 2. For Tucker decompositions, Frandsen and Ge (2022) exhibit descent of order 2, 3 and 4 when 2, 3 and 4 cores must move.

Lemma A.20 (Several learned components). proven new *Let E_+ be the set of bonds e with $\mathrm{rank}(W_S + U_r)^{\langle e \rangle} > \mathrm{rank}(W_S)^{\langle e \rangle}$. (i) E_+ is connected. (ii) If $E_+ = \emptyset$, a single core can produce U_r, so $\nabla L(\theta_S) \neq 0$ and θ_S is not a saddle.*

Proof. (i) If the restriction of U_r to one side of a bond lies outside the learned column space, so does its restriction to any larger side; hence every bond between two bonds of E_+ is in E_+. (ii) Walking from any core toward a branch on which U_r is new, the walk cannot return, so it ends at a core all of whose branches are old; that core alone can produce U_r. □

With several learned components, adding r exactly needs $k_{\mathrm{ex}} = |E_+| + 1$ cores, and $k^* \le k_{\mathrm{ex}}$ derived. The inequality is strict when the learned space of a bond holds a superposition of learned patterns that overlaps $U_r|_{H_f}$ without containing it; a cheaper route then first learns the projection, which adds a plateau at the best fit of an intermediate rank profile (*partial learning*, Remark A.27). When $k^* = k_{\mathrm{ex}}$ the coupling is $c_r = s_r \prod_{f\in\partial\mathcal{C}} \|T_f^\top U_r|_{H_f}\|$ in the balanced gauge derived (checked against exact Hessians at saddles with one and several learned components).

Saddle type.

Proposition A.21 (Hessian at a balanced learned saddle). proven *(at θ_S)* new (that two-core descent gives a strict saddle is Furman et al., 2026, Lemma D.9; deep linear networks: Achour et al., 2024) *(a) The Hessian at θ_S is block diagonal between the old entries and the new entries. The old block (the* fixed-rank block*) is the Hessian of the problem with the bond ranks fixed at r_e: Gauss–Newton term plus the piggyback couplings of Proposition A.22. (b) The new block (the* rank-increasing block*) vanishes except on routes with two cores: a route $\{u, v\}$ through the bond e contributes $\left(\begin{smallmatrix} 0 & -B \\ -B^\top & 0 \end{smallmatrix}\right)$ with $B = \mathcal{R}_{\{u,v\}}$, for each of the c_e new channels, hence eigenvalues $\pm\sigma_k(B)$ with multiplicity c_e. With one learned component the negative one is $-\lambda$,*

$$\lambda = \sigma_{\max}(\mathcal{R}_{\{u,v\}}) = s_r\, s_j^{(K-2)/K}. \tag{37}$$

(c) Hence θ_S is a strict saddle iff the fixed-rank block has a negative eigenvalue or a route with $k^* = 2$ exists. If $k^* = 1$, θ_S is not critical (with one learned component $\|\nabla L(\theta_S)\| = s_r q_j^{K-1}$). If $k^* \ge 3$ and the fixed-rank block is positive semidefinite, the Hessian has no negative eigenvalue: θ_S is a higher-order saddle. (d) The gauge orbit through θ_S has dimension $\sum_e[\chi_e^2 - (\chi_e - r_e)^2]$, and the kernel of the Hessian has dimension $\sum_e r_e^2 + \dim\theta_{\mathrm{new}} - 2\sum_{k^*=2 \text{ routes}} c_e$, where $\dim\theta_{\mathrm{new}}$ counts the new entries.

Proof sketch. An old–new cross term of $\langle \mathcal{R}, W_{\mathcal{J}}\rangle$ needs an old index to meet a new one, which gives zero, and the Gauss–Newton cross term vanishes by Lemma A.16(i). On the new entries the only second-order terms are $-\langle \mathcal{R}, W_{\{u,v\}}[\delta]\rangle$ for adjacent pairs, and with one learned component $\mathcal{R}_{\{u,v\}} = q_j^{K-2}\sum_r s_r U_r|_{\{u,v\}}$ by (35). The gauge generators supported on the new–new blocks act trivially at θ_S, so the stabilizer has dimension $\sum_e(\chi_e - r_e)^2$. □

Checked at exact saddles in 21 (tree, D) cases: $\|\nabla L\| = s_r q_j^{K-1}$ for $k = 1$, λ to six digits with multiplicity $\chi - 1$ for $k = 2$, no negative eigenvalue for $k \ge 3$ in the correct order, and the kernel and orbit dimensions of (d). The negative curvature of a $k = 2$ route is the same λ that sets its escape rate (Proposition A.25).

Proposition A.22 (Piggyback block and piggyback graph). proven new (for general trees; deep linear networks: Achour et al., 2024, Thm 7; odeco components are local optima of rank-one approximation: Anandkumar et al., 2014) *Let $S = \{j\}$. A* piggyback direction *keeps the learned bond index and rotates the physical vectors of one terminal core u toward a site pattern w on S_u. (i) For one residual component r with $m_{jr} = 2$ and terminal cores u, v the rotation block is $\left(\begin{smallmatrix} q_j^{2(K-1)} & -s_r q_j^{K-2} \\ -s_r q_j^{K-2} & q_j^{2(K-1)} \end{smallmatrix}\right)$, with*

Table 6: Saddle type at a balanced saddle with one learned component j and one residual component r (Propositions A.21 and A.22; proven at the saddle). With several residual components use the criterion $\lambda_{\max}(\Gamma) > s_j$, which can make the correct-order cases strict.

route data	$s_j > s_r$ (correct order)	$s_j < s_r$ (wrong order)
$k = 1$	not critical	not critical
$k = 2$ (then $m = 2$)	strict, $\lambda_{\min} = -s_r q_j^{K-2}$	strict, $\lambda_{\min} = -s_r q_j^{K-2}$
$k \geq 3$, $m = 2$	higher order (order k)	strict via piggyback, $\lambda_{\min} = -(s_r - s_j) q_j^{K-2}$
$k \geq 3$, $m \geq 3$	higher order	higher order

eigenvalues $q_j^{K-2}(s_j \mp s_r)$*; for* $m_{jr} \geq 3$ *it is* $q_j^{2(K-1)} I \succ 0$*. (ii) In general the rotation block is*

$$H_{\text{rot}} = q_j^{K-2}(s_j I - \Gamma), \tag{38}$$

where Γ *is the weighted adjacency matrix of the* piggyback graph*: its nodes are the pairs* (u, w) *of a terminal core and a rotated site pattern, with one edge between* $(u, \pi_r|_u)$ *and* $(v, \pi_r|_v)$*, of weight* s_r*, for every residual component* r *with* $N(D_{jr}) = \{u, v\}$*. Hence* $\lambda_{\min}(H_{\text{rot}}) = q_j^{K-2}(s_j - \lambda_{\max}(\Gamma))$*, and* θ_j *is a strict saddle iff* $\lambda_{\max}(\Gamma) > s_j$ *or a route with* $k^* = 2$ *exists.*

Proof. Rotating core u by ε_u toward w creates the mixed tensor (pattern w on S_u, pattern of j elsewhere) of norm $q_j^{K-1}\varepsilon_u$. Mixed tensors of different nodes are orthonormal, so the Gauss–Newton part is diagonal, with entries $q_j^{2(K-1)} = s_j q_j^{K-2}$. A residual r with terminals $\{u, v\}$ couples the two rotations through $-\langle \mathcal{R}, W_{\{u,v\}} \rangle = -s_r q_j^{K-2} \varepsilon_u \varepsilon_v$; residuals with $m_{jr} \geq 3$ add no quadratic term, and U_r itself appears only at order m_{jr}. The radial part (rescaling the learned component) is positive semidefinite. □

A *star* (residuals $r_1, \ldots, r_n$ sharing one node) has $\lambda_{\max}(\Gamma) = (\sum_i s_{r_i}^2)^{1/2}$, and a triangle of equal weights s has $\lambda_{\max}(\Gamma) = 2s$. So θ_j can be a strict saddle although every residual is smaller than s_j and every route has $k^* \geq 3$; with $d = 2$, any two residuals that differ from j on the same single-site core share a node. The criterion was checked to six digits in five strict cases and three sub-critical controls. Dynamics empirical: for $j = 0000$ (weight 2) and $r = 1010$, $r' = 1001$ (weight 1.6 each) on the MPS with $N = 4$, $\lambda_{\max}(\Gamma) = 2.263 > 2$ and $\lambda_{\min} = -0.3716$; gradient flow passes near j, forgets it (a_j falls to 0.01–0.10), learns r and r', and relearns j (5 of 5 seeds), whereas with weights 1.2 ($\lambda_{\max}(\Gamma) = 1.70 < 2$) it plateaus at j (3 of 3 seeds). Learning-order experiments with three or more components should therefore compute $\lambda_{\max}(\Gamma)$ for their teachers.

Remark A.23 (Binary tree with $N = 4$). proven known (Saxe et al., 2014; Jacot et al., 2022; Achour et al., 2024) Write the bottom cores as matrices $\tilde{A}, \tilde{B} \in \mathbb{R}^{d^2 \times \chi}$ (sites $(0, 1)$ and $(2, 3)$ as rows) and the root as a $\chi \times \chi$ matrix. Then $W^{\langle e \rangle} = \tilde{A}\, G_{\text{root}} \tilde{B}^\top$ on both bonds, and L is exactly the loss of a depth-3 linear network fitting the $d^2 \times d^2$ matrix $(W^*)^{\langle e \rangle}$. Its saddles are SVD truncations (Jacot et al., 2022); critical points that select non-top singular values are strict, while balanced ones that select the top values are not (Achour et al., 2024, Thm 7; their full-rank assumption is not met by our low-rank targets, which the computation above covers). Every statement of this appendix about this tree is therefore a statement about deep linear networks.

Remark A.24 (After a wrong first escape). derived new Two-component teachers with the smaller component j learned first; ε is the piggyback amplitude and b the route amplitude when the plateau starts. (1) $k \geq 3$, $m = 2$ (matrix-like pair): the learned bond index rotates from j to r

after $\Delta t \approx \ln(1/\varepsilon)/((s_r - s_j)s_j^{(K-2)/K})$, the loss plateaus at $\frac{1}{2}s_j^2$ with $a_j \approx 0$, and j is relearned through its route after a time $\propto \sigma_{\text{eff}}^{-(k-2)}$. (2) $k = 2$: r is added directly, at the larger rate $s_r s_j^{(K-2)/K}$, and j is kept. (3) $m \geq 3$ (tensor-like pair): no rotation; r is added through its route after $\Delta t \approx b^{-(k-2)}/((k-2)s_r s_j^{(K-k)/K})$ and a_j stays at 1. Hence a correction through a strict saddle exists only for matrix-like pairs, exactly where the wrong order becomes rare as $\sigma \to 0$ (Proposition A.15). Checked on binary and MPS trees with $N = 4$ (rotation times within 9% of the prediction; escape at $t = 4999.97$ against 5000), in runs made before the random streams of teacher and initialization were separated.

Escape times. Let the route of component r have $k = |\mathcal{C}|$ cores and amplitudes b_v, $v \in \mathcal{C}$ (its new entries: the pattern of r on the physical legs, a new channel on the internal bonds, channel 1 on $\partial\mathcal{C}$). With the old entries frozen and one learned component, put $A := q_j^{K-k}$; (32) and Lemma A.16 give the closed route dynamics

$$\dot{b}_v = A \prod_{u \in \mathcal{C}\setminus v} b_u \Big(s_r - A \prod_{u \in \mathcal{C}} b_u\Big), \qquad a_r = \frac{A}{s_r} \prod_{u \in \mathcal{C}} b_u, \tag{39}$$

a depth-k mode with target s_r/A that conserves the differences $b_u^2 - b_v^2$.

Proposition A.25 (Escape times with head start). derived new (for $k \geq 3$; the $k = 2$ exponent is known: Zhang et al., 2026, Thm 4, and Kunin et al., 2025; the trichotomy in k: Rawal and DeWeese, 2026, Thm 4.2) *Let the initialization be aligned: every core of the learned component j and of the route starts at amplitude σ, and the hanging cores are aligned with the shared pattern. Let $\Delta t_2 = t(a_r = \frac{1}{2}) - t(a_j = \frac{1}{2})$. As $\sigma \to 0$,*

$$k = 2: \quad \Delta t_2 = \frac{(1 - s_r/s_j)\ln(1/\sigma) + O(1)}{s_r\, s_j^{(K-2)/K}}, \tag{40}$$

$$k \geq 3: \quad \Delta t_2 = \frac{\sigma^{-(k-2)}\,(1 - s_r/s_j)}{(k-2)\, s_r\, s_j^{(K-k)/K}}(1 + o(1)), \tag{41}$$

and $\Delta t_2 = O(1)$ (no plateau) for $k = 1$. For $k \geq 3$ the route thus enters the plateau with the effective amplitude $\sigma_{\text{eff}} = \sigma(1 - s_r/s_j)^{-1/(k-2)}$. For $k = K$ the route never sees j, and r escapes at the absolute time $\sigma^{-(K-2)}/((K-2)s_r)$.

Derivation. The route exists from the start, with the then-small learned cores in its hanging part, so it grows during the rise of j with the instantaneous coupling $c(t) = s_r q_j(t)^{K-k}$, where $\dot{q}_j \simeq s_j q_j^{K-1}$ while q_j is small. For $k = 2$, $\ln b$ grows by $\int c\,\mathrm{d}t = \int s_r q_j^{K-2}\,\mathrm{d}q_j/(s_j q_j^{K-1}) = (s_r/s_j)\ln(1/\sigma) + O(1)$, so the route enters the plateau with $\ln b_{\text{eff}} = -(1 - s_r/s_j)\ln(1/\sigma) + O(1)$ and needs a further time $\ln(1/b_{\text{eff}})/\lambda$, with λ from (37). For $k \geq 3$ the balanced route obeys $\dot{b} = c(t)\, b^{k-1}$, i.e. $\frac{\mathrm{d}}{\mathrm{d}t} b^{-(k-2)} = -(k-2)c(t)$, so that at the end of the rise of j

$$b_{\text{eff}}^{-(k-2)} = \sigma^{-(k-2)} - (k-2)\int c\,\mathrm{d}t = \sigma^{-(k-2)}(1 - s_r/s_j) + O(1),$$

and the plateau lasts $b_{\text{eff}}^{-(k-2)}/((k-2)s_r s_j^{(K-k)/K})$. The integrand $q_j^{K-k}\mathrm{d}t = q_j^{1-k}\mathrm{d}q_j/s_j$ is largest at $q_j \approx \sigma$: for $k \geq 3$ the head start accrues during the first plateau, while the learned cores are still of order σ. □

The factor $1 - s_r/s_j$ is the head start. Without it the slope of Δt_2 against $\ln(1/\sigma)$ for $k = 2$ would be $1/\lambda$ instead of $(1 - s_r/s_j)/\lambda$, which separates the two pictures experimentally (Section 4.4). Checked: with aligned initialization, $k = 2$ slopes 1.0067 and 0.78408 against 1.0000 and 0.78405, and Δt_2 divided by (41) tends to 1.0000 for $k = 3, 4, 5$ down to $\sigma = 3 \cdot 10^{-5}$; with random fan-in initialization, $k = 2$ slopes 1.05 and 1.066 against 1.00 and 1.065, and per-seed exponents -0.97 to -1.02 ($k = 3$) and -1.99 to -2.00 ($k = 4$). For an unbalanced route with positive product, $\Delta t = \int_0^\infty \mathrm{d}\beta/(2c\prod_{v\in\mathcal{C}}(b_v^2 + \beta)^{1/2})$ derived, as in Proposition A.13; if the product is negative, the smallest $|b_v|$ first passes through zero, which lengthens the plateau. With random initialization the prefactors are not predicted by σ alone: they are set by the route amplitudes at the start of each plateau, which vary between seeds by factors above 20 for $k \geq 4$ empirical.

Proposition A.26 (Signatures of the route degree)**.** derived known (the exponent $2 - 2/k$ of a balanced product of k factors: Arora et al., 2019; Razin et al., 2022; new: its use to read k off TTN plateaus) *(i) Per seed (fixed initialization direction, σ varied), $\Delta t_2 \propto \sigma^{-(k-2)}$ for $k \geq 3$. (ii) On a balanced route,* (39) *gives $\dot{a}_r = kA^{2/k} s_r^{2-2/k} a_r^{2-2/k}(1 - a_r)$, so the local growth exponent is*

$$\gamma(a_r) = \frac{\mathrm{d}\ln \dot{a}_r}{\mathrm{d}\ln a_r} = 2 - \frac{2}{k} - \frac{a_r}{1 - a_r}, \tag{42}$$

that is $1, \frac{4}{3}, \frac{3}{2}, \frac{8}{5}$ for $k = 2, \ldots, 5$ as $a_r \to 0$. The route is balanced only once $b^2 \gg |b_u^2 - b_v^2| = O(\sigma^2)$, so k is read in the window $a_r \in [10^{-4}, 10^{-2}]$; a smaller γ at smaller a_r indicates an unbalanced route, not a smaller k. (iii) At a plateau point, where the route amplitudes are small but nonzero, each active route contributes one negative Hessian direction modulo gauge, with

$$\lambda_{\min} \simeq -s_r s_j^{(K-2)/K} \quad (k = 2), \qquad \lambda_{\min} \simeq -(k-1)\, s_r A\, b^{k-2} = O(\sigma_{\mathrm{eff}}^{k-2}) \quad (k \geq 3), \tag{43}$$

where $b = (s_r a_r/A)^{1/k}$. The count of one negative eigenvalue per route holds for $k \leq 3$ only: at $k = 4$, new-index pairs that connect to learned structure add indefinite terms of order σ^2, the same order as the route eigenvalue.

In (iii), the Hessian on the route block is $-s_r A b^{k-2}(\mathbf{1}\mathbf{1}^\top - I)$; its $k - 1$ positive directions rescale the amplitudes at fixed product, which is a gauge motion. Checked: $\gamma = 0.995$–1.000, 1.334–1.344, 1.490–1.500 for $k = 2, 3, 4$, and $|\lambda_{\min}|/\lambda_{\max}$ at plateau midpoints scaling as σ^0, σ^1, σ^2 for $k = 2, 3, 4$ (MPS, $N = 4$). For the diagnostics of Section 2.7 this means that the fixed relative threshold 10^{-3} calls $k = 3$ plateaus strict for $\sigma \gtrsim 0.01$–0.04 (depending on the seed); the saddle type should be read from the scaling of $\lambda_{\min}$ with σ (constant: strict; $\propto \sigma^{k-2}$: higher order) or from γ.

Remark A.27 (The whole sequence). derived at LO for each step; conjecture (heuristic) as an algorithm. new (1) The first escape ends near a critical point of the bond-rank-one problem; it is a single component j only if θ_j is a local maximum of the spherical problem (Proposition A.10, with the piggyback graph when several partners share a core); a star with $\lambda_{\max}(\Gamma) > s_j$ sends the flow past j toward a superposition. (2) At every later saddle, if the fixed-rank block has a negative eigenvalue (wrong order, stars), the next event is a rotation of the learned structure or a $k = 2$ route, whichever is more unstable, in time $\propto \ln(1/\sigma)$, and Proposition A.18 does not apply. Otherwise the next event is the route of smallest k^* (logarithmic beats power law, smaller powers beat larger ones); ties are decided by the coupling and the head start, $c\, b_{\mathrm{eff}}^{k-2}$, which for $k \geq 3$ are dominated by the random route amplitudes. (3) The escape raises the bond rank by one on the bonds inside the route, and the next saddle is the best approximation of W^* with the new rank profile (for pattern teachers without superpositions, the partial sum of the learned components). (4) Plateau levels are the optimal values of these rank-constrained problems.

Because k^* depends on the teacher, the tree and the learned set, *locality beats weight*: a smaller component with a smaller k^* is learned first, and the same teacher gives different sequences on different trees. Examples (checked numerically): for the teacher 0000 (weight 1), 0110 (0.3), 1001 (0.6), the MPS with $N = 4$ learns 0000, then 0110 ($k = 2$, logarithmic gap), then 1001 ($k = 4$, gap $\propto \sigma^{-2}$), while the binary tree learns 0000 and then 1001 and 0110 ($k = 3$ each, gaps $\propto \sigma^{-1}$). For 0000 (1), 1100 (0.6), 1111 (0.4) on the MPS with $N = 4$ a cheaper $k = 2$ route learns part of 1111 first (partial learning): the levels are 0.76, 0.26, 0.08, 0.0569, 0, and every gap after the first is logarithmic, although adding 1111 exactly would need three cores. The number of cores that must change at once plays the role of the leap of Abbe et al. (2023); what is specific to the TTN is that it is set by the tree.

Proposition A.28 (Conservation-law trap). derived *(mechanism, in a reduced model)* new *Let a route end at a leaf core. On the bond e of that end (child v, parent p) let ν be the new channel used by the route and $Q_{\nu\nu} = \nu^\top Q_e \nu$. Let x be the leaf's new column, and split the parent's slice along ν into its route entry y_{route} and the* visible *entries y_{vis}, which connect through the other side to learned structure. Conservation gives $\|x\|^2 - \|y_{\text{vis}}\|^2 - \|y_{\text{route}}\|^2 = Q_{\nu\nu}$. On the plateau the Gauss–Newton term makes x and y_{vis} decay together, at a rate of order $|Q_{\nu\nu}| = O(\sigma^2)$, so the leaf column decays to zero iff the end is* unfavourable,

$$Q_{\nu\nu} + \|y_{\text{route}}\|^2 < 0. \tag{44}$$

The route's drive on x is proportional to the product of the other route amplitudes, so while another end is favourable it pins x at a positive value and the route escapes, slowly. A permanent stall ($x \to 0$ exponentially, down to roundoff) therefore requires every end of the route to be unfavourable.

The dead state is linearly stable if $\gamma_1\gamma_2 > \kappa^2$, with decay rates $\gamma_i \sim |Q_i| \sim \sigma^2$ and mutual drive $\kappa \sim s_r A \prod_{\text{inner}} b \sim \sigma^{k-2}$ (conjecture, heuristic): stable as $\sigma \to 0$ for $k \geq 5$, never for $k = 3$, undecided for $k = 4$. On HT leaf bonds ($\chi_v = d = 2$ below a parent bond of dimension 3), fan-in initialization gives $Q_{\nu\nu} = \sigma^2(X_2/2 - X_6/4)$ with independent chi-squared variables X_n of n degrees of freedom (if ν is independent of the initialization), so $\mathbb{E}Q_{\nu\nu} = -\sigma^2/2$, $P(Q_{\nu\nu} > 0) = (3/2)^{-3} \approx 0.30$, and about half of the two-ended routes have both ends unfavourable (conjecture, heuristic). Data empirical (HT, $N = 4$, the teacher 0000, 0110, 1001 above, $\sigma = 0.2$, nine fan-in initializations): every run whose route ends were all unfavourable stalled with dead leaves (6 runs), each run with one favourable end had the other end pinned (2 runs), and one run had no stall. The trapped point is not a local minimum, but gradient flow does not move along its flat directions. Gauge-balanced initialization ($Q_e = 0$, same W_0) kept the routes alive (3 of 3 runs); that it always does, and that the stalled fraction tends to P(all ends unfavourable) for $k \geq 5$ as $\sigma \to 0$, are conjectures. The trap is a property of unbalanced initialization, not of the landscape; experiments on HT trees should report the stalled fraction and $Q_{\nu\nu}$ at every route end (Section 4.4).

A.4 Teachers that are not orthogonally decomposable

For a general teacher the saddles are no longer partial sums of components. Notation: $\mathcal{M}_{\mathbf{r}}$ is the manifold of tensors with bond ranks exactly $\mathbf{r} = (r_e)_e$ (Holtz et al., 2012b; Uschmajew and Vandereycken, 2013). For $W \in \mathcal{M}_{\mathbf{r}}$ and $e = e_v$, $p = p(v)$, U_e and V_e are the column and row spaces of $W^{\langle e \rangle}$; the environment spaces of e are $E_e^- = (\bigotimes_{c \in \text{ch}(v)} U_c) \otimes \mathbb{R}^{d_v}$ and $E_e^+ = (\bigotimes_{c \in \text{ch}(p) \setminus \{v\}} U_c) \otimes \mathbb{R}^{d_p} \otimes V_p$, with $d_v = \prod_{i \in S_v} d_i$ (no V_p factor at the root); and $\widetilde{W}_e^* := P_{E_e^-} (W^*)^{\langle e \rangle} P_{E_e^+}$ is the *environment-projected teacher*.

Proposition A.29 (Critical points of the rank-constrained problem). proven known (in substance: the fixed-point conditions of HOOI, De Lathauwer et al., 2000; Eldén and Savas, 2011, and of one-site ALS,

Holtz et al., 2012a; tangent spaces: Uschmajew and Vandereycken, 2013) *$W \in \mathcal{M}_{\mathbf{r}}$ is a critical point of L restricted to $\mathcal{M}_{\mathbf{r}}$ iff, for every bond e,*

$$W^{\langle e \rangle} = P_{U_e} \widetilde{W}_e^* = \widetilde{W}_e^* P_{V_e}, \tag{45}$$

that is, iff U_e is spanned by r_e left singular vectors of $\widetilde{W}_e^*$ (not necessarily the top ones) and $W^{\langle e \rangle}$ is the corresponding SVD truncation; equivalently, iff W is a fixed point of one-site alternating least squares. Padded with zeros, such a point is a critical point of the parameterized loss for every $\chi_e \geq r_e$ (Lemma A.2).

Proof sketch. At a parameterization with full bond ranks the differential with respect to G_v spans the local space $\mathcal{V}_v = E_e^- \otimes V_e$ ($e = e_v$), so W is critical iff $P_{\mathcal{V}_v} W^* = W$ for every v. For $e = e_v$ this reads $\widetilde{W}_e^* P_{V_e} = W^{\langle e \rangle}$ at core v and $P_{U_e} \widetilde{W}_e^* = W^{\langle e \rangle}$ at core p. If $P_U A = A P_V = X$ has rank r for subspaces U, V of dimension r, then $P_U A A^\top = X X^\top$ is symmetric, so P_U commutes with $A A^\top$ and U is spanned by left singular vectors of A. □

Proposition A.30 (When is the hierarchical SVD critical?). proven *(a, b)* derived *(c)* new (the ε^4 gap; the HSVD is Grasedyck, 2010) *Let W_H be the HSVD truncation of W^* at ranks $\mathbf{r}$. (a) W_H is critical iff its frames satisfy (45); for generic teachers they do not, because the parts of W^* discarded at other bonds leak into the environment of each bond. (b) If the best rank-r_e approximation of W^* across every bond e is one and the same tensor W (consistent truncations: for example an odeco teacher truncated to the same components on every bond, or the binary tree with $N = 4$, whose two bonds induce the same bipartition), then $W_H = W$ is critical and is the global minimizer over bond ranks at most $\mathbf{r}$. (c) If $W^* = W + \varepsilon E$ with W consistent and E generic, the nearby critical point W_c satisfies $\|W_H - W_c\|_F = O(\varepsilon^2)$ and $L(W_H) - L(W_c) = \frac{1}{2}\|W_H - W_c\|_F^2 \, (1 + O(\varepsilon)) = O(\varepsilon^4)$.*

For (b), $\|W^* - X\|_F \geq \|W^* - W\|_F$ for every X with a single bond rank at most r_e. For (c), both sets of frames are first-order rotations of those of W with the same driving terms, and the residual $W^* - W_c$ is normal to $\mathcal{M}_{\mathbf{r}}$ at W_c while $W_H - W_c$ is tangent up to curvature terms. Checked: criticality defects of the HSVD between 0.004 and 0.18 for random TTN teachers, independent of σ; in (c) the exponents hold on every draw and the ratio $(L(W_H) - L(W_c))/(\frac{1}{2}\|W_H - W_c\|_F^2)$ is within 3% of 1 for $\varepsilon \leq 0.2$.

The rank-one plateau. derived new By Section A.2 the origin plateau ends along a KKT direction of g. The global maximum of g has bond rank one on every bond (Proposition A.8); that local maxima do too is a conjecture, true at every rank-one plateau computed. The next plateau therefore sits at a critical point of the bond-rank-one problem, not at an HSVD truncation. If every core carries at most one site (MPS, HT), bond rank one means $W = \bar{s} \bigotimes_i w_i$ with unit vectors w_i: the plateau is a local best rank-one approximation of W^*, with $\bar{s} = \langle W^*, \bigotimes_i w_i \rangle$ and level $L_1 = \frac{1}{2}(\|W^*\|_F^2 - \bar{s}^2)$. On the binary tree with $N = 4$ it is the top singular pair of the matricization (Remark A.23), a unique maximum. In the cases computed the visited plateau lies below the HSVD truncation at the same ranks (by 6–10% of its level) empirical, but a plateau in a minor basin is a worse critical point, so this is not general.

Proposition A.31 (Two non-orthogonal components). proven *(stationarity; threshold among symmetric critical points)* derived *(fold)* new (elementary) *Let $W^* = s_1 U_1 + s_2 U_2$ with N sites, $s_1 \geq s_2$ and overlaps $\langle u_1^{(i)}, u_2^{(i)} \rangle = \cos\phi_i \in [0, 1)$. Every rank-one critical point has w_i in the plane of $u_1^{(i)}, u_2^{(i)}$; with ψ_i the angle of w_i from $u_1^{(i)}$ its value is $f(\psi) = s_1 \prod_i \cos\psi_i + s_2 \prod_i \cos(\phi_i - \psi_i)$. For equal weights and equal overlaps $\cos\phi$, the symmetric point $\psi_i = \phi/2$ has the Hessian eigenvalue*

$-2\cos^{N-2}\frac{\phi}{2}\,[\cos^2\frac{\phi}{2}-(N-1)\sin^2\frac{\phi}{2}]$ *along* $(1,\dots,1)$ *and* $-2\cos^{N-2}\frac{\phi}{2}$ *on its complement (per unit weight), so there are two local maxima, one near each component, iff*

$$\cos\phi < \frac{N-2}{N}, \tag{46}$$

and one merged maximum otherwise (asymmetric maxima were excluded numerically). For $\varrho = s_2/s_1 < 1$ *the maximum near* U_2 *disappears at a fold,* $\cos\phi_{\text{fold}} \approx \frac{N-2}{N-1}\big(\frac{\varrho}{N-1}\big)^{1/(N-2)}$ *for small* ϱ *(1–13% below the exact values). The rank-one plateau is never one of the components.*

Proposition A.32 (Rates at the rank-one saddle of an MPS)**.** derived *(rates)* proven *(bound)* new *Let the student be an MPS (*$K = N$*, core* k *carries site* k*) at the balanced, zero-padded rank-one critical point* $W_1 = \bar{s}\bigotimes_i w_i$*, a local maximum of the rank-one problem, and let* $M_e = W^* \times_{i\notin\{k,k+1\}} w_i$ *be the* $d\times d$ *matrix of the bond* e *between cores* k *and* $k+1$*. Then* (w_k, w_{k+1}) *is the top singular pair of* M_e*, with singular value* $\bar{s}$*, and the two-core route on* e *contributes the Hessian eigenvalues* $-\bar{s}^{(N-2)/N}\sigma_m(M_e)$*,* $m = 2,\dots,d$*, each with multiplicity* $\chi_e - 1$*. The escape rate of bond* e *is*

$$\gamma_e = \bar{s}^{(N-2)/N}\,\sigma_2(M_e), \qquad \text{with} \quad \sigma_m(M_e) \le \sigma_m\big((W^*)^{\langle e\rangle}\big) \tag{47}$$

by interlacing, since M_e *is a compression of* $(W^*)^{\langle e\rangle}$ *by isometries; equality holds when the teacher's Schmidt vectors are products with* w *on the sites away from the bond.*

The route block of e is $\bar{s}^{(N-2)/N}(M_e - \bar{s}\,w_k w_{k+1}^\top)$, whose singular values are those of M_e without the top one. For two components with $d = 2$, $\det M_e = s_1 s_2 a_e b_e \sin\phi_k \sin\phi_{k+1}$ with $a_e = \prod_{i\neq k,k+1}\cos\psi_i$ and $b_e = \prod_{i\neq k,k+1}\cos(\phi_i - \psi_i)$, so $\sigma_2(M_e) = \det M_e/\bar{s}$: *the bond that grows first is the one whose two sites differ most while the other sites agree most*, not the one with the largest teacher Schmidt value. Checked: Hessian eigenvalues at the exact saddle to four digits; at gradient-flow plateaus $\lambda_{\min} = -0.1474$ against $-\gamma_e = -0.1474$.

Corollary A.33 (Second escape)**.** derived new (Proposition A.25 with $s_j \to \bar{s}$ and $s_r \to \sigma_2(M_e)$) *Let* $\kappa_e = \sigma_2(M_e)/\bar{s}$*. During the first escape the route of bond* e *grows by the factor* $\sigma^{-\kappa_e}$*, so the first bond to gain rank is* $e^* = \arg\min_e h_e$ *and*

$$t_{e^*} - t_1 = h_{e^*}\ln(1/\sigma) + O(1), \qquad h_e = \frac{1-\kappa_e}{\gamma_e}. \tag{48}$$

On an MPS $\arg\min_e h_e = \arg\max_e \sigma_2(M_e)$*.*

Here t_e is the first time the second singular value across e exceeds a fraction $f \in [0.03, 0.1]$ of the teacher's; the 50%-of-final rule of Section 2.7 can fire only after later escapes, because the intermediate saddles have second singular values well below the final ones. Checked: slopes 5.853 and 17.255 (two components, equal overlaps 0.5 and 0.3) measured within 1%; for overlaps $(0.2, 0.5, 0.7, 0.3)$ the teacher's second Schmidt values are nearly tied $(0.567, 0.566, 0.548)$ while $\sigma_2(M_e) = (0.127, 0.036, 0.059)$, and bond 0 grows first in every run.

Proposition A.34 (Later escapes on an MPS: an accumulated-utility race)**.** derived *(as* $\sigma\to 0$*, under (A-carry))* known (the rule: alternating gradient flows, Kunin et al., 2025; new: the TTN routes and their rates) *Give every bond* e *that has not yet grown the log-amplitude* $\ell_e = \ln x_e/\ln(1/\sigma) \in [-1, 0]$ *of its route and use the clock* $\vartheta = (t - t_1)/\ln(1/\sigma)$*. Then* $\ell_e(0) = -1 + \kappa_e$*; while the flow sits at the saddle* S_n*,* $\mathrm{d}\ell_e/\mathrm{d}\vartheta = \gamma_e(S_n)$*, the rate of the route at the exact critical point* S_n*; bond* e *escapes when* $\ell_e = 0$*, after which the flow moves in time* $O(1)$ *to* S_{n+1}*, the critical point with the rank of* e *raised by one. (A-carry): at each transition a pending route's amplitude projects onto its new unstable direction with a* σ*-independent factor* $p \neq 0$*.*

Event times are $t_e - t_1 = \vartheta_e \ln(1/\sigma) + C_e + o(1)$ with offsets C_e that depend on the initialization (measured up to about 50 time units), so near-tied later escapes come in a seed-dependent order at accessible σ (conjecture, heuristic, at finite σ). Unlike greedy low-rank learning (Li et al., 2021), the race keeps memory across saddles. Checked (two components, overlaps $(0.2, 0.5, 0.7, 0.3)$, 8 seeds): slopes 6.823, 12.178, 14.467 for the three bonds, confirmed to 0.5% at $\sigma \leq 10^{-6}$ in 7 of 8 seeds; (A-carry) holds with $p = 0.775$ at one transition.

Generic teachers: dependence on the tree. derived new At any critical point, the local residual of an adjacent pair $\{u, v\}$ through the bond e annihilates the learned subspaces on both sides, so it lives on complements of dimensions $\mathsf{d}_u - r_e$ and $\mathsf{d}_v - r_e$, where d_u is the dimension of the legs of u other than e (physical legs and learned outer bonds). For a generic teacher it is nonzero there: *a route with $k^* = 2$ (strict saddle, logarithmic escape) exists whenever some adjacent pair has $\mathsf{d}_u > r_e$ and $\mathsf{d}_v > r_e$.* Consequences: (i) on an MPS every core has a physical leg, so a generic teacher gives one polynomial plateau (the origin plateau) and then only logarithmic ones, with ranks growing bond by bond (consistent with the plateaus and abrupt rank growth seen with gradient descent on tensor-train factorizations, Milanesi et al., 2021); (ii) on binary and HT trees at the rank-one saddle the internal cores carry no site and have rank-one outer legs, so the cheapest routes join two sibling bottom cores through their parent ($k = 3$): later escapes are polynomial even for generic teachers, and the root bipartition grows only after both halves have (bottom-up order); (iii) odeco and pattern teachers suppress the two-core routes by orthogonality, so the power-law plateaus of Proposition A.25 are a property of teachers whose components are orthogonal on the relevant site blocks, not of TTNs in general. Checked on one random CP teacher: on the MPS with $N = 4$, $\lambda_{\min} = -0.129, -0.132, -0.136$ at $\sigma = 0.05, 0.025, 0.0125$ (strict) with logarithmic gaps; on the binary tree $\lambda_{\min} \propto \sigma$ and gaps $\propto \sigma^{-1}$.

Teacher on another tree. proven known (standard; cf. Hackbusch and Kühn, 2009) A student bond e must reach $\operatorname{rank}(W^*)^{\langle e \rangle}$, which for a TTN teacher is at most the minimum, over cuts of the teacher's tree that separate the two sides of e, of the product of the cut dimensions (bond dimensions and physical dimensions). For example, a binary student with $N = 8$ of an MPS teacher with bond dimension r needs rank $\min(d^2, r^2)$ on the bonds of sites $\{2, 3\}$ and $\{4, 5\}$, i.e. about r^2 rank increments, each a separate saddle. A binary student with $N = 4$ is the deep linear network of Remark A.23 for any teacher: its plateaus sit at $\frac{1}{2}\sum_{i>m} \sigma_i((W^*)^{\langle (01\vert 23) \rangle})^2$, every escape has degree three ($\propto \sigma^{-1}$), and the number of plateaus is the rank of the matricization. The same teacher thus has different plateau levels and exponents on an MPS and on a binary tree. These predictions are tested in Sections 6.4 to 6.6.

A.5 Input covariance

Let $\Sigma = \Sigma_1 \otimes \cdots \otimes \Sigma_N$ (independent sites; cross-site correlations are treated at the end). Let Ξ act on every core by multiplying the physical leg of each site $i \in S_v$ by $\Sigma_i^{1/2}$, leaving bond legs and cores without sites unchanged. Ξ is linear, symmetric and invertible, and $W(\Xi\theta) = \Sigma^{1/2} W(\theta)$ because a leg-wise linear map commutes with the contraction.

Proposition A.35 (A product covariance is a preconditioned whitened problem)**.** proven new (for TTNs) *(a) $L(\theta) = \tilde{L}(\Xi\theta)$ with $\tilde{L}(\tilde{\theta}) = \frac{1}{2}\|W(\tilde{\theta}) - \Sigma^{1/2} W^*\|_F^2$. (b) In the variables $\tilde{\theta} = \Xi\theta$ gradient flow reads $\dot{\tilde{\theta}} = -\Xi^2 \nabla \tilde{L}(\tilde{\theta})$ from the anisotropic initialization $\Xi\theta_0$: a whitened problem with teacher $\Sigma^{1/2} W^*$ and the preconditioner Σ_i on every physical leg. (c) $\nabla L(\theta) = \Xi \nabla \tilde{L}(\Xi\theta)$ and $\nabla^2 L(\theta) = \Xi\, \nabla^2 \tilde{L}(\Xi\theta)\, \Xi$ at every θ. Critical points, critical values, Morse indices and saddle types are therefore those of the*

whitened problem with teacher $\Sigma^{1/2}W^*$ (Sylvester's law of inertia), but escape rates are eigenvalues of $\Xi^2\nabla^2\tilde{L}$, not of $\nabla^2\tilde{L}$. (d) The gauge group acts on bond legs only, so Proposition 2.5 holds unchanged in θ: small initialization is balanced in θ, not in $\tilde{\theta}$. (e) Plain whitened gradient flow is recovered iff $\Xi^2 \propto I$, i.e. iff every $\Sigma_i = \kappa_i I$ and $\prod_{i\in S_v}\kappa_i$ is the same for every core; on binary and HT trees, whose internal cores have factor 1, this forces that product to equal 1. For $\Sigma_i = \kappa_i I$, $L = (\prod_i \kappa_i)L_{\Sigma=I}$, so time is rescaled by $\prod_i \kappa_i$.

Proof. (a) follows from $W(\Xi\theta) = \Sigma^{1/2}W(\theta)$; (b) and (c) are the chain rule through the symmetric linear map Ξ. □

Checked: the preconditioned whitened flow reproduces $\Sigma^{1/2}W(\theta(t))$ to 10^{-8} on binary, MPS and HT trees with $N = 4$, while the naive whitened flow (same teacher and initialization, no preconditioner) deviates by 4–100%.

Proposition A.36 (Aligned covariance: exact clocks)**.** proven known (Saxe-type modes: Saxe et al., 2014; depth two: Gidel et al., 2019; new: every tree) *Let the teacher be odeco with factors that are eigenvectors of the site covariances, $\Sigma_i u_r^{(i)} = \lambda_{r,i} u_r^{(i)}$, and let $\Lambda_r = \prod_i \lambda_{r,i}$. On the aligned family of Proposition 2.4: (a) $L = \frac{1}{2}\sum_r \Lambda_r(\Pi_r - s_r)^2$; (b) the family is invariant and $\dot{q}_{v,r} = \Lambda_r(s_r - \Pi_r)\prod_{u\neq v} q_{u,r}$, so each component evolves as in the whitened problem with its own clock, $\Pi_r(t;\Sigma) = \Pi_r(\Lambda_r t; I)$, with the same factor Λ_r for every core, with or without sites; (c) from a balanced start q_0, $t_{1/2,r} = q_0^{-(K-2)}/((K-2)\Lambda_r s_r) + \delta_K/(\Lambda_r s_r^{2-2/K}) + O(q_0^2)$, with δ_K of Proposition A.6; (d) with the set S learned the loss is $\frac{1}{2}\sum_{r\notin S}\Lambda_r s_r^2$, and the events of j and then r are separated by $(1 - \frac{\Lambda_r s_r}{\Lambda_j s_j})/((K-2)\Lambda_r s_r q_0^{K-2})$.*

Proof. $\langle U_r, \Sigma U_{r'}\rangle = \delta_{rr'}\Lambda_r$ and $\Sigma(W - W^*) = \sum_r \Lambda_r(\Pi_r - s_r)U_r$; the rest is the proof of Proposition 2.4 with this residual. □

Three orderings therefore coexist derived new: by escape speed $\Lambda_r s_r$ (which component is learned first), by loss drop $\frac{1}{2}\Lambda_r s_r^2$, and the whitened order by s_r. With $s_1 > s_2$, component 2 escapes first iff $\Lambda_2/\Lambda_1 > s_1/s_2$, and in the *mismatch window* $s_1/s_2 < \Lambda_2/\Lambda_1 < (s_1/s_2)^2$ the faster component is the smaller loss drop, so the first plateau $(\frac{1}{2}\Lambda_1 s_1^2)$ lies above the reduced-rank-regression optimum $(\frac{1}{2}\Lambda_2 s_2^2)$. The visited saddles are then not the reduced-rank solutions of Gidel et al. (2019), whose Corollary 2 states no ordering condition. With power-law spectra $\lambda_k \propto k^{-\beta}$ and component r on eigenvector k_r at every site, $\Lambda_2/\Lambda_1 = (k_1/k_2)^{\beta N}$: the natural pairing $k_r = r$ reinforces the whitened order.

Proposition A.37 (Random initialization: what replaces the teacher)**.** derived new (TTN form of the covariance-weighted first escape of Jacot et al., 2022; Atanasov et al., 2022; Rawal and DeWeese, 2026) *Near the origin $L = L_0 - \langle W, \Sigma W^*\rangle + \frac{1}{2}\|W\|_\Sigma^2$, and the fan-in initialization law of θ does not depend on Σ. Hence: (a) the escape from the origin (direction, time and learning order of Section A.2) is that of the whitened problem with the* escape tensor *$\Sigma W^* = \mathbb{E}[f^*(x)\Phi(x)]$ in place of W^*; for aligned covariance $s_r \to \Lambda_r s_r$, and $\mathrm{d}\ln t_1/\mathrm{d}\ln\Lambda = -1$ exactly, for every K, tree and split of Λ over the sites; (b) the landscape (critical points, plateau levels, saddle types) is that of $\Sigma^{1/2}W^*$ (Proposition A.35); for aligned covariance $s_r \to \Lambda_r^{1/2} s_r$; (c) the problems (Σ, W^*) and $(I, \Sigma W^*)$, which have the same escape tensor, have the same escape at LO and differ only by a saturation offset of order one: for an aligned component the half-times, each relative to its own target, differ by $-\delta_K s_r^{-(2-2/K)}(\Lambda_r^{-(2-2/K)} - \Lambda_r^{-1})$.*

Plain gradient flow on the whitened teacher without the preconditioner predicts the exponent $-1/2$ instead of -1. Checked: $\mathrm{d}\ln t_1/\mathrm{d}\ln\Lambda = -1.0000$ in 18 fits (binary and MPS trees, three splits,

three seeds) and -1.0005 to -1.0011 on the HT tree; same-escape-tensor offset -0.2218 against -0.2219. Random initialization keeps the random learning order of Proposition A.14, at the rate ratio $\Lambda_2 s_2/(\Lambda_1 s_1)$.

Proposition A.38 (Piggyback block under aligned covariance). proven *(at the saddle)* new *Let j be learned (balanced, $q_j = s_j^{1/K}$) and let r differ from j on $D = D_{jr}$, with two terminal cores c_1, c_2. With $\mu_x = \prod_{i\in D\cap S_{c_1}} \lambda_{x,i}$, $\nu_x = \prod_{i\in D\cap S_{c_2}} \lambda_{x,i}$ for $x \in \{j, r\}$ and $\lambda_{\mathrm{sh}} = \prod_{i\notin D} \lambda_{j,i}$, the Hessian on the two rotations (per unit parameter displacement) is*

$$H_{\mathrm{pb}} = \frac{\lambda_{\mathrm{sh}}}{s_j^{2/K}} \begin{pmatrix} s_j^2 \mu_r \nu_j & -s_r s_j \mu_r \nu_r \\ -s_r s_j \mu_r \nu_r & s_j^2 \mu_j \nu_r \end{pmatrix}. \tag{49}$$

It has a negative eigenvalue iff $\Lambda_r s_r^2 > \Lambda_j s_j^2$, whatever the split of the eigenvalues over the cores. If c_1, c_2 are adjacent ($k = 2$), its rate is below the add-route rate $b = \Lambda_r s_r s_j^{(K-2)/K}$, so r is added directly: a rotation needs $m = 2$ and $k \geq 3$.

Proof. At the learned amplitude, and up to constants, $L = \frac{1}{2} s_j^2 \lambda_{\mathrm{sh}} (\mu_j \cos^2\varphi + \mu_r \sin^2\varphi)(\nu_j \cos^2\psi + \nu_r \sin^2\psi) - s_j(\Lambda_j s_j \cos\varphi\cos\psi + \Lambda_r s_r \sin\varphi \sin\psi)$, where φ, ψ rotate the two terminal cores and decouple from the amplitude at second order; $\det H_{\mathrm{pb}} \propto \Lambda_j s_j^2 - \Lambda_r s_r^2$. For adjacent terminals $\lambda_{\mathrm{sh}} \mu_r \nu_r = \Lambda_r$, so the off-diagonal entry is $-b$; a symmetric 2×2 matrix with positive diagonal h_1, h_2 and off-diagonal $-b$ has $\lambda_{\min} > -b$, because $(\frac{h_1+h_2}{2} + b)^2 - (\frac{h_1-h_2}{2})^2 - b^2 = h_1 h_2 + b(h_1+h_2) > 0$. □

In the mismatch window, conditionally on the faster component winning the first escape derived: on the binary tree with $N = 4$ ($m = 2$, $k = 3$) the learned term rotates from the faster to the other component at bond rank one, the loss plateaus at $\frac{1}{2}\Lambda_2 s_2^2$, and the faster component is relearned later; on the MPS with $N = 4$ and on every tree with $m \geq 3$ there is no rotation, and the loss plateaus at the higher level $\frac{1}{2}\Lambda_1 s_1^2$. Checked: rotation rates 0.1263, 0.1130, 0.0548 for three splits of the same Λ, as predicted.

Misaligned covariance. proven (identities and inequality) derived (order, to leading order in the cross overlaps) new For every CP teacher, exactly, $\Sigma W^* = \sum_r s_r \Lambda_r^{\mathrm{esc}} \widetilde{U}_r$ with $\widetilde{U}_r = \bigotimes_i \Sigma_i u_r^{(i)} / \|\Sigma_i u_r^{(i)}\|$ and $\Lambda_r^{\mathrm{esc}} = \prod_i \|\Sigma_i u_r^{(i)}\|$, while $\Sigma^{1/2} W^* = \sum_r s_r (\Lambda_r^{\mathrm{loss}})^{1/2} \widehat{U}_r$ with $\Lambda_r^{\mathrm{loss}} = \prod_i \langle u_r^{(i)}, \Sigma_i u_r^{(i)}\rangle$. By Cauchy–Schwarz $\Lambda_r^{\mathrm{esc}} \geq \Lambda_r^{\mathrm{loss}}$, with equality iff every $u_r^{(i)}$ is an eigenvector of Σ_i. Components escape in the order of $s_r \Lambda_r^{\mathrm{esc}}$ along $\widetilde{U}_r$ and then realign to U_r without an extra plateau (the TTN form of the early alignment of Atanasov et al., 2022), while the loss drops are $\frac{1}{2} s_r^2 \Lambda_r^{\mathrm{loss}}$; misalignment alone can reverse the order. Misalignment also opens routes of degree below k^*, with couplings of order $\epsilon^{n_{\mathrm{out}}}$ at per-site misalignment ϵ, n_{out} being the number of sites outside the route.

Cross-site correlations. Consider the one-factor model $x^{(i)} = \sqrt{1-c}\, z_i + \sqrt{c}\, z_0$, with independent $z_0, \ldots, z_N \sim \mathcal{N}(0, I_d)$: every marginal is standard and the covariance between two sites is cI. (i) proven new (the model is the standard one-factor Gaussian model; new: its tree contraction) Given z_0 the sites are independent with second moments $M_i(z_0) = (1-c)I + c z_0 z_0^\top$, so $L = \mathbb{E}_{z_0}[\frac{1}{2}\|W - W^*\|^2_{\otimes_i M_i(z_0)}]$, a polynomial of degree $2N$ in z_0; tensor Gauss–Hermite quadrature with $N+1$ nodes per dimension evaluates it exactly with $(N+1)^d$ product-metric tree contractions. A site-aligned component has the effective weight $\Lambda_{\mathrm{eff}}(c, N) = \sum_k \binom{N}{k} (1-c)^{N-k} c^k (2k-1)!!$. (ii) derived The first plateau survives for every $c < 1$, because the origin is a degree-K critical point for any input law, and the bond-rank-constrained critical points remain critical (Lemma A.2). (iii) proven (rates at the saddle) new On an MPS with even N and a site-aligned odeco teacher, the correlations open strict

two-core routes at the product saddle on every bond, with rate $\lambda_2 \approx s_2 c^{N-2}((N-3)!!)^2$ at small c. Whitened power-law plateaus therefore become logarithmic once c exceeds a crossover $c^* = O(\sigma)$, up to logarithms (conjecture, heuristic, for $N \geq 6$; checked for $N = 4$ and on one example with $N = 6$). (iv) derived On the MPS with $N = 4$ this escape has two stages: it passes through the first plateau point of the binary tree (a point with junk pairing components, which is a higher-order saddle on the binary tree and a strict one on the MPS), and the second plateau grows by $\ln 2\,[(1-\kappa_c)/\lambda_2 + 1/\lambda'']$ per halving of σ, where λ'' is the rate at the intermediate saddle and $\kappa_c \in (0,1)$ the head start accrued during the first escape (2.128 predicted, 2.13 measured, at $c = 0.4$). These predictions are tested in Sections 6.1 to 6.3.

A.6 Discrete steps and gradient noise

Odeco teacher with $s_1 > s_2 > \ldots$, whitened Gaussian inputs, $q_r = s_r^{1/K}$; gradient descent (GD) and online SGD as in Section 2.5, recorded at $t = \eta \times$ (number of steps).

Proposition A.39 (Hessian at aligned global minima: the pattern rule). proven known (binary $N = 4$, a deep linear network: Ghosh et al., 2025, Lemma 1; new: general trees) *At a global minimum $H = J^\top J$ with $J = \partial W/\partial\theta$. At an aligned minimum (Proposition 2.4 with $\Pi_r = s_r$), an elementary perturbation of core w changes W by a gain times a unit rank-one* pattern*: the chosen vectors at the sites of w, the teacher vector ℓ_c at every site below a child c whose bond index is ℓ_c, and the teacher vector ℓ_p at every site outside the subtree of w. The gain is $\prod_{u \neq w} q_{u,\ell(u)}$, with $\ell(u)$ the label of the branch of w that contains u (zero if a label exceeds R). Distinct patterns are orthonormal, so*

$$\operatorname{spec}(H) \setminus \{0\} = \Big\{ \lambda_P = \sum_{w \in \operatorname{prod}(P)} \operatorname{gain}_w(P)^2 \;:\; P \text{ a pattern} \Big\}, \tag{50}$$

where $\operatorname{prod}(P)$ is the set of cores that can produce P. The radial mode of component r (pattern U_r, produced by every core) has $\lambda = s_r^2 \sum_v q_{v,r}^{-2} \geq K s_r^{2-2/K}$, with equality iff the component is balanced. At the balanced minimum every pattern has $\lambda_P \leq |\operatorname{prod}(P)|\, q_1^{2(K-1)}$, hence

$$\lambda_{\max}(H) = K\, s_1^{2-2/K}, \tag{51}$$

with a nondegenerate eigenvector if $s_2 < s_1$.

Consequences derived new: GD is linearly stable at the balanced minimum iff $\eta < \eta_c := 2/(K s_1^{2-2/K})$, the stability limit tested in Section 5.5. Imbalanced aligned minima are sharper (AM–GM; proven; for deep linear networks: Mulayoff and Michaeli, 2020), and with $\chi_e = R$ so is every gauge-equivalent minimum (derived; random GL gauges were at least 1.43 times sharper), so for $\eta > \eta_c$ no global minimum is linearly stable (derived for $\chi_e = R$, conjecture for $\chi_e > R$). Because the conservation laws fix the imbalance, the minimum reached by gradient flow and its sharpness are functions of $Q_e(0)$; a strongly imbalanced weaker component can then carry the top mode. If that minimum is too sharp for the step ($2/\eta$ below its sharpness), GD oscillates, the imbalance contracts by Lemma A.3, and GD converges to a flatter, more balanced minimum with sharpness just below $2/\eta$ (empirical on binary $N = 4$; known (scalar and deep linear networks: Kreisler et al., 2023; Wang et al., 2022; Ghosh et al., 2025)). Checked: autograd Hessians against (50) to $6 \cdot 10^{-15}$ on five trees with $K = 3$–7.

Sharpness along the saddle-to-saddle path. derived new On the first plateau the Hessian of the degree-K polynomial W at $\|\theta\| \sim \sigma$ is indefinite and $O(\sigma^{K-2})$. During the event of a balanced

component of amplitude a, its radial eigenvalue $(2K-1)a^{2K-2} - (K-1)s_r a^{K-2}$ rises to $Ks_r^{2-2/K}$ without overshoot, and the $K-1$ transverse eigenvalues $(s_r - a^K)a^{K-2}$ stay below it. At a later plateau that has learned the set S, the radial mode of the largest learned component is an exact eigenvector, so the sharpness is $K \max_{r\in S} s_r^{2-2/K}$. A finite initialization adds a relative excess $O(\sigma^4)$: balance minimizes the radial eigenvalue, so the excess is quadratic in an imbalance of order σ^2. Hence, at a fixed step η, a plateau is linearly unstable iff it has learned a component with $s_r > s_c(\eta) = (2/(\eta K))^{K/(2K-2)}$: GD first becomes unstable at that component's event.

Edge of stability. derived known (the scalar map: Ghosh et al., 2025; new: the period-two amplitude) For $\eta_c < \eta < 2/\lambda_2(H)$ only the radial mode of component 1 is unstable, and GD restricted to it (exactly for aligned initialization) is the balanced scalar map $x \mapsto x + hx^{K-1}(1-x^K)$, $x = q/s_1^{1/K}$, $h = \eta s_1^{2-2/K}$, which doubles its period at $hK = 2$, i.e. at $\eta = \eta_c$. Expanding the map to third order around $x = 1$, the period-two orbit $1 \pm \delta_{\mathrm{p2}}$ has $\delta_{\mathrm{p2}}^2 \simeq 3(hK-2)/(4(K-1)(5K-4))$, and the time-averaged loss is

$$L_{\mathrm{EoS}} \simeq \frac{3K^2 s_1^2}{4(K-1)(5K-4)}\Big(\frac{\eta}{\eta_c} - 1\Big). \tag{52}$$

Later components still converge and later events still lower the loss. Checked (binary, $N = 4$, weights $(1, 0.6, 0.3)$, $\eta = 1.05\eta_c$): late loss $1.45 \cdot 10^{-2}$, against $1.452 \cdot 10^{-2}$ from the map and $1.53 \cdot 10^{-2}$ from (52); the full network diverges between $1.5\eta_c$ and $1.65\eta_c$.

Proposition A.40 (Shift of event times under gradient descent)**.** derived new (for TTNs; the modified flow is known: Barrett and Dherin, 2021) *To order η, GD follows the modified flow $\dot\theta = -(I + \frac{\eta}{2}H)\nabla L$. Its component along the gradient-flow path reparametrizes time, and since $\frac{\mathrm{d}}{\mathrm{d}t}\ln\|\nabla L\| = -\nabla L^\top H \nabla L/\|\nabla L\|^2$ along gradient flow,*

$$t_{\mathrm{GD}} - t_{\mathrm{GF}} = \frac{\eta}{2}\ln\frac{\|\nabla L(t_{\mathrm{event}})\|}{\|\nabla L(0)\|} + (\textit{transverse part}) + O(\eta^2). \tag{53}$$

For aligned initialization the path is exact, and the event of component r moves by

$$\frac{\eta}{2}\ln\frac{f(a_1)}{f(a_0)}, \qquad f(a) = a^{K-1}(s_r - a^K),$$

with a_0 and a_1 the per-core amplitudes at initialization and at the event: plateau exponents and prefactors are those of gradient flow, and no saddle is skipped or reordered while the gradient-flow gaps exceed the differences of the shifts.

For generic initialization $\|\nabla L(0)\| \propto \sigma^{K-1}$, so $\Delta t_1/\eta \approx \frac{K-1}{2}\ln(1/\sigma) + \mathrm{const}$ asymptotically (conjecture, heuristic: seeds with a long alignment phase are pre-asymptotic). At a strict saddle the growth per step is $1 + \eta|\lambda_-|$ instead of $e^{\eta|\lambda_-|}$, so $t_{\mathrm{GD}}/t_{\mathrm{GF}} = \eta|\lambda_-|/\ln(1+\eta|\lambda_-|) \in [1, 1.82]$ for $\eta|\lambda_-| < 2$. Checked: aligned shifts within 1.3%; generic slopes 0.69–0.70 per halving of σ on binary $N = 4$ (predicted $\frac{K-1}{2}\ln 2 = 0.69$).

SGD on the first plateau. derived new (framework: stochastic differential equations for SGD, Li et al., 2017, 2019, and Type-II saddles, Ziyin et al., 2024; consistent with Corlouer et al., 2026) For online SGD with batch B, the per-sample gradient is $\langle W - W^*, \Phi(x)\rangle J^\top \Phi(x)$. Near the origin $J = O(\|\theta\|^{K-1})$ and $W - W^* \approx -W^*$, so the gradient covariance is $O(s_1^2 3^N \|\theta\|^{2K-2})/B$ while the drift is $O(s_1\|\theta\|^{K-1})$: the noise vanishes at the origin at the same order as the drift, and in the clock $\mathrm{d}\tau = \|\theta\|^{K-2}\mathrm{d}t$ its amplitude tends to zero, so noise cannot drive the first escape. For one aligned balanced

component, $\mathrm{d}a = s_1 a^{K-1}\mathrm{d}t + \sqrt{\eta/B}\, s_1 \kappa_{\mathrm{n}}^{1/2} a^{K-1}\mathrm{d}\mathcal{B}_t$, with $\mathcal{B}_t$ a standard Brownian motion and $\kappa_{\mathrm{n}} = 3^N - 1 + \sum_{r\geq 2} s_r^2/s_1^2$; its hitting time is inverse Gaussian to leading order:

$$\mathbb{E}t_1^{\mathrm{SGD}} - t_1^{\mathrm{GF}} \simeq \frac{\eta}{2}(K-1)\Big(1+\frac{\kappa_{\mathrm{n}}}{B}\Big)\ln\frac{a_{1/2}}{a_0} > 0, \qquad \mathrm{sd}(t_1^{\mathrm{SGD}}) \simeq \Big(\kappa_{\mathrm{n}}\frac{\eta}{B}t_1^{\mathrm{GF}}\Big)^{1/2}, \tag{54}$$

with $a_{1/2} = (s_1/2)^{1/K}$. Both corrections vanish relative to $t_1^{\mathrm{GF}} \propto a_0^{-(K-2)}$, so the plateau exponent $K-2$ is unchanged. Checked with 160–2000 seeds: within 10–15% for aligned initialization; for generic initialization the spread is about 1.2 times the formula and the delay smaller.

SGD at the minimum. derived (exact mean-square threshold) proven (the inequality) new (the reduction; general criteria are known: Ziyin et al., 2024; exact mean square, Mulayoff and Michaeli, 2024; one-dimensional, Wu et al., 2018; top Lyapunov exponent, Chemnitz and Engel, 2025) In the eigenbasis of (50), linearized SGD at the minimum acts on one coordinate per pattern, with the products $Z_P(x) = \prod_i (t_{P,i}\cdot x^{(i)})$ of the pattern's site vectors. Its second-moment operator maps diagonal matrices to diagonal ones (since $\mathbb{E}[Z_R^2 Z_P Z_Q] = 0$ for $P \neq Q$) and preserves the positive semidefinite cone, which gives the exact mean-square threshold

$$\eta_{\mathrm{ms}}^*(B) = \frac{2}{\lambda_{\max}(G_B)}, \qquad G_B = \Big(1-\frac{1}{B}\Big)\Lambda_H + \frac{1}{B}\Lambda_H^{1/2}\mathcal{K}\Lambda_H^{1/2}, \qquad \mathcal{K}_{PQ} = \prod_i m(t_{P,i}, t_{Q,i}), \tag{55}$$

with $\Lambda_H = \mathrm{diag}(\lambda_P)$ and $m = 3$ if the two site vectors coincide, 1 if they are orthogonal (Gaussian sites). Jensen's inequality gives $\eta_{\mathrm{as}} \geq \eta_{\mathrm{ms}}^*$ for the almost-sure divergence threshold; computed ratios are about 2 for $N = 4$ and 8–21 for the binary tree with $N = 8$ empirical. Full nonlinear SGD started near the minimum diverges at η_{as}, not at η_{ms}^*.

Proposition A.41 (SGD lengthens later plateaus)**.** conjecture *(heuristic law)* known (mechanism: stochastic collapse, Chen et al., 2023) *At a later saddle with a route of degree $k \geq 3$ the population Hessian vanishes on the route and the per-sample gradients vanish at zero route amplitude. Zero-mean multiplicative noise coupling the route entries to mixed learned entries gives the log-amplitude of the coherent route a negative drift $-\gamma_c$, $\gamma_c = c_\gamma \eta/B$. Adding it to the gradient-flow route equation, $\dot{b} = s_r b^{k-1} - \gamma_c b$; for $k = 3$, with $T_{\mathrm{GF}} = 1/(s_r b_0)$,*

$$\frac{T_{\mathrm{SGD}}}{T_{\mathrm{GF}}} = \frac{-\ln(1-x)}{x}, \qquad x = \gamma_c T_{\mathrm{GF}}, \tag{56}$$

and the route is trapped (it decays) for $x \geq 1$; for general k escape needs roughly $(k-2)\gamma_c T_{\mathrm{GF}} < 1$. Since b_0 scales with the initialization, at fixed η/B SGD stalls on a later degenerate plateau once σ is small enough. Strict saddles escape if $|\lambda_-| > \gamma_c$.

Solving $\dot{w} = -s_r + \gamma_c w$ for $w = 1/b$ gives (56). Checked on binary $N = 4$ only (four noise levels, 21 runs): delays 1.02–2 times T_{GF}, trapping for $x \geq 1$, no escape faster than gradient flow, $c_\gamma \approx 1.35$ (range 1.0–2.5) empirical, against 1.08 from a commuting-pair estimate. Near the stability limit SGD breaks the saddle sequence instead of compressing it: no learned component was erased for $\eta \leq 0.25\eta_{\mathrm{ms}}^*$, while at 0.7–1.2 η_{ms}^* components were repeatedly erased and relearned, or the run diverged empirical. Tested in Section 5.5.

A.7 Regime boundaries

Here $d_i = d$, and Θ denotes the tensor-space tangent kernel $\Theta = \sum_v \eta_v J_v J_v^\top$ on $\mathbb{R}^{dN}$, $J_v = \partial W/\partial G_v$, which drives the function-space dynamics of (15); for whitened inputs the probe-set kernel of Table 2

is $\Phi(x_p)^\top \Theta \Phi(x_{p'})$. Recall $W^{\langle e_v \rangle} = T_v C_v^\top$ (Section 2.1); the initialization is (11) with scales σ_v and $\alpha = \prod_v \sigma_v$. A bond e is *full* if $\chi_e = \min(d^{n_e}, d^{N-n_e})$, its cut dimension, with n_e the number of sites below e.

Proposition A.42 (Tangent kernel at initialization). proven new (exact at every bond dimension and on every tree; Gaussian-process and NTK limits of MPS and TT models: Guo and Draper, 2021; Wesel and Batselier, 2025) *(a)* $J_v J_v^\top = \bigotimes_{c \in \mathrm{ch}(v)} T_c T_c^\top \otimes I_{d_v} \otimes C_v C_v^\top$ *(legs in site order). (b) For all bond dimensions,* $\mathbb{E}\,\Theta = \frac{\alpha^2}{d^N} \sum_v \frac{\eta_v F_v}{\sigma_v^2} I$ *and* $\mathbb{E}[W_0 \otimes W_0] = \frac{\alpha^2}{d^N} I$. *With* $\sigma_v = \sigma$ *and* $\eta_v = 1$ *the kernel scale is* $\bar{\lambda} := \mathbb{E}\,\mathrm{tr}\,\Theta / d^N = \sigma^{2(K-1)} \sum_v F_v / d^N$, *and core* v *carries the share* $\eta_v F_v \sigma_v^{-2} / \sum_u \eta_u F_u \sigma_u^{-2}$ *of* $\mathrm{tr}\,\Theta$. *(c) If the bond dimensions are consistent (each* χ_e *at most the product of the other leg dimensions of both of its cores) and the initialization is generic,*

$$\mathrm{rank}\,\Theta = \sum_v F_v \chi_v - \sum_e \chi_e^2 \quad (\textit{capped at } d^N), \tag{57}$$

the dimension of the manifold $\mathcal{M}_\chi$ *of tensors with bond ranks at most* χ_e *(for inconsistent bond dimensions, first reduce each* χ_e *to its effective value).* $\mathrm{rank}\,\Theta = d^N$ *iff every bond is full.*

Proof sketch. (a) W is linear in G_v. (b) The cores are independent, T_c and C_v involve disjoint cores, and the law of the initialization is invariant under orthogonal maps of each physical leg, which act irreducibly; the traces follow by induction from the leaves as in Proposition 2.2. (c) $\mathrm{rank}\,\Theta$ is the dimension of the tangent space of $\mathcal{M}_\chi$ at W_0: parameters minus the dimension of the gauge group, as for hierarchical Tucker tensors (Uschmajew and Vandereycken, 2013). □

Consequences derived: with a bond that is not full, the linearized (lazy) dynamics fits only the projection of the target on the range of Θ and stops at $\frac{1}{2}\|P_\perp W^*\|_F^2$, with $\mathbb{E}\|P_\perp W^*\|_F^2 / \|W^*\|_F^2 = (1 - \chi/d^2)^2$ for the binary tree with $N = 4$ (compare Chizat et al., 2019, Thm 2.5, and Ghorbani et al., 2019). $\Theta \approx \bar{\lambda} I$ needs every $T_c T_c^\top$ and $C_v C_v^\top$ to be nearly isotropic, i.e. every χ_e far above its cut dimension; at capped full bonds $\lambda_{\min}(\Theta)/\bar{\lambda} \sim 10^{-4}$ empirical, so lazy runs need uncapped $\chi \geq 2d^2$ (Section 5.1). At practical χ the lazy regime therefore learns along the eigenvectors of a random, tree-dependent kernel.

Proposition A.43 (Lazy boundary of the centred model, $N = 4$). derived new (the TTN constants; the criterion is that of Chizat et al., 2019; a coherent second-order kernel change is known in wide networks: Bordelon and Pehlevan, 2022) *Binary tree with* $N = 4$, *standard parameterization, centred model; let* $\xi = \|W^*\|_F / \alpha$, *assume* $\chi \gg d^2$ *and keep the leading order in* d^2/χ. *Then*

$$\delta_\Theta^2 \approx (c_1 \xi)^2 + 2\zeta c_1 c_2 \xi^3 + (c_2 \xi^2)^2, \qquad c_1 = \frac{4d^3\sqrt{d^2+1}}{\chi^3}, \quad c_2 = \frac{2d^4\sqrt{2d^2\nu + 2}}{\chi^3}, \tag{58}$$

with $\nu = \sum_r s_r^4 / (\sum_r s_r^2)^2$ *and a seed-dependent correlation* $\zeta \in [-1, 1]$. *The first-order term is linear in* W_0 *and has zero mean; the second-order term is coherent, proportional to* $(W^*)^{\langle e \rangle} \big((W^*)^{\langle e \rangle}\big)^\top$, *because the bottom cores drift toward the singular directions of the target. The boundary* $\delta_\Theta = \delta_*$ *lies in* $[\xi_c^+, \xi_c^-]$ *with* $\xi_c^\pm = (\mp c_1 + \sqrt{c_1^2 + 4c_2\delta_*})/(2c_2)$, *and for large* χ *(*$c_1^2 \ll c_2\delta_*$*)*

$$\sigma_{\mathrm{lazy}} = \Big(\frac{\|W^*\|_F^2 c_2}{\delta_*}\Big)^{1/6} \propto \chi^{-1/2} \tag{59}$$

($6.50\,\chi^{-1/2}$ *for* $d = 3$, *weights* $(3, 2, 1)$, $\delta_* = 0.1$*), not* χ^{-1} *as the first-order term alone would give. In the two-copy model the first-order term cancels,* $\delta_\Theta \approx \frac{1}{4} c_2 \xi^2$, *and* $\sigma_{\mathrm{lazy}}^{(2)} = 2^{-1/3} \sigma_{\mathrm{lazy}}$.

Checked empirical: at $\chi = 32$ the predicted $\xi_c \in [2.17, 2.81]$ against 2.25 and 2.82 in gradient flow; at $\chi = 64$, $[6.67, 7.31]$ against 6.73 and 7.53; two-copy model at $\chi = 32$, 4.94 against 4.96 and 5.15. The boundary moves with the threshold as $\sigma_{\text{lazy}} \propto \delta_*^{-1/(Ks)}$, where $s = \mathrm{d}\ln\delta_\Theta/\mathrm{d}\ln\xi \in [1, 2]$ is the local slope (measured 1.5–1.8). Under the standard parameterization the root can move by more than 100% while $\delta_\Theta < 0.1$ (it acts as a trained readout of random features), so per-core changes should be reported next to δ_Θ.

Deficient bonds. derived new If some bond is not full, $W^* + W_0$ generically has more bond rank than the student can carry, so the centred model is unrealizable at every σ. At small σ the best reachable point is $W^* + P_T W_0 + O(\|W_0\|_F^2)$, with P_T and $P_\perp$ the tangent and normal projectors of $\mathcal{M}_\chi$ at W^*, so the loss floors at $\frac{1}{2}\|P_\perp W_0\|_F^2$, with mean, by Proposition A.42(b),

$$\frac{\mathbb{E}L_{\text{end}}}{L(0)} \approx \Big(1 - \frac{\dim \mathcal{M}_\chi}{d^N}\Big)\frac{\sigma^{2K}}{\|W^*\|_F^2}. \tag{60}$$

A typical run meets the convergence rule $L < 10^{-3}L(0)$ only below $\sigma_{\text{floor}} = [10^{-3}\|W^*\|_F^2/(1 - \dim \mathcal{M}_\chi/d^N)]^{1/(2K)}$: 0.56, 0.60, 0.71, 1.02 for the binary tree with $N = 4$, $d = 3$, weights $(3, 2, 1)$ and $\chi = 3, 4, 6, 8$, and 0.74 for $N = 8$, $d = 3$, $\chi = 3$. At large σ the floor tends to $\|P_\perp W^*\|_F^2/\|W^*\|_F^2$, with $P_\perp$ now the normal projector at W_0. Runs above σ_{floor} should be labelled stalled (unrealizable). In the two-copy model a deficient bond gives a lazy plateau at that level and then an escape by rotation of the bond subspaces: adiabatic elimination of the fast antisymmetric part gives an escape time $t_{\text{esc}} = 2\vartheta\sigma^2/\|W^*\|_F^2$, with a seed-dependent $\vartheta \approx 0.5$, i.e. a kernel time $\propto \sigma^{2K}$, and no saddle: the loss first falls linearly in t (compare the mixed regimes of Tu et al., 2024 and Lyu et al., 2024).

Proposition A.44 (Saddle-to-saddle boundary for aligned initialization)**.** proven *(given the exact aligned reduction)* known (mode equation: Saxe et al., 2014; threshold exponent $K/(K-2)$: Gissin et al., 2020, Thm 2; new: the constants for the criterion of Section 2.7 and the exact $K \to \infty$ limit) *Let every component of an odeco teacher start at product amplitude $u_0 = q_0^K$, let $I_K(v_0, v_1) = \frac{1}{K}\int_{v_0}^{v_1} v^{-2+2/K}(1-v)^{-1}\mathrm{d}v$ and $\omega_K = I_K(0.1, 0.9)$. Then $t_r = s_r^{-(2-2/K)} I_K(u_0/s_r, \frac{1}{2})$ and $w_r = s_r^{-(2-2/K)}\omega_K$, so, with $\varrho_i = s_{i+1}/s_i$,*

$$\rho_i = \frac{I_K(u_0/s_{i+1}, \frac{1}{2}) - \varrho_i^{2-2/K} I_K(u_0/s_i, \frac{1}{2})}{\omega_K} \simeq \Big(\frac{u_0}{s_{i+1}}\Big)^{-(1-2/K)} \frac{1-\varrho_i}{(K-2)\,\omega_K} \quad (u_0 \to 0), \tag{61}$$

and the boundary $\rho_i = 3$ lies at $u_{0c} \simeq s_{i+1}[(1-\varrho_i)/(3(K-2)\omega_K)]^{K/(K-2)}$. As $K \to \infty$, $KI_K(v_0, v_1) \to \frac{1}{v_0} - \frac{1}{v_1} + \ln\frac{v_1}{v_0} + \ln\frac{1-v_0}{1-v_1}$ and $K\omega_K \to \frac{80}{9} + 2\ln 9 = 13.283$: the function-space boundary u_{0c} stays finite ($1.306 \cdot 10^{-2}$ for weights $(3, 2, 1)$), while the per-core boundary $a_{0c} = u_{0c}^{1/K} = 1 + \ln u_{0c}/K + O(K^{-2})$ tends to 1.

Values: $\omega_3 = 2.009$, $\omega_5 = 1.612$, $\omega_7 = 1.318$; for weights $(3, 2, 1)$, $a_{0c} = 0.066$, 0.324, 0.480 at $K = 3, 5, 7$. Aligned gradient flow reproduces (61) to four decimals at $K = 3$ and 7.

Random initialization. derived (t_1) conjecture (heuristic: later gaps) empirical (boundary) By Corollary A.5 and Proposition A.6, $t_1 = \sigma^{-(K-2)}\hat{t}_1(Z)/s_1 + O(1)$. Later gaps scale the same way if the dormant amplitudes entering each plateau stay proportional to σ (exact for aligned initialization), so $\rho_i \approx R_i\sigma^{-(K-2)}$ with seed-dependent R_i, and the boundary is a distribution over seeds: median per-seed $\sigma_c = 0.27$–0.31 (quartiles 0.20–0.39) for the binary tree with $N = 4$, $d = \chi = 3$, weights $(3, 2, 1)$, and 0.71 (quartiles 0.62–0.74) for $N = 8$, $d = 2$, $\chi = 16$, weights $(3, 2)$. Writing random

initialization as aligned initialization with per-core amplitude $\kappa_K\sigma$ gives $\kappa_3 \approx 0.22$ and $\kappa_7 \approx 0.69$; conjecture: $\kappa_K \to 1$, so that $\sigma_c(K) \to a_{0c}(K) \to 1$. For matrix-like pairs the transient overlap of a component not yet learned peaks at a value $\propto \sigma^{2(1-\varrho)}$ (conjecture, heuristic), which is why events are timed by their final rise (Section 2.7). With χ equal to the teacher rank and $K \geq 5$, the last component needs a route through all K cores, and with fan-in initialization the child-heavy root bonds (Section A.1) can let that route die before it escapes, as in Proposition A.28 (empirical: 2 of 4 runs at $K = 5$ and 2 of 2 at $K = 7$, $d = 3$; gauge-balanced initialization removed the dead routes at $K = 5$ but not the long plateau at $K = 7$; spare channels, $\chi \geq 2R$, avoided stalls in all runs).

Large bond dimension. derived new (the spectral condition is Yang et al., 2024; factor-wise rules for structured layers: Qiu et al., 2024; Potapczynski et al., 2024) The input of a core is always a product of vectors, so the norm that controls its forward and backward signals is its injective norm, not the spectral norm of its matricization. Imposing the spectral condition with the injective norm keeps the fan-in initialization of every non-root core, multiplies the root by $(\sum_\ell m_\ell)^{-1/2}$, where m_ℓ are the dimensions of its child legs (physical and bond), and sets the learning-rate multipliers $\eta_v \propto \chi_v/F_v$ ("μP-TTN"). At fixed ξ ($N = 4$) this gives $\delta_\Theta \propto \chi^0$, against χ^{-3} for the standard and χ^{-1} for the NTK parameterization (compare Guo and Draper, 2021 for the MPS NTK); the factor-wise rule applied to TTN cores makes the initial kernel vanish like χ^{-2} and slows the initial fit about linearly in χ empirical.

Imbalance. derived (aligned mode) new (compare two-layer linear networks: Kunin et al., 2024) With the root at scale $\lambda_{\text{root}}\sigma$ and the other cores at σ (fixed α), the root bonds have $\mathbb{E}Q_e = \sigma^2(1 - \lambda_{\text{root}}^2/\chi_e)I$, balanced at $\lambda_{\text{root}} = \sqrt{\chi_e}$ (default fan-in initialization is root-light). A small root ($\lambda_{\text{root}} \ll 1$) makes training lazy, with $\delta_\Theta \propto \lambda_{\text{root}}^2$, but only with full bonds; with a deficient bond it is lazy and then rich. A large root freezes and leaves a degree-$(K-1)$ model, which is not lazy for $K \geq 3$, unlike the two-layer case. These boundaries are tested in Sections 3.2 to 3.4 and 5.4.

B Experimental details

[Skeleton. For every experiment: trees, dimensions, teachers, initialization, dynamics and tolerances, sweeps, seeds, time budgets, compute cost.]

B.1 Common settings

[Double precision; Dormand–Prince 5(4) with relative and absolute tolerances 10^{-8} and 10^{-12} unless stated; recording grid; conservation-drift threshold 10^{-6}; event detection; Hessian and kernel computations.]

B.2 Saddle-to-saddle experiments

[Settings of the experiments of Section 4 (s2s-core, locality, order).]

B.3 Regime-map experiments

[Grids, seeds and time budgets of the phase diagrams of Section 3.]

B.4 Covariance and teacher experiments

[Settings of Section 6.]

B.5 Optimizer and sample-size experiments

[Learning rates, batch sizes, training-set sizes and seeds of Section 5.5.]

B.6 Optimizers used in practice

Settings of Section 5.6. Experiment folder `experiments/optimizers/` (one YAML configuration per sweep in `configs/`); findings and claims withdrawn after review in `notes/optimizers.md`, sections 2, 3 and 5.

Problems. Whitened Gaussian inputs, exact population loss (empirical loss in the finite-sample sweep), double precision. Teacher seed 100; initialization seeds 0–5, from a stream independent of the teacher's.

- *binary4*: binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$; odeco teacher with weights $(3, 2, 1)$ and Haar-random site bases.
- *binary8*, *MPS8*: $N = 8$ ($K = 7$ and 8), the same odeco teacher, $d = 3$, $\chi = 4$.
- *E8* on binary8 and MPS8: pattern teacher with $d = 2$, $\chi = 3$ and patterns $\pi_0 = (1, \dots, 1)$ (weight 1), π_A equal to 2 at sites 4 and 5 (weight 0.4) and π_B equal to 2 at sites 2 and 7 (weight 0.7), 1 elsewhere. The cores that carry the differing sites span $k = 2$ (π_A) and $k = 6$ (π_B) cores on the MPS and $k = 5$ for both on the binary tree.

Initializations. Fan-in (11) with per-core scale σ; *balanced*: the same W_0 in the gauge with $Q_e = 0$ on every edge. All deep-tree baselines use the balanced version, because fan-in gradient flow converges by $t = 2 \cdot 10^6$ in none of the 12 MPS8 odeco runs (the conservation-law trap of Proposition A.28, verified at the route ends in 11 of the 12 stalls), against 10 of 12 (binary8 odeco), 11 of 12 (binary8 E8) and 11 of 12 (MPS8 E8) fan-in runs. Methods whose iterates depend on the gauge were run from both versions; re-gauged GD, Riemannian GD, ALS and DMRG depend only on W_0 and were run from the balanced one.

Dynamics. All methods stop at $L < 10^{-6} L_0$ (finite samples: empirical loss below 10^{-8} times its initial value) or at their budget.

- *Gradient flow*: SciPy LSODA with relative and absolute tolerances 10^{-9} and 10^{-14} and the exact Hessian as Jacobian (Radau as fallback), instead of the default integrator of Section 2.5; the crossings $a_r \in \{0.1, 0.5, 0.9\}$ are located by root finding, so event times are exact. Budgets: $t_{\max} = 8/\sigma$ (binary4), $2 \cdot 10^6$ (deep trees), $4 \cdot 10^6$ (binary8 scan), 10^6 (finite samples).
- *GD and heavy ball* (PyTorch): $\eta = 0.02$ and 0.002 with $\beta = 0.9$ on binary4 ($0.13\,\eta_c$ for both, counting heavy ball's effective step $\eta/(1-\beta)$); $\eta = 0.01$ and 0.001 on the deep trees (0.23–$0.27\,\eta_c$), at $\sigma = 0.8$ only, because gradient-flow plateaus at $\sigma = 0.3$ last 10^4–10^6 time units. Budgets $8/\sigma$ (binary4) and $2 \cdot 10^4$ (deep) gradient-flow time units, at most $5 \cdot 10^5$ steps.
- *Adam* (PyTorch, bias-corrected): $\beta_1 = 0.9$, $\beta_2 = 0.999$ (also 0.99 and 0.9 at $\eta = 10^{-3}$ on binary4), $\epsilon = 10^{-8}$ (10^{-30} in the ϵ sweep), $\eta \in \{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 10^{-2}\}$ on binary4 and $\{10^{-3}, 10^{-2}\}$ on the deep trees. *RMSProp*: $\alpha = 0.99$, $\epsilon = 10^{-8}$, $\eta \in \{10^{-4}, 10^{-3}\}$. Budgets 10^5–$3 \cdot 10^5$ (binary4), $3 \cdot 10^5$ (deep), $5 \cdot 10^5$ (binary8 scan) and $2 \cdot 10^5$ (ϵ sweep) steps.
- *Re-gauged GD*: $\eta = 0.02$ (binary4) and 0.01 (deep); QR sweeps from the leaves to the root bring the cores to the root-centred canonical form every 1 or 50 steps; $5 \cdot 10^4$, 10^5 and $2 \cdot 10^5$ (binary8 scan) steps.

- *Riemannian GD*: $\eta \in \{0.1, 1\}$; tangent projection by least squares with the dense Jacobian of $W(\theta)$ (relative cutoff 10^{-12}), so only small d^N are feasible; leaves-to-root HSVD retraction; 2000 steps.
- *Gram-preconditioned GD*: $\eta \in \{0.01, 0.1, 0.5\}$; each Kronecker factor of a core's Hessian block is damped by 10^{-3} times its mean eigenvalue (scale-free) or, at $\eta = 0.1$, by an absolute floor 10^{-4} or 10^{-2}; no divergence guard, because the first step overshoots by up to $10^{34} L_0$ before the loss returns; $5 \cdot 10^4$ (binary4; $2 \cdot 10^5$ with a floor) and 10^5 (deep) steps.
- *ALS* (one site) and *DMRG* (two sites, then SVD truncation to at most χ values with relative cutoff 10^{-12}): exact least squares; the orthogonality centre moves by QR along a depth-first tour of the tree; at most 50 sweeps.

Sweeps and run counts. Table 7 lists the sweeps; six initialization seeds per setting, 2529 runs in total.

Table 7: Sweeps of the optimizer experiment (Section 5.6), one configuration file per sweep in `experiments/optimizers/configs/`.

sweep	problem	σ	methods	runs
`scaling_b4`	binary4, fan-in	$0.002 \cdot 2^j$, $j = 0, \ldots, 6$	19 settings: gradient flow, GD, heavy ball, Adam (6), RMSProp (2), re-gauged GD (2), Riemannian GD (2), Gram (2), ALS, DMRG	798
`precond_b4`	binary4, fan-in	as above	Gram: 3 step sizes with scale-free damping, 2 absolute floors	210
`deep`	binary8, MPS8; odeco, E8; balanced and fan-in	0.3, 0.8	gradient flow, GD and heavy ball ($\sigma = 0.8$), Adam (2), RMSProp, Gram; from the balanced W_0 only: re-gauged GD (2), Riemannian GD (2), ALS, DMRG	864
`later_b8`	binary8 odeco, balanced	0.25, 0.3, 0.35, 0.4, 0.5, 0.6, 0.8	gradient flow, Adam (2; every step recorded for the first 2000 steps), re-gauged GD	168
`eps_mps8`	MPS8 odeco, balanced	0.01, 0.015, 0.02, 0.03, 0.05, 0.1, 0.2, 0.3	Adam: $\eta \in \{10^{-3}, 10^{-2}\}$, $\epsilon \in \{10^{-8}, 10^{-30}\}$	192
`implicit_bias2_b4`	binary4, $n \in \{45, 60\}$, data seeds 200–203	0.01, 0.1	gradient flow, Adam (2); pairs share draw and initialization	288
`adam_probe`	binary4, binary8, MPS8 E8	single runs	Adam and RMSProp internal state (illustration only)	9

Events and classification.

- Event r is the last upward crossing of $a_r = \frac{1}{2}$ after which a_r stays at least $\frac{1}{2}$; transient crossings, which the first-crossing rule of Section 2.7 would count, are not events. Width: final rise from 0.1 to 0.9. Saddle-to-saddle: every separation ratio $\rho_i > 3$ (Table 3).
- Discrete methods record loss and overlaps at every step for the first 2000 steps, then every $\max(1, n_{\text{steps}}/(5 \cdot 10^4))$ steps. A first escape or gap shorter than two recording intervals is unresolved and excluded from fits (and counted in the result tables). Two components that cross $\frac{1}{2}$ in the same step or micro-step have no resolved order. For Gram-preconditioned GD, events are defined only after the loss has returned below L_0; a component already above $\frac{1}{2}$ at that point has no resolved event.

- Exponents: per seed, a least-squares fit of $\ln T$ against $\ln \sigma$ over the σ at which the seed follows its most common learning order (for plateau durations, not for times to reach a loss level); mean $\pm$ standard error over seeds. A logarithmic law fitted in the same space serves as comparison (gradient flow on binary4: root-mean-square residual of $\ln T$ 0.005 for the power law, 0.62 for the logarithmic law).
- Orders are compared with gradient flow only where both are resolved, for the same problem, σ and seed. Re-gauged GD, Riemannian GD, ALS and DMRG give one trajectory per seed at every σ, so their orders are counted over the six independent seeds (at the smallest σ).
- ϵ *test* (20): $T_{\mathrm{GF}}(\sigma)$ is the seed's exact gradient-flow first-escape time at $\sigma = 0.3$ (`deep` sweep, same balanced initial point up to scale) times $(0.3/\sigma)^6$, and n_{sign} the median over σ of the same seed's escape with $\epsilon = 10^{-30}$ at the same η. Runs without an escape within $2 \cdot 10^5$ steps enter only through the check that the prediction exceeds the budget.
- Convergence for cost and time-to-loss tables: $L < 10^{-3} L_0$. Runs that hit their budget are reported as budget-limited, never as stalls. RMSProp hovers above $10^{-6} L_0$ on the deep trees within $3 \cdot 10^5$ steps; its events are used, its convergence flags are not.

Excluded runs. Runs replaced after review round 1 are kept in `superseded/` subfolders of the raw data and ignored by the analysis: gradient-flow runs whose event times were interpolated on an output grid (errors up to 1.7%), Gram-preconditioned runs stopped by a $10^6 L_0$ loss guard after one step, binary8 Adam runs recorded only every 10 steps, and the first finite-sample sweep (`implicit_bias_b4`, 288 runs, one data draw per initialization). The pilots (`pilot_calib`, `pilot_trap`) support no claim. Raw trajectories (one file per run, 2.2 GB) are not in the repository; Section C says how to regenerate them.

C Reproducibility

[Skeleton. How every number and figure in this report is regenerated.]

C.1 Software and data

[Repository layout, environment and where raw outputs live.]

C.2 Figures

[Table: figure, experiment folder, script and command, config, raw data needed.]

Table 8: Figures of the report: the command that draws each one (from the repository root) and the sweeps whose raw runs it reads. Figures are copied into `report/figures/` by `report/collect_figures.sh` from the list in `report/figures.txt`.

figure	command	raw data (sweeps; Section B)
2	`uv run python -m experiments.optimizers.make_figures --results-dir experiments/optimizers/results b4`	`scaling_b4` (Table 7)
3	as above, with target `eps`	`scaling_b4`, `eps_mps8`, and `deep` for the gradient-flow reference times

C.3 Building the report

The report is built by `report/build.sh` (pdflatex, BibTeX, then pdflatex until the cross-references are stable), after `report/collect_figures.sh` has copied the figures listed in `report/figures.txt` from `experiments/*/results/` into `report/figures/`. `report/README.md` describes both scripts.